Part B

History of Changes

The following is a list of changes that have been implemented from the submitted proposal to the DoA:

DoA_v1_01.05.2015
Put into DoA template.
Addressed Grant Agreement preparation comments numbers 4, 5, 6, 7, 9, 13.
Deliverables Table completed (a lot of it was missing in the proposal).

DoA_v2_15.05.2015
Addressed Grant Agreement preparation comments numbers 3, 8.
Corrected Gantt (there were a number of inconsistencies with the work plan which have been corrected in the Gantt).

DoA_v3_06.07.2015
Correct DoA structure and headings implemented.
A brief overview of the "other direct costs" for partners 1, 2, 3, 5, 6 and 15 has been added in 3.4.

DoA_v4_09.12.2015
Partner 14, BPC communicated their departure from the consortium as of M1, and they will be replaced with Northern England Process Industry Cluster (NEPIC). As such, we have included NEPIC as Partner 16, whereby they have substituted BPC throughout the DoA. A Partner Description has been added for NEPIC in the *Section 4.1 Participants,* and the *Table 3.4b:'Other direct cost' items (travel, equipment, other goods and services, large research infrastructure),* has been updated in *Section 3.4. Resources to be committed* to remove the Other costs for BPC and to include those for NEPIC.
As of M2, a key investigator from UNIVLEEDS, Dr. Andrew Heyes, has transferred to the University of Strathclyde (USTRATH). UNIVLEEDS have agreed to transfer some of their participation in the project to USTRATH, and they have been included in the consortium as Partner 17, whereby we have added them throughout the DoA. In addition: their Partner Description has been added in the *Section 4.1 Participants* and the UNIVLEEDS partner description has been modified accordingly; USTRATH has been included in the *Table 3.4b:'Other direct cost' items (travel, equipment, other goods and services, large research infrastructure)*, and the UNIVLEEDS table has also been modified accordingly.

DoA_v5_27.03.2018
Changes in the beneficiaries for deliverables 7.9 (DECHEMA -> ISL) and 2.3 (ZHAW -> USTRATH)
Changes in budget and efforts after transfer of standardization tasks from DECHEMA to ISL.

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1. Excellence

1.1. Objectives

"*One company's waste and by-products, can be another company's opportunity and cost saving. We need to put those two companies in touch so that we don't just throw away valuable materials but make the most of them."*

Janez Potočnik, European Commissioner for the Environment

Resource efficiency offers major economic opportunities for the European Process Industry, both in terms of cost savings as well as opportunities to offer greener products and services. According to a recent estimate, the EU could realistically reduce the total material requirements of its economy by 17% to 24%, boosting GDP and creating between 1.4 and 2.8 million jobs[1].

In order to migrate to the energy and resource efficient zero-waste process industry of the future, the SPIRE work programme calls for "*solutions for more efficient processing and energy systems for the process industry, including* ***industrial symbiosis***". Industrial symbiosis (IS) is the use by one company or sector of by-products, including energy, water, logistics and materials, from another. IS networks have proven successful not only in diverting waste from landfill, but also in contributing to the preservation of resources and moving waste up the value chain. They have also been an accelerator of innovation and creation of green jobs. The European Resource Efficiency Platform has championed IS as a mechanism for reducing carbon, preserving critical resources and securing business sustainability. However, **inadequate business-to-business information on, for example, what resources a product or process contains hinders efficient material flows and the creation of value in the circular economy**.

Objectives

In order to pave the way forward for industrial symbiosis as a "*solution for more efficient processing and energy systems for the process industry*"[2], this project will develop a **secure platform for the flexible management of shared process resources** (SHAREBOX- figure 1) that will provide plant operations and production managers with the robust and reliable information that they need in real-time in order to effectively and confidently share resources (plant, energy, water, residues, and recycled materials) with other companies in an optimum symbiotic eco-system.

The Platform will consist of a module for facilitating the identification and realisation of new cross-sectorial interactions ('synergies'), that draws upon the expertise of partner ISL who have designed a standalone database software from partner ISL (SYNERGie®) in commercial use around the world supporting facilitated industrial symbiosis. This module will be embedded in the SHAREBOX Controller, which will consist of a suite of newly developed analysis and optimisation tools for flexible energy use and material flow integration for optimising symbiosis among companies**.** These tools will be based on input-output (IO) modelling for resource (waste and energy) supply-demand matching and process efficiency analysis (to understand physical and technological conditions), game theoretical (GT) approach for integrating company behaviour in cost-, benefit-, and resource-sharing (to understand economic conditions), and agent-based modelling (ABM) for designing the (economic, environmental, and social) optimal symbiotic network (to have the holistic optimum). The outputs from the SHAREBOX controller will provide plant and operations managers with commands for actions to be taken and/or recommendations for decision support. It will be ensured that all commands and recommendations a) fulfil plant operations requirements, b) are within the constraints of any contractual obligations, c) are in compliance with all regulatory thresholds, and d) deliver optimal impacts in terms of cost/savings and ecological footprints. The historical data that is generated by the SHAREBOX platform will be processed by data mining tools that will provide the production/process BIG DATA for symbiotic shared resources optimisation.

Once the potential synergies, i.e. resources that can be shared, have been identified by means of the SYNERGie® module, the companies (players) have to establish contractual deals in terms of shared resources, costs and benefits.

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1 EUROPEAN COMMISSION STAFF WORKING DOCUMENT. Progress Report on the Roadmap to a Resource Efficient Europe. COM(2014) 398} {SWD(2014) 211. Available at: http://ec.europa.eu/environment/circular-economy/pdf/Progress-report-roadmap.pdf

2 Pablo Tello & Ron Weerdmeester. SPIRE Roadmap http://ec.europa.eu/research/participants/portal/desktop/en/opportunities/h2020/topics/2167-spire-08-2015.html

The SHAREBOX mathematical tools -based predominantly on game theory and implemented as easy-to-use "wizards"- will assist them to find win-win optimal deals.

Once the deals have been signed and translated into mathematical constraints/conditions to the operational variables - through another set of "wizards"-, SHAREBOX will generate operative commands -recommended and mandatory actions- according to real-time optimal decisions automatically generated by means of Theory of Scheduling algorithms.

In an inner holistic layer, a self-learning artificial intelligence sub-module -based on some data mining tools like multivariate analysis and neural networks routines and case-based reasoning- collects and analyses the data according to a multi-way approach in order to unveil complex hidden trends and relationships which cannot be rationalized by deterministic reasoning. These trends/relationships can be integrated into the core algorithm of SHAREBOX to enable further improvements according to a predefined Core Algorithm Maintenance Schedule.

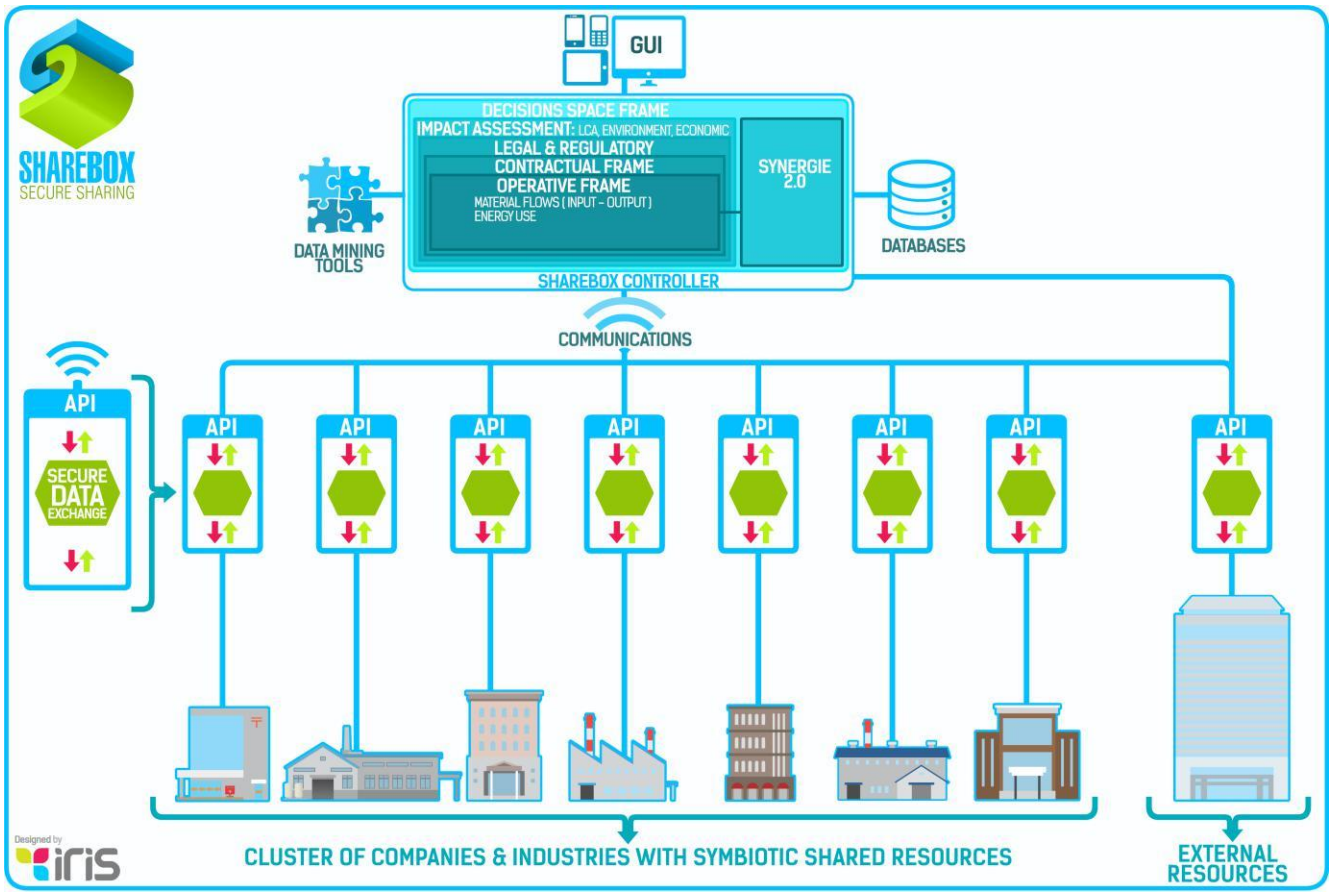

Figure 1- Overview of the SHAREBOX platform

In order to test and validate the SHAREBOX platform using realistic industrial streams and process conditions, and to demonstrate its impacts of reduced carbon emissions and cost, SHAREBOX partners include **a representative group of industrial demonstration sites** from Germany, Spain, UK and Turkey involving SPIRE process industries (chemical, ceramics, petroleum and rubber industries, to ceramics minerals, metal, mining, etc.), and other sectors (including automotive, food, energy, among others) for cross-sector working. SHAREBOX will be used to optimise industrial symbiosis in these industrial parks and zones, as well as to strengthen cross-sectoral interaction.

1. **Chemie-Cluster Bayern** (partner **CCB**), which links over 260 chemical companies from the region of Bavaria in Germany, SHAREBOX will be used to optimise industrial symbiosis among selected sites from the Cluster, as well as to strengthen cross-sectoral interaction.
2. The **North East of England Process Industry Cluster** (partner **NEPIC**) is a process industry cluster with over 720 active participants from Chemical, Petrochemical, Polymer, Specialty Chemical, Fine Chemical, Pharmaceutical, Biotechnology, Composites, Bioresources, Biofuels and Renewable Materials and Energy

companies which is aimed at the rejuvenation and growth of these key hi-tech advanced manufacturing industries while leading them into a lower carbon sustainable future.

3. The **Eskişehir Organized Industrial Zone** (partner **ECI**) home to over 322 companies present from chemical, petroleum and rubber industries, to ceramics minerals, metal, mining, food and energy, among others.
4. **Ceramics Cluster** (partners **AICE, KEROS, KERAFRIT, ISSA, GUZMAN**). SHAREBOX will be used to instigate industrial symbiosis among the Ceramics cluster in Castellon and with the wide industrial tissue of the region, which will be managed by partner **AICE.** This new symbiosis commences with a kernel set of companies, partners **KEROS** and **KERAFRIT**, ceramics companies from the Nules area, as well as partner **ISSA** producing helicoidally suspensions for the automotive industry and GUZMAN, a raw material supplier for mineral, rubber, polymer, chemical processors, amongst others.

The ShareBOX platform will be co-created, implemented and tested with these demonstration sites **using realistic industrial streams and process conditions**. The platform will optimise interdependencies and the identification of technology components allowing for a breakthrough in difficult cross-company collaboration, resulting in a cost-effective reduction in resource use. Production, quality and operations managers will be reassured that resource sharing does not involve trade-offs in relation to access to sensitive product or process information by competitors (which will be ensured through the encryption of data), nor in their ability to have a secure and failsafe supply of quality resources (which will be guaranteed through the powerful predictive capability of the system to forecast drops, as well as peaks, in supply). The Platform will be validated for its potential to optimise energy and resources supply and demand and to reduce the overall costs in energy intensive industries by at least 15%, to ensure we fulfil an expected impact of the call topic.

The wide deployment of the SHAREBOX platform will have a major impact for the competiveness, resource efficiency and sustainability of the European process industries and other sectors in that it will a) facilitate industrial symbioses through the identification of new synergy opportunities, and b) optimise existing Industry Symbiosis or synergies among multiple companies on a single industrial production site.

In order to achieve the overall goal of building, testing and validating the world's first energy and resource management platform for improved efficiency in the process industries (figure 1), a number of **specific objectives** that will be fulfilled over the course of this 48-month Research and Innovation Action have been defined. These objectives are S.M.A.R.T- i.e. **S**pecific, **M**easurable, **A**ttainable, **R**ealistic and **T**ime bound:

Table 1- Specific Objectives of the SHAREBOX project

1	**To define the requirements of the participating industry sites and to draw up the specifications of the SHAREBOX platform** *(linked to WP1)*
A Kano Survey will be performed with the participating industry sites and the Quality Function Deployment method will be implemented to ensure a systems approach is used to establish the industry requirements that will feed into the derived specifications for the SHAEBOX Platform. ***Performance Indicator(s):*** By M7, strategies for the industry sites will be defined, along with the derived system specifications for the SHAREBOX Platform. Other potential indicators: number of companies attracted in the proposal, number of resource (waste) types included in the database.	
2	**To establish the Industrial Symbiosis methodologies** *(Linked to WP2)*
We will build on over 10 years of practical experience to move ICT to the next level whereby knowledge and opportunities will be readily available to companies together with on line advice as to how to proceed. The ICT implementation will learn from experience and apply the learning to seeking out opportunities which will then be delivered to the companies without the company having to spend time searching for the opportunity. Various smart delivery methods will be implemented so that the information will be easily accessible together with stepwise guides showing how the companies should progress identified opportunities. ***Performance Indicator(s):*** By M6, IS methodologies are defined in a comprehensive backlog that will enable their coding into the SHAREBOX Platform software and taxonomies for next generation ICT will be defined and aligned with international standards.	
3	**To research the necessary infrastructures that would be required for realising Industrial Symbiosis**

(Linked to WP2)

Having identified the opportunities for IS between companies on an industrial park we will also consider, design and cost the necessary additional infrastructure required to make exchange of materials and energy possible. These may include energy conversion technologies for waste heat to electricity and state of the art storage technologies for time-shifting for increased utilisation of resources by better synchronisation between processes. Furthermore, we will incorporate the opportunity presented by demand management and reflect the impact of this opportunity on the design of new energy conversion/storage/transportation systems.

Performance Indicator(s): By M12, A Database of energy transfer, storage and conversion technologies parameterised by their input/output requirements and efficiencies as a function of key variables such as waste stream temperature and scale will be produced. By M24 with a cost model for new energy technologies in terms of capex and opex on per installed capacity and per unit output basis and for end-use-energy-demand (export and import) opportunities to inform management strategies associated with new energy technologies. By M30, optimised methodologies for designing and managing new energy infrastructure will be ready to be converted into an energy technology wizard under SHAREBOX.

4 To develop intelligent decision support tools that will be integrated into the SHAREBOX platform *(Linked to WP3)*

SHAREBOX will consist of a comprehensive series of tools as an integrated model of energy and materials use across multiple process industries. On the one hand, tools will dynamically map the input-output of material flows within a plant. Material flow analysis (MFA) is an analytical method of quantifying flows and stocks of materials or substances in a well-defined system. MFA is an important tool to assess the physical consequences of human activities and needs in the field of industrial ecology, where it is used on different spatial and temporal scales. On the other hand, tools will analyse energy-related business cases based on end-user energy demand management knowledge and energy storage technology that allows time shifting and delivery of waste energy from one process to another at the most opportune moment. As such, a series of mathematical tools will be developed based on input-output (IO) modelling, game theory (GT), and agent-based modelling (ABM) for enabling flexible energy use and material flow integration for holistic resource management in process industries. A number of data mining techniques (Multivariate Analysis, Artificial Neural Networks,...) and/or some Artificial Intelligence reasoning paradigm (Case-Based Reasoning, ...) will be used to improve recommendations and/or strategies for synergies decisions. All the decision support tools will be integrated in WP4 into the SHAREBOX software platform.

Performance Indicator(s): By M10 optimum waste supply-demand quantity matches are ready, the energy scheduling and conversion tools, followed by the benefit-sharing schemes which will be produced by M18. ABM-based optimal designs will be produced by M30. So, by M30 algorithms will be ready of incorporation into the SHAREBOX system software, along with flow diagrams and analytical routines. Any shortcomings will be addressed after testing the algorithms in the real symbiosis cases and the models will be re-visited (by M40).

5 To develop the SHAREBOX system software and to build the platform architecture. *(Linked to WP4)*

The Platform architecture will be drawn up based on system specifications defined from WP1. The system controller core routines will be developed. The SYNERGIES software will be upgraded. The Platform backend and frontend infrastructure will be developed, whereby the mathematical tools, data mining tools, or artificial intelligence reasoning paradigms will be integrated, and the databases, APIs, communications will be implemented. An ergonomic, user centred Graphical User Interface will respond to end-user tastes and requirements.

Performance Indicator(s): By M12, the integrated Platform will have been successfully tested in a virtual simulator confirming that the on-site pilot trials can commence. An alpha release will be made, and the platform will continue development to a beta version.

6 To test and validate the SHAREBOX platform in the field using real industrial streams and process conditions from industry partners *(Linked to WP5 and WP6)*

Through an IS implementation team, the SHAREBOX platform will be set up and will be used, on the one hand, for optimising industrial symbiosis among selected sites and to strengthen cross-sectoral interaction with partners CCB, NEPIC, and ECI (WP5). On the other hand, SHAREBOX will be used to instigate industrial symbiosis among the Ceramics cluster in Castellon and with the wider industrial community of the region (WP6). These demonstration sites will first be engaged in WP1 Kano survey of user needs, and commence (WP5) with an early alpha version of the platform to verify its performance, and to ensure the versatility of the platform to the specific conditions in real

production units. Feedback from the sites will be fed back to the development team to guide the various iterations of the Platform. Verification of the impacts of SHAREBOX on resource efficiency, energy efficiency and the emission performance, and cost-effectiveness will be performed.
Performance Indicator(s): By M24, engagement with 1,000 companies, resource and process data received from 250 companies, 1,000 synergy opportunities identified and impacts quantified.

7	**To verify the impacts of the SHAREBOX platform on resource efficiency, energy efficiency and the emission performance, and cost-effectiveness** *(linked to WP6)*

An LCA will be performed to verify the energy, resource savings as well as the economic savings in a holistic LCA approach. ***Performance Indicator(s):*** By M45, the SHAREBOX impacts will be confirmed.

8	**To perform data mining of the historical data generated from the industry sites for further symbiotic shared resources optimisation** *(linked to WP3)*

Data mining capability will be a feature of the SHAREBOX platform. To this end, the latest in data mining technology will be used to develop the data analytics capability of the platform to mine the historical data (Big Data) that is being gathered by the platform in order to further optimise the symbiotic shared resources. A Case-Based Reasoning (CBR) approach will be used to increase the competence of the recommendation process through the learning from past experiences to improve recommendation decisions.
Performance Indicator(s): By M36, the data collected from the two demonstration sites will have been mined to validate the data mining functionality of the platform and recommendations for IS optimisation will be reported.

9	**To develop a business model for monetising the SHAREBOX Platform.** *(linked to WP7)*

The results protection routes will be evaluated and all IPRs defined, and the consortium will build on the initial exploitation and business plans that have been agreed by all partners and presented in Section 2.2. To support a broad exploitation and implementation of the platform, it will be crucial to develop some scenarios of how SHAREBOX can support the business development of industry park operators, including the use of SHAREBOX for site marketing strategies. The business model for the SHAREBOX Platform will consider certain types of industrial parks (integrated parks, "Verbund"-production, heterogeneous industrial areas) and include tailor-made offers for the integration of SHAREBOX into the management processes of targeted site operators. With a view to gearing up for commercial adoption of Platform and the intensified modules in the industry, standardisation and regulatory issues will be addressed to ensure compliance with process conditions. ***Performance Indicators(s):*** By M24, an analysis of target customers and users among industrial park operators and a draft of the business plans outlining the business model for commercial exploitation of the results is in place and the protection procedures for the projects results defined. By M48 a final business plan is in place.

10	**To maximise the innovation impacts of the project through effective communication, dissemination and contribution to standards** *(linked to WP7).*

A communication and dissemination campaign will be implemented to raise awareness about the project and its activities among the industry and to stimulate future market demand, to outreach to and engage with policy makers and the general public, and to secure 'buy-in' for the project. The knowledge generated during the project will feed into training and education programmes to upskill and bridge knowledge gaps in relation to IS. ***Performance Indicators***: By M48, wide dissemination of project results to the process industry, plant and equipment providers, policy makers, educational bodies, the research community and the general public is reported. Training outputs will be available for up-skilling in the European Process industry as well as for education programmes. Business and exploitation plans are finalised.

The European process industry is uniquely positioned and represents the economic roots of the European economy (by transforming raw materials into intermediate and end-user products but also using energy, water, and possibly producing by-products, emissions and wastes)[2]. It thus sits at the core of most industrial value chains via discrete manufacturing into a whole host of sectors[2]. The process industry is characterised by having a high dependence on resources (energy, raw materials and water) in production. Optimising the effectiveness of process control systems and strategies is **vital to improve quality, reliability and efficiency[3], as well as the sustainability of operations**.

3 http://www.spire2030.eu/mediaroom/36/21/Process-Control-Industrial-Automation-Summit-2014

1.2. Relation to the work programme

SHAREBOX mobilises a multidisciplinary team who will work together to respond to the specific challenge of topic ***SPIRE-06-2015: Energy and resource management systems for improved efficiency in the process industries***, whereby we propose to develop a highly scalable, secure platform for:

a) **Identification of new synergies.** The upgraded SYNERGie® module that will be integrated into the SHAREBOX platform will enable next generation IS delivery through the 'smart' identification and realisation of new cross-sectorial interactions that will provide the foundations to enable 'a more general cross-sectorial interaction' by providing the next generation of IS delivery.

b) **Optimising IS implementation.** The SHAREBOX intelligent decision support tools for the flexible management of shared process resources will provide plant operations and production managers with the robust and reliable real-time needed to optimise symbiotic connections (plant, energy, water, residues, and recycled materials) with other companies in a symbiotic eco-system.

The SHAREBOX project also develops '*new approaches that perform cost-saving optimisation of energy and resources supply and demand'.* Based on the input-output approach, SHAREBOX will implement the optimum IS as a pre-design model for future symbiotic networks (SNs) and use it as the theoretical optimum that the existing SNs should reach. Based on a game theory approach SHAREBOX will construct the benefit-sharing schemes for involved actors integrating the potential cooperation strategies in the negotiation phase of IS. Through an agent-based modelling approach SHAREBOX will enable the optimum IS design embedding the company behaviour into the operations of IS.

This '*holistic*' integrated platform will be a highly scalable and will '*integrate into a single management system" all the* necessary *'environmental, energy and economic factors*', as well as regulatory and contractual requirements, to ensure optimised industrial symbiosis. Consequently, it will be a key platform for improving the resource and energy efficiency of the process industries.

As identified by SPIRE-06, a key challenge that is preventing the development of technical and non-technical interactions and exchanges is that of '*poor understanding of each other's processes'.* Behind this challenge is a **lack of trust among industry to cooperate at a level which could provide visibility on how their operations are run**. Indeed, lack of cooperation and trust, as well as lack of information, are frequently cited among the key barriers to industrial symbiosis[4,5]. This is a common and longstanding phenomenon in competitive industrial environments.
However, nowadays, with the advancement in information and communication technology, information sharing can be more conceivable[5]. **To** this end, one way to overcome the willingness to share information, and in particular potentially sensitive process information, is to develop an information sharing platform that **offers outstanding security and anonymity prior to the agreement to progress an idea**. We propose to do this through APIs for secure data exchange between companies. Moreover, the platform will properly encrypt or anonymize the data that is being fed into the system so that it does not reveal itself to other end-users, rather it is used intelligently by the platform's models and tools that will process the data for the purpose of '*flexible energy use and material flow integration for holistic resource management in process industries*'. The highly flexible and easy to use platform will be '*suitable both for small and large scale in a flexible approach*'. It may be implemented for the supply being generated by a park operator at first step and then being extended to the entire resource flows of a production site; or it may be used by a single tenant, then inviting neighbouring companies to join gradually. This way, SHAREBOX can be used as a nucleus for the continuous development of site integration advantages.

As required by the call scope, we will develop an open standards-based platform, whereby we will on the one hand ensure that there is no proprietary programming languages, development environments, middleware services, or back-

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[4] Artem Golev, Glen D. Corder and Damien P. Giurco (2014). Barriers to Industrial Symbiosis. Journal of Industrial Ecology. Article first published online: 14 JUL 2014. DOI: 10.1111/jiec.12159

[5] Zahra Lotfi, Muriati Mukhtar, Shahnorbanun Sahran, Ali Taei Zadeh (2013). Information Sharing in Supply Chain Management Procedia Technology. Volume 11, 2013, Pages 298–304. 4th International Conference on Electrical Engineering and Informatics, ICEEI 2013. doi:10.1016/j.protcy.2013.12.194

end connectors or adapters- i.e. no vendor lock-in. On the other hand, we will utilise common and accepted official and unofficial Industry Standards such as JavaScript, CSS, HTML5, J2EE, CSS, REST, SOAP, XML, and industry standard web services.

Our approach is based on '*rapid transfer from lab-scale and conceptual design into testing at demonstration sites, using realistic industrial streams and process conditions'*, whereby pilot tests will be carried out at existing established organised production sites of CCB in Germany, ECI in Turkey, and NEPIC in the UK to optimise symbiosis among and between SPIRE industry companies, and to establish industrial symbiosis among the ceramics cluster in Castellon, Spain (managed by AICE and with KEROS, KERNFRIT, ISSA and GUZMAN as the kernel set. Not only will these demonstration sites serve to verify the performance of the '*integrated solutions and tools to the specific conditions in real production units'*, they will also "*facilitate future industrial symbiosis between different sectors"* (e.g. ceramics and automotive sector) *"by integrating energy and material flows within existing industrial parks*" (the Nules industrial park in Castellon, Spain). Information on eco-innovation solutions will be collected by partners in the pilot phase. SHAREBOX will stimulate demand for eco-innovative solutions to resource challenges, resulting in qualitative and quantitative benefits for participating companies.

Finally, the SHAREBOX consortium already have an initial business model concept drawn up for monetising the SHAREBOX Platform and have ambitious exploitation plans (discussed in Section 2.2) to ensure that this European developed platform can grow to become a globally recognised Energy and resource management platform for improved efficiency in the process industries, and as an enabling tool for industrial symbiosis. Our robust business and exploitation planning will ensure the future deployment of the platform in Europe (and indeed globally, leveraging ISL's world-leading IS delivery across Europe, North and South America, Asia and Africa), whereby it will make a pragmatic contribution to increasing IS in the European Process industry by providing the secure, confidential and reliable B2B information that will eliminate any of the risks (perceived or real) associated with energy and resource sharing among industry.

Post-project, partners will scale-up, deploy and roll-out the SHAREBOX Solution in markets across Europe and around the world. Moreover, our budget is within scope (< €5 M) and the outputting of the world's first Secure Management Platform for Shared Process Resources, along with 3 validated resource sharing case studies within and between SPIRE industry sectors [chemical, ceramics, engineering, minerals, non-ferrous metals, steel and cement] in real industrial settings, as well as cross-sectoral industry engagement delivering real impact, is considered **excellent value for high impact.**

1.3. Concept and approach

The overall concept underpinning the project

Industrial symbiosis (IS) has been defined as engaging "engaging "diverse organisations in a network to foster eco-innovation and long-term culture change. Creating and sharing knowledge through the network yields mutually profitable transactions for novel sourcing of required inputs, value-added destinations for non-product outputs, and improved business and technical processes."[6]

In fact, the term industrial symbiosis has its origins in the Danish municipality of Kalundborg, where the first recognized and most well-known example of an industrial symbiosis network as an intricate network of material and energy exchanges is located[7]. In 1989 the seminal article by Frosch and Gallopoulos[8] in Scientific American envisioned "industrial ecosystems" in which "the consumption of energy and materials is optimized and the effluents of one process serve as the raw material for another process." Shortly after, a cluster of companies from different industries that were intensively sharing resources in Denmark came to light[9,10]. This symbiosis in Kalundborg, Denmark began

[6] Lombardi DR and PT Laybourn. 2012. Redefining Industrial Symbiosis: Crossing Academic-Practitioner Boundaries. *Journal of Industrial Ecology* 16(1): 28-37. Both authors, at ISL, are closely involved with the Sharebox proposal

[7] Zahra Lotfi, Muriati Mukhtar, Shahnorbanun Sahran, Ali Taei Zadeh (2013). Information Sharing in Supply Chain Management Procedia Technology. Volume 11, 2013, Pages 298–304. 4th International Conference on Electrical Engineering and Informatics, ICEEI 2013. doi:10.1016/j.protcy.2013.12.194

[8] Frosch, R. A. and N. E. Gallopoulos. 1989. Strategies for manufacturing. Scientific American 261(3):144–152

[9] Knight, P. 1990. A rebirth of the pioneering spirit. The Financial Times, sec.1, p. 15, 14 November.

because of the low availability of groundwater and the need for a surface water source which, once identified, became a key part of the resource exchange network there[11]. The first exchanges were in the 1970s, and, by the late 1980s, at least ten additional exchanges had begun across multiple firms[12] (Symbiosis Institute, 2003). Yet, until some local high school students prepared a science project in 1989 in which they made a scale model of all the pipelines and connections in their small community, the unique aspects of the project went largely unnoticed. The industrial symbiosis network in Kalundborg provided a concrete realization of the industrial ecosystems Frosch and Gallopoulos theorized[11]. The Kalundborg symbiosis emerged from self-organization initiated in the private sector to achieve certain goals, such as cost reduction, revenue enhancement, business expansion, and securing long-term access to water and energy. Once a revelation was made, a coordinative function was found to be helpful in organizing more exchanges and moving them forward. In Kalundborg, for example, managers belonged to an Environment Club and a coordinative organization, the Symbiosis Institute, was launched in 1996 as part of Kalundborg's industrial development agency, specifically working to accelerate the number and complexity of new exchanges[11]. In short, the network that has evolved since the early 1970's was first recognized for its environmental efficiency in the late 80's. By sharing ground water, surface water, steam and fuel as well as exchanging by-products to serve as input materials in other firms' processes, **both high financial and environmental efficiency have been achieved**[11].

Industrial symbiosis since has gained increasing attention in the production economics as the pressure on companies increases for the reduction of waste emissions as well as of primary resources consumption[13]. It is fast becoming the most talked about approach for delivering the circular economy[14]. Simple examples include the use of food waste from the catering sector to feed farm animals, or the use of non-toxic industrial waste to produce energy through incineration[15]. However, in a developed economy with many industrial activities, many different by-products are generated, and the range of potential uses for them can be equally diverse[15].

Many technical solutions for waste and by-product material, water, and energy reuse between neighbouring industries (so-called synergies) have been discovered and applied in the IS examples from all over the world[16]. The industrial area of Händelö in Norrköping, Sweden, for example, has received some attention in the last few years for its example of a renewable energy driven industrial symbiosis network[17,18]. This industrial symbiosis network has evolved spontaneously **because of profitability in each connection**. A central connection in this network is the steam delivery from the CHP plant managed by heat provider E.ON Värme to large scale ethanol producer Lantmännen Agroetanol. E.ON has realized the potential business and environmental value of this connection and have ambitions to develop similar synergistic connections in the industrial area of Northern Harbour in Malmö, Sweden where they also operate a natural gas fuelled CHP plant.

However, such **industrial systems rarely organize themselves by individual initiative only** and many factors need to be considered in order to effectively implement them. While economic convenience is generally considered the most important driver for IS[11], it is recognised that trust plays an important role in sustaining cooperative relationships[19]. Indeed, the potential for uptake of new synergies is often limited by a range of non-technical barriers identified and detailed in the literature, which include lack of cooperation and trust between industries in the area, lack of information sharing, environmental regulation, and economic barriers[20,21,22,23].

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10 Barnes, H. 1992. Fertile project exploits recycled wastes. Survey on Denmark. The Financial Times, 8 October, 5
11 Chertow, M.R. (2007), "Uncovering" Industrial Symbiosis. Journal of Industrial Ecology, 2007. Volume 11, Number 1. Page 12-30.
12 Lowe, E. A., S. R. Moran, and D. B. Holmes. 1995. A fieldbook for the development of eco-industrial parks. Report for the U.S. Environmental Protection Agency. Oakland, CA: Indigo Development International
13 Devrim Murat Yazan, Vincenzo Alessio Romano, Vito Albino. How Far is an Input-Output Production System from Perfect Industrial Symbiosis? University of Twente and Politecnico di Bari
14 European Industrial Symbiosis Association: http://www.eur-isa.org/
15 http://ec.europa.eu/environment/ecoap/about-eco-innovation/experts-interviews/20140127_industrial-symbiosis-realising-the-circular-economy_en.htm
16 Artem Golev, Glen D. Corder and Damien P. Giurco (2014). Barriers to Industrial Symbiosis. Journal of Industrial Ecology. Article first published online: 14 JUL 2014. DOI: 10.1111/jiec.12159
17 Baas, L. (2011). "Planning and uncovering industrial symbiosis: Comparing the Rotterdam and Östergötland regions", Business Strategy and the Environment, vol. 20, no. 7, pp. 428-440.
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19 Hewes, A.K., Lyons, D. (2008). The humanistic side of eco-industrial parks: champions and the role of trust, Regional Studies, Vol. 42, pp. 1329-1342.
20 Chertow, M. R. 2003. Evaluating the success of ecoindustrial development. In Eco-Industrial Strategies: Unleashing Synergy Between Economic Development and the Environment, edited by J. M. E. Cohen-Rosenthal. Sheffield, UK: Greenleaf Publishing.
21 Gibbs, D. C. 2003. Trust and networking in interfirm relations: The case of eco-industrial development. Local Economy 18(3): 222–236

This is where **facilitated industrial symbiosis programmes have come into play**, involving a form of brokering to bring companies together in innovative collaborations, finding new ways to use resources, and thus to increase revenues while reducing waste and the associated costs[15]. The most successful implementations of IS throughout the world have been delivered with the support of facilitation to companies. This facilitation aids the capture of resources and assists the companies in identifying and then progressing opportunities to valorise those resources. When compared with more passive, non-facilitated implementations such as on line waste exchanges, the facilitated model provides a process that is both replicable and scalable across markets and situations.

The delivery most cited in European policy is the facilitated National Industrial Symbiosis Programme (NISP) model that was devised by partner ISL. The NISP model involves expert facilitators **gathering information from companies, making expert assessments, then identifying and facilitating synergy opportunities through to completion.** Based on NISP data, 90% of successful IS opportunities (called 'synergies') move the resource up the waste hierarchy, and in 20% of activity, it involves innovation (either new application of existing information, or new research and development). The success of NISP has inspired many parties to implement IS across Europe: current activity largely comprises regional efforts supported by public-sector investment. It is noteworthy that NISP has informed the SPIRE Roadmap and has engaged successfully with all SPIRE sectors. This holistic approach has demonstrated *in practice* verified positive impacts on materials, water and energy use, innovation, knowledge transfer and best practice sharing, capacity utilisation, and job creation.

Inspired by NISP and through the collaboration of ISL, other facilitated IS Programmes have been developed, such as the Iskenderun Bay Industrial Symbiosis project in Turkey, and has resulted in the Ministry of Development and its regional agencies adopting industrial symbiosis as a tool for regional development. This project was supported by Baku-Tbilisi-Ceyhan (BTC) Petrol Pipeline Company and included forming of the infrastructure that is required for the industrial symbiosis in the region, the implementation of pilot applications and development of proposals for a national program. ISL has further supported regional IS delivery in Finland and South Africa, where regional programmes scaled up to national in 1-2 years. ISL also has European IS activity in Belgium, Netherlands, Italy, Poland, Hungary, Romania, Denmark, and a new national programme launching imminently in France – all modelled on NISP®.

Finance continues to present a potential barrier to full scale implementation of facilitated IS across Europe and beyond. Facilitation is expensive to implement.

The approach that underpins SHAREBOX centres on a logical work flow (Figure 2) that covers from the identification of new symbiotic synergies right through to optimised connections among companies and organisations in established symbiotic relationships. This logical series of steps is facilitated by the state-of-the-art SHAREBOX ICT platform and its tools.

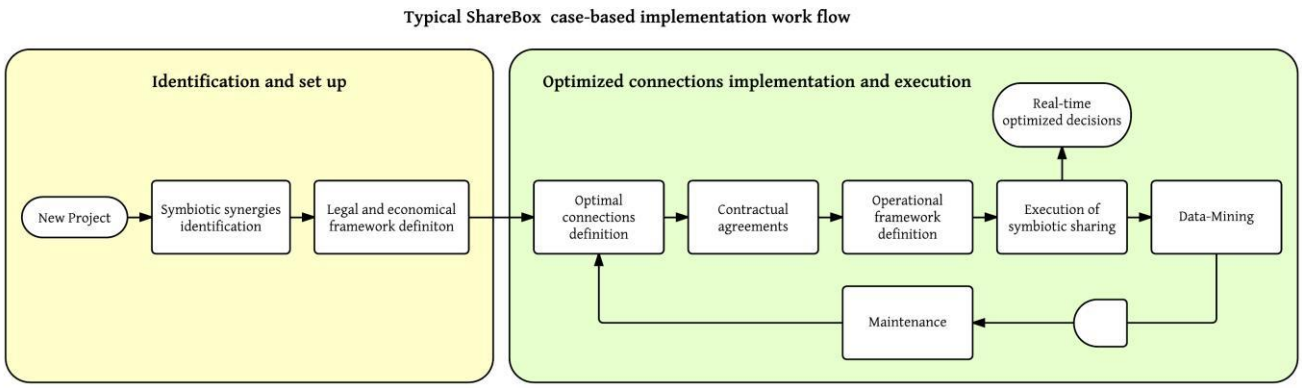

Figure 2- Typical SHAREBOX case-based implementation work flow

[22] Ehrenfeld, J. and N. Gertler. 1997. Industrial ecology in practice: The evolution of interdependence at Kalundborg. Journal of Industrial Ecology 1(1): 67–79.
[23] Mirata, M. 2004. Experiences from early stages of a national industrial symbiosis programme in the UK: Determinants and coordination challenges. Journal of Cleaner Production 12(8–10): 967–983

As can be seen from Figure 2, SHAREBOX will enable next generation IS delivery, through a ground-breaking new, single integrated information module (SYNERGie® 2.0) to facilitate the identification and uptake of commercial IS opportunities for mainstream businesses by lowering the requirement for IS expertise to progressing IS.

Through its mathematical tools, SHAREBOX will also enable robust and intelligent energy and resource management. Our project frames IS networks as complex adaptive systems and proposes the implementation an agent-based simulation model (ABM) consisting of N agents (companies), each of them available to create a symbiotic relationship with other agents, and characterized as an input-output process. IO modelling and GT precede the implementation of the ABM to ensure that the physical, technological, and economic pre-requisites are reached. To this end, through our platform, we will bring a global innovation for effectively running an optimised, systematic symbiosis among process companies in a cluster, whereby they can incorporate symbiotic exchanges with the companies and organisations from the cluster into the operational planning of their entire production processes, as well as draw on new exchanges with external companies as required through the SYNERGie® 2.0 module.

The consortium's experience combines excellence in expert-supported IS delivery drawing on an unparalleled engagement with companies of all sizes and sectors (ISL), IO modelling, GT, ABM (UT), energy scheduling and conversion (UNILEEDS), purification and waste treatment technologies and Life Cycle Assessment (ZHAW), ICT and software (IRIS), regulatory and standards expertise (DECHEMA), expertise in industry park operations (CCB with its Competence Center for Industrial Site Development, EIS, NEPIC), chemical cluster (CCB), ceramics cluster (AICE, KEROS, KERAFRIT), raw materials supplier to the process industry (GUZMAN), cross-sectoral cases (e.g. ISL, ISSA-automotive sector). The strong presence of industry partners from the process industries and other sectors (KEROS, KERAFRIT, GUZMAN and ISSA) is very important for this project as we aim to securely enable information about each other's processes to be fed into the platform to enable integrated energy and material flows between companies that will share resources. These end-user industry partners will work closely with the development teams in defining their requirements and expectations for the platform, as well as providing the demonstration sites, whereby the alpha version of the SHAREBOX platform can be trailed using realistic industrial streams and process condition to enable practical feedback to be fed back to the development teams and incorporated into the various iterations of the platform. These industry partners are also early adopters of the platform and their use cases will serve as powerful showcases of the value of the SHAREBOX platform for facilitating industrial symbiosis for improved efficiency in the process industries, and cross-sectors. In short, this consortium is uniquely qualified to develop, pilot and subsequently exploit the innovative and sustainable SHAREBOX Platform.

Positioning of the project

In keeping with the expected scope of projects within SPIRE-06, we will ensure rapid transfer of the SHAREBOX tools for flexible energy use and material flow integration from lab-scale and conceptual design. In relation to the Technology Readiness Levels (TRLs) of the **different flexible energy use and material flow integration tools** that will feed into the overall SHAREBOX platform, these are as follows:

Although the most of the tools to be used for implementing the SHAREBOX Platform have different TRLs –typically ranging from 4 to 9- the combined use for the purpose of the SHAREBOX Platform has been only assayed at laboratory scale. The equilibrated combination of mathematical foundations and expertise on implementation and management of industrial symbiotic networks within a reliable software platform will assure delivering, by the end of the Project, to a TRL6-rated SHAREBOX Platform which can be further upgraded to a commercial-grade product according to the expected recommendations derived from the validation in real conditions.

By the end of the project, the integrated unique SHAREBOX ICT platform, through which enterprises will securely exchange the data via APIs, will be pilot tested at demonstration sites, using industrial partners' real production units. The SHAREBOX Platform will be developed to a beta version that will be hardened post-project, whereby the immediate post-project phase will centre on implementing additional case studies to further develop the platform itself and to serve as powerful endorsers for gaining market confidence.

National or international research and innovation activities

Industrial symbiosis eco-systems exist in Europe and beyond. SHAREBOX is not decoupled from this eco-system and the SHAREBOX consortium is uniquely positioned to leverage beneficial exchange between our project and ongoing

efforts of the numerous IS-related networks, initiatives and projects. It is noteworthy that partner International Synergies Limited (ISL) was the initiator leading to the establishment of the European Industrial Symbiosis Association (EUR-ISA) to support the implementation of industrial symbiosis across Europe, and provides a single point of contact with the multiple industrial symbiosis projects in different Member States. EUR-ISA (letter of support provided in Annex I) aims to promote cross-border opportunities for the 20,000+ businesses currently participating in the networks, and for the thousands of companies expected to engage with industrial symbiosis in the near future. Undoubtedly, EUR-ISA and SHAREBOX will benefit from a close cooperation, especially in the context of engaging with members and mobilising them to be involved at various stages of the research and development work, from the communication of their requirements and expectations in relation to energy and resource management systems, as well as being involved as beta testers of the platform, to benefitting from training sessions and workshops, etc. Moreover, ISL's National Industrial Symbiosis Programme (NISP) is recognised for eco-innovation by the OECD, European Technology Action Programme, Worldwatch Institute, and Worldwide Fund for Nature (WWF), amongst others, and SHAREBOX will build on NISP's credentials in fostering eco-innovation.

Significant national or international research and innovation activities and initiatives related to SHAREBOX are summarised in Table 2.

Table 2- National and international research and innovation activities that will be linked with the project

Project	Description	Partners involved	Opportunity for carryover to SHAREBOX
National Industrial Symbiosis Programme (NISP)	Most successful example of industrial symbiosis that has achieved wide-scale engagement with industry and significant, independently audited results.	ISL	NISP identifies mutually profitable transactions between companies so that underused or undervalued resources (including energy, waste, water and logistics) are brought into productive use. NISP members comprise of micros, small and medium businesses (SMEs) and multinational/corporates from every industry sector. Through ISL, SHAREBOX will have access to the NISP team knowledge and insight, for creating a Platform that will reduce the requirement for highly skilled support.
POLFREE (FP7-EN 2012.6.3-2):	Constructing a theoretical framework for resource efficiency, with detailed comparison of the trends and policies at EU and Member State level, cross-country analysis, and an analysis of business barriers to resource efficiency; through an exploration of new concepts and paradigms to bring about a radical increase in resource efficiency, new policy mixes, business models and global governance mechanisms are proposed.	ISL	SHAREBOX will build on this research to significantly increase the uptake of industrial symbiosis. The IS WP of the POLFREE project is part of business challenges and business models, detailing the barriers and enablers to more resource efficient business models.
Industrial Symbiosis as an Innovative Method in Tackling Climate Change – ISIM-TCC (LIFE08/ENV/H/000291) (LIFE+)	Demonstration of industrial symbiosis as an innovative tool for tackling climate change and fostering prudent management of natural resources and the potential for sustainable development of the Hungarian economy. Through IS, the project achieved significant reduction of the	ISL	SHAREBOX will seek to increase market penetration and impacts by providing a commercial ICT Platform applied in multiple locations.

	GHG associated with businesses.		
Novel materials for the Built Environment	The project evaluates the use of data from the UK Environment Agency, IPPC directive, and Waste Interrogator to identify waste streams with the potential to be low-cost, and low-carbon, alternative raw materials (ARM) for major infrastructure (construction) projects. This project tests the feasibility of using environmental data to develop a management tool that signposts project ARMs that deliver economic and environmental savings via integration of real-time waste transfer data into current synergy management software.	ISL	Benefiting from this knowledge, SHAREBOX will explore the integration of various databases (including material substitution data, public waste transfer data, and research) into an ICT tool to support IS implementation for those users without IS experience
Global Green Growth Forum (3GF)	3GF convenes governments, businesses, investors and international organisations to act together for inclusive green growth	ISL	ISL leads the 3GF Platform for industrial symbiosis and can convene stakeholders from around the world to share case studies and impacts deriving from the SHAREBOX project.
CEOPS project (NPM3-SL-2012-309984, FP 7 http://www.ceops-project.eu/)	The CEOPS project develops a sustainable approach for the production of methanol from CO2 at large CO2 emitting sites in order to initiative new and sustainable chemical production schemes.	CCB	While CEOPS is developing a chemical approachoflow-carbonproduction, SHAREBOX may provide a management and monitoring tool for the practical application of these solutions.
Site Development Program of CCB	In cooperation with German national authorities (GIZ, BMBF, BMU), CCB initiates partnerships for the creation ofEU-internationaleco-efficient industry parks in particular in emerging third countries.	CCB	The results of the SHAREBOX project may boost the development and set-up of eco-efficient industrial parks also in third countries and will definitely be integrated into the Site Development Program.
Valorization of wastes	Different projects have been developed in order to use different type of waste in ceramic production and also to use ceramic wastes in other industries. Zero waste in ceramic tile manufacture (Life CERAM. LIFE12 ENV/ES/230). Valorisation of ceramic wastes in cement manufacture (CTM2004-06619-C02-02)	ITC	Establish the possibilities of valorisation of the wastes considered in the project
Sustainability in the ceramic production	Projects: Reduction of carbon dioxide emissions in the ceramic manufacturing process (REDUCER. SI2.639876). Environmental sustainability in the ceramic industry (IMIDIC/2009/3). Reduction of company energy costs with cogeneration by means of hydrogen generation (FIT-120000-2006-1)	ITC	Methodology for the footprint calculation in ceramic processing.
The Finnish Industrial	A nationally coordinated, regionally	ISL	SHAREBOX will seek to increase market

Symbiosis System	delivered IS programme in Finland, currently using SYNERGie for data management		penetration and impacts by providing a commercial ICT Platform applied in multiple locations.
SYMBIOSE	A facilitated IS programme in Belgium currently using SYNERGie for data management	ISL	SHAREBOX will seek to increase market penetration and impacts by providing a commercial ICT Platform applied in multiple locations.
SILVER	A facilitated IS programme in Limburg, Netherlands using SYNERGie for data management	ISL	SHAREBOX will seek to increase market penetration and impacts by providing a commercial ICT Platform applied in multiple locations.
WISP, GISP and KISP	Three regional IS programmes in South Africa currently using SYNERGie for data management	ISL	SHAREBOX will seek to increase market penetration and impacts by providing a commercial ICT Platform applied in multiple locations.

It is noteworthy that partner ISL is also working with BAYER, in the industrial symbiosis theme of the Climate KIC (Knowledge Innovation Community)[24], in addressing the challenges of decoupling the economy from environmental degradation and climate change through innovation and extracting the learning from each country that implements IS to inform the ongoing policy process. Through ISL we will benefit from link in to this initiative. Engagement with these projects, and others, such as LOCIMAP- Low Carbon Industrial Manufacturing Parks (www.locimap.eu), a major European industrial symbiosis project, will be managed through our networking task in WP7 (Task 7.2).

Fundamentally for this project, continuous links will be ensured with the SPIRE PPP, in that it aims at integrating, demonstrating and validating systems and technologies capable (across all SPIRE sectors) and the SHAREBOX consortium will leverage SPIRE to reach the European Process Industry. Links will also be built with other European initiatives such as the Resource and Energy Efficiency Partnership "REP", which is a group comprising a broad range of major European Industry Associations (such as the European Engineering Industry Association, the Industrial Minerals Association, Glass for Europe, the European Coil Coating Association, the European Copper Institute, the European Aluminium Association, Eurométaux, the Confederation of European Paper Industries, the European Ceramic Association, the European Cement Association and the Nickel Institute, PlasticsEurope among others) and European Technology Platforms (such as the European Steel Technology Platform, the European Platform for Sustainable Chemistry, European Regions Research and Innovation Network, among others).

Moreover, in the context of the European Structural and Investment Funds through the Smart Specialisation Strategies (RIS3), the SHAREBOX consortium will engage with policymakers to communicate the benefits of industrial symbiosis and to demonstrate the positive benefits and impacts achieved during the project, as well as its replicability across Europe, in order to mobilise investment of resources which would stimulate knowledge-driven growth. ISL has already successfully engaged with DGs Environment, Enterprise, Regions, and Research and Innovation as well as cross-DG European Resource Efficiency Platform to cite NISP® in policy; furthermore, following on ISL's engagement in Turkey, the Turkish regional development agencies are incorporating IS into their strategies.

We will target these projects (as well as other projects from Horizon 2020's PPPs- SPIRE, FOF, etc.), networks and initiatives reaching key players and stakeholders in the European process industries and value chain through a professional communication campaign (discussed in Section 2.2 and executed through Task 7.4). We will mobilise their active involvement at critical points along the project work plan, especially when defining the platform architecture and specifications, as well as during trials and demonstration work, as well as for their participation in knowledge transfer and training activities.

The overall approach and methodology

In relation to our overall approach to the project, Figure 3 below has been the centerpiece for mapping out our work

24 http://www.climate-kic.org/themes/industrial-symbiosis/

plan (presented in Section 3.1, in line with our project objectives (presented in Section 1.1). The project has essentially 4 broad inter-related activities and each is discussed below:

a) Co-creation: Industry Requirements: As the participating process industry demo sites (focusing on the SPIRE sectors and a cross-sectoral engagement with industry partners directly representing over 800 businesses with networks accessing over 10,000 through ISL's NISP in Europe) are a pivotal component of this project, an "Industrial Symbiosis Implementation Team" will be established consisting of the academics, and IS experts from among the consortium, along with the experts from the participating process industry demo sites. A co-creation approach will be implemented between the IS Implementation Team, the development teams and the industries from the demonstration sites in order to define user requirements and to establish the system specifications for the SHAREBOX Platform. To this end, at the beginning of the project (through WP1), the Quality Function Deployment (QFD) method will be implemented in order to enable "the voice of the customer to be translated into the voice of the engineer"[25]. By implementing QFD, in combination with the KANO Survey[26], we will prioritise the spoken and unspoken customer wants and needs, translate these needs into technical characteristics and specifications, and build and deliver a quality product by focusing everybody toward customer/end-user satisfaction. Ultimately the goal of QFD is to translate often subjective quality criteria into objective ones that can be quantified and measured and which can then be used to design and manufacture the product. It is a method for determining how and where priorities are to be assigned in product development. The intent is to employ objective procedures in increasing detail throughout the development of the product[27]. QFD is a very powerful tool for managing client expectations in an efficient and effective way right from the conceptual stages of the development. The QFD method will also form the basis for quality management throughout the project. Following this, a **systems development management approach** will be implemented based on the KANBAN agile development methodology, which has been designed for managing knowledge work with an emphasis on just-in-time delivery while not overloading the team members. In this approach, the process, from definition of a task to its delivery to the customer, is displayed on a KANBAN board for participants to see and team members pull work from a queue. This is highly compatible with the SCRUM agile methodology that will be used for the Platform smart systems developments (as discussed below in Point c). IRIS will establish a KANBAN Board (using TRELLO development boards) for the implementation of the development tasks (the Building Blocks for the Platform-see below) and IRIS can train the consortium on this highly effective method.

b) Methodologies for Industrial Symbiosis and Mathematical tools for flexible energy use and material flow integration

Our project frames IS networks as complex adaptive systems and proposes **the implementation an agent-based simulation model to reach optimum design of symbiotic networks (SNs).** To this end, first, the optimal connections definition is conducted and the SN is modelled as an input-output system[28] having stochastic final demands where the impact of process efficiencies and final demand fit between involved companies are analysed. Two main reasons drive this analysis: First, waste is not produced upon demand, but appears as a secondary output whose quantity depends on the final demand for the principal product. When the waste substitutes the primary input of another production process, then, the fit between the final demands of both production processes is critical for both the collective and individual economic benefit. Second, there is also a relationship between the technological process efficiencies of both processes, which is also decisive for the waste emission and primary input use quantities. To this end, a unique resource sharing platform will be created as a secure tool for seamlessly managing the inputs and outputs of a network of companies and organisations (agents). The platform will contain the periodical final demands for the principal products of supply-side and demand-side processes and the associated waste emission coefficients and traditional primary input coefficients. So, we can both understand the impact of the efficiency and final demand fits and the consequences on the traditional suppliers. Accordingly we can propose dematerialization, waste reduction, or technology change via R&D activities for achieving the physical (related to final demands) and technological (related to

25 http://www.ciri.org.nz/downloads/Quality%20Function%20Deployment.pdf

26 Kano, N, Seraku, N, Takahashi, F and Tsuji, S (1984) Attractive quality and must-be quality, The Journal of the Japanese Society for Quality Control 14 No 2 pp 39e48

27 Reilly, Norman B, The Team based product development guidebook, ASQ Quality Press, Milwaukee Wisconsin, 1999

28 Albino, V., Yazan, D.M. (2013). Economic and environmental benefits of industrial symbiosis: an enterprise input-output analysis, 21st International Input-Output Conference, Kitakyushu, Japan, July 7-12

process efficiency) optimal conditions. It is here that we will be able to assess the potential benefit of new infrastructure for the exchange of waste between actors particularly in the form of energy.

The new software and modelling tools developed under the SHAREBOX programme will identify opportunities for synergy between industrial actors in the context of an industrial park. Realising those synergies may exploit existing infrastructure but a greater range of opportunities would be open to the users if investment in new infrastructure could be contemplated. As such the SHAREBOX tools will be expanded to identify enabling technologies and to build an economic case for investment in them by either one of the industries already resident or perhaps by an additional actor (in the form of an energy service company (ESCO), for example), who might also act as the IS facilitator.

The sharing of energy resources provides a particular set of challenges. Process integration, using pinch point analysis for example, is well established but, as discussed, IS is often impeded by a lack of understanding of processes and practises across businesses. Nevertheless, the same ethos that drives process integration within a business can drive integration between businesses. What then remains is to establish the physical means of sharing resources and in the context of energy this often means waste heat. There are established ways of distributing waste heat using heat exchangers and pipe networks. Retro-fitting the necessary infrastructure to an existing park can be expensive and for example is often cited as an impediment to district heating systems in cities. If the necessary investment is to be forthcoming a clear business case must be made. The SHAREBOX tools will enable this and will also identify the potential owner/operator of the resources whether they be existing or additional actors.

In the context of shared energy resources, the evolving energy landscape has thrown up new opportunities that must be considered and integrated into the SHAREBOX tool. New technologies have emerged and will continue to emerge for the storing and transport of energy. This is crucial because it would allow sharing across a wider area and time-shifting so that waste is collected and consumed at the convenience of the generator/consumer without impinging on the production schedules of either. New and emerging technologies also exist for the conversion of waste energy resources – low-grade waste heat to electricity for example. These would allow the sharing of energy via its most convenient vector – electricity – and should be factored in. Finally, the move towards renewable energy sources and the evolution of smart grids presents the motivation and the means for new business opportunities in the form of end use energy demand in support of the system operators.

Energy conversion, storage and transmission capability, managed by existing or an additional actor in the form of an ESCO will increase the range of potential exchanges in terms of form, distance and time and will ameliorate the scheduling problems associated with variations in final demand.

Second, in developing opportunities for waste-based business, behaviour of companies shape the negotiation phase as no certain benefit-sharing schemes are available. Even though potentially saveable costs (e.g. discharge cost, purchasing cost) are promising, to operate symbiotically, additional costs (e.g. recycling cost, transportation cost) must also be sustained. Then, three questions arise: (1) 'Who sustains which cost?; (2) Who performs which task?; and, (3) Who gets what percentage of the total benefit?'. This is a game played between (at least) two actors who have to share a benefit that is created by both. Such games are crucial for operating IS and become more complex when the number of involved actors increases.

Converging interests of companies render waste-pricing a difficult task to accomplish as waste is not produced upon demand. Several following cases may appear[29]:

- Companies take irrational decisions such as not cooperating even though profitability and environmental benefits are promising.
- One of the firms take advantage of power asymmetry and the benefit is unfairly shared.
- Imposed waste recovery regulations drive waste providers' decisions (e.g. paying for supplying waste).

Then, we intend to take a step further to identify the impacts of factors influencing company behaviour during negotiations and propose benefit-sharing schemes adopting GT approach. The structure of game(s) played between

[29] Albino, V., Dangelico, R., Yazan, D.M., 2010 Cooperation in reverse supply chains: environmental and economic impacts of actor decisions, 18th International Input-Output Conference, Sydney, Australia, June 20-25.

involved firms shapes the impacts of the aforementioned factors which influence the bargaining power of both supply and demand sides leading to different levels of willingness-to-cooperate[30]. Therefore, an in-depth analysis of the following game structures will be performed in order to measure this willingness: (i) sequential vs. simultaneous games, (ii) cooperative vs. non-cooperative games; and (iii) one-to-one vs. consortia games. Finally, the SN is represented as a system of agents interacting among themselves and with the environment (e.g. governments, third-party logistics players) using ABM which produces the best set of solutions, e.g. the minimum CO_2 scenario, maximum profit scenario, or maximum employment scenario. By this approach, we also have the advantage of tackling with the multi-use character of wastes, which can be processed also by multi-pathways. Once the models are tested in the real cases, the feedback will assist us to address any shortcomings and to re-feed the initial data used for input-output matching. Hence, we have critical input from real-life cases to design the future SNs.

c) SHAREBOX Platform Software and Architecture development: The initial Platform Architecture is presented in Figure 1 and this will be validated and adapted based on the House of Quality from the Living Labs I. The methodology that will be used to develop the platform will be an agile methodology called SCRUM. SCRUM is a highly proven agile development methodology which was initially created for software developments, but which works well for any complex innovative scope of work. Very briefly, the Scrum framework operates as follows[31]: A product owner (i.e. the end-users of the Platform, who will be represented by the Project Manager in IRIS, Dr. Mirta Pinilla) creates a prioritized wish list called a product backlog (this will be extracted from the House of Quality above). During sprint planning, the Scrum team (the IRIS software developers, which are identified in the IRIS participant profile in Section 4.1) pulls a small chunk from the top of that wish list, a sprint backlog, and decides how to implement those pieces. The team has a certain amount of time — a sprint (usually two to four weeks) — to complete its work, but it meets each day to assess its progress (daily Scrum). Along the way, the Scrum Master (Mr. Sergi Casbas from IRIS) keeps the team focused on its goal. At the end of the sprint, the work should be potentially shippable: ready to hand to a customer, put on a store shelf, or show to a stakeholder. The sprint ends with a sprint review and retrospective. As the next sprint begins, the team chooses another chunk of the product backlog and begins working again. IRIS, who will be developing the Platform infrastructure has operated Scrum Teams for the past 3 years and has their own SCRUM team and Sprint Room infrastructure.

d) IS Demonstration Sites: This activity will center on testing of the Platform at Demonstration sites using realistic industrial streams and process conditions provided through our industry partners. Our demonstration work is centered on two IS demonstrations sites. The pilot tests will help to generate standards of how SHAREBOX can be implemented at an integrated production site. Depending on the individual framework conditions, it may be implemented by a site operator in order to optimize its services towards the tenants or by an individual company, being gradually extended to neighboring production units. The Living Labs will help to identify optimized ways of implementation and distribution of management tasks among the local production and facility officers. The aim is for an alpha version of the SHAREBOX platform to be available by M12, so that early tests can be initiated with these sites with the early alpha version of the platform to verify its performance, and to ensure the versatility of the platform to the specific conditions in real production units. Feedback from the sites will be fed back to the development team to guide the various iterations of the Platform. Verification of the impacts of SHAREBOX on effectively establishing new synergies and optimizing existing one, and an analysis of impacts on resource efficiency, energy efficiency and the emission performance, and cost-effectiveness will be performed.

e) Development of the Business Model: A very important activity in this Innovation Action will be the development of a business model for monetising the SHAREBOX Platform for ensuring from its outset that the concept is commercially viable and profitable. This is vital for ensuring the sustainability of the outputs of this SHAREBOX project and for ensuring that the participating technology SMEs in the project can further develop the SHAREBOX platform from a beta version to a viable commercial platform so that it can be widely deployed post project. The participating Process Industries and Clusters are committed to being early adopters and co-creators of a Platform that can have lasting impacts on their businesses, and indeed they will develop exploitation plans for the post-project roll out of the

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[30] Yazan, D.M., Clancy, J., Lovett, J.C., 2012b. Supply chains, techno-economic assessment and market development for second generation biodiesel. In: Luque R., Melero J.A. Advances in biodiesel production. Second generation processes and technologies, Cambridge, Woodhead Publishing.

31 Source: The Scrum Alliance. Available at: https://www.scrumalliance.org/why-scrum

IS activities that they establish during the project through their involvement as Demo Sites. Indeed "Business" is the order of the day for all involved in the SHAREBOX project and by making it about doing business all partners, including the academic partners, have approached the project from a highly pragmatic and real world approach, whereby each has been very eager to lock down IPR issues, from the expected results, ownership of the results, access rights, Background, exploitation intentions, etc. The Business Model activity of the project will build on the outline approach that has been agreed by all partners and that is presented later in this document.

How sex and/or gender analysis is taken into account in the project's content

In relation to **gender analysis**, the SHAREBOX consortium understands that inventions and developments that incorporate the smartest aspects of gender can open new markets and enable innovation in products, processes, services, or infrastructures. Indeed, products that meet the needs of complex and diverse user groups enhance global competitiveness and sustainability[32]. Inspired by how taking gender into account has yielded new research results and sparked creativity, opening new avenues for future research for those who give it serious attention[33], during the work plan, **gender analysis will be addressed in particular in relation to the KANO survey, which will ensure a good participation of female professionals in order to allow for a gender analyses to be incorporated into the early user requirement definition and in this way we will ensure that any gender-specific requirements and expectations are automatically picked up.** As the Platform is being developed, aspects such as the Graphical User Interface will consider gender aspects, whereby graphic designs of the interface will be consulted with both men and women during the design phase, and posterior alpha and beta versions will be trialled by both men and women.

Moreover in relation to **business behaviour** gender analysis will also be important, especially in view of growing trends and commitments to see more women occupy high level and decision-making positions in industry. For example, in Europe, Norway a decade ago mandated a 40% quota for female director's at large companies, and according to reports has now achieved or even exceeded this target. Spain is aiming for a similar target by 2015[34]. In June 2011, Belgium's parliament voted to force state-owned businesses and stock market-listed firms to award 30% of managing board seats to women. Large companies have five years to comply, SMEs eight years. In 2012, the parliament of the Netherlands was poised to debate similar rules. By 2017, at least 40% of managing and supervisory board members at companies listed on France's CAC-40 stock exchange and all other firms with sales of more than €50m and at least 500 employees must be female. Germany and the UK established voluntary targets for female representation in 2011[34]**Error! Bookmark not defined.**. All of this points towards a strong movement for change.

Indeed, with more young women than ever now leaving universities with chemistry degrees[34] the consortium, will benefit greatly from effectively addressing gender analysis issues in that there is a distinct European trend in process industries, and in particular the chemical industry, in **boosting role of women decision makers**. Some examples from the chemical industry include, among others: Rotterdam-based petrochemicals giant Lyondell Basell, appointed Karyn Ovelmen as chief financial officer in 2011; in October 2010, Sandra Peterson was named CEO of Bayer Crop Science and BASF has pledged to increase the proportion of women in what it terms 'solid-line' German management to 15% by the end of 2020, from just below 10% in 2010; up to 2015, the Bayer group aims to 'gradually' increase the share of women in senior management from about 21% at present 'toward a target of 30%' and its goal for the supervisory board is 20%; German chemicals firm Merck made two key appointments in December 2011 with Friederike Rosch, being named head of corporate accounting and Isabel De Paoli, head of corporate strategy.

In relation to the **gender dimension of the SHAREBOX consortium**, the team of researchers, technologists, industrialists and policy influencers has a significant cohort of both sexes (ca. 40% female; 60% male). Moreover, a recruitment policy is in place designed to ensure a reasonable level of balance between the sexes. This is important in ensuring the education and training of the next generation of scientists, technologists and others needed in the EU and elsewhere. It is noteworthy that a number of the key management and decision-making positions within the project organisational structure have been assigned to highly qualified women, not least the role of the Project Coordinator (Dr. Mirta Pinilla), the IPR manager, Ms. (Rachel Lombardi), the Dissemination Manager (Dr. Renata Körfer).

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32 http://ec.europa.eu/research/science-society/gendered-innovations/index_en.cfm

33 Londa Schiebinger (2008). Gendered Innovations in Science and Engineering. ISBN: 978080475814

34 http://www.soci.org/Chemistry-and-Industry/CnI-Data/2012/2/Closing-the-gender-gap

Interestingly, additional females are among the key drivers of this INTENSO proposal (Oonagh Mc Nerney, Dr. Cristina Serrano, Dr. Agnes Zoller, Dr. Encarna Bou Solsona, etc.).

1.4. Ambition

Advance beyond the state-of-the-art

The ambition of SHAREBOX is to bring to market **a secure ICT platform that integrates analysis and optimisation tools for flexible energy use and material flow integration to enable a holistic approach for resource management in process industries**, suitable both for small and large scale in a flexible approach, and that lowers barriers, and dramatically increases the mainstream adoption of IS. The platform will greatly advance the state-of-the-art in energy and resource management systems, and will overcome key limitations in industrial symbiosis delivery.

Limitations in industrial symbiosis delivery: Inefficient resource use by Europe's industries has been identified by the European Commission[35] as a clear market failure creating additional and unnecessary costs that constrain growth, contribute to greenhouse gas (GHG) emissions, and further exploit scarce natural resources.

The European Resource Efficiency Platform recently published its final recommendations for developing a resource-efficient economy, citing SHAREBOX partner ISL's NISP® and urging all EU member states to 'foster industrial symbiosis' as one of the key strategies for doubling resource productivity by 2030[36]. Indeed it states: "*A number of schemes show the benefits of increased information flows, and the payback from providing advice or bringing firms together in National Industrial Symbiosis Platforms (NISP): Based on the performance of the UK National Industrial Symbiosis Programme, improving the re-use of raw materials through greater 'industrial symbiosis' across the EU could save €1.4bn a year and generate €1.6bn in sales*".

DG Enterprise & Industry identified that the 'information market failure' in relation to resource efficiency can be addressed through mechanisms to improve information flow between businesses[37], and recommended enhanced IS as a policy instrument to overcome this information market failure[38]. An integrated comprehensive software is required to advance IS by engaging mainstream businesses. Part of the challenge of IS in practice is that the majority of opportunities for a given company lie outside that company's sector.[39] Decision-makers in industry today were not educated in sustainable development, resource efficiency, industrial symbiosis, or circular economy thinking – many will have experience deriving primarily from a single industry. They know intimately the benchmarks, best practice, technologies, processes and jargon of their own sector. Going outside their sector, and their traditional supply chain, is not standard operating procedure for many mainstream industries and thus a barrier to mainstream IS. At some point in the future, when these concepts are standard in educational curricula[40] it will be much easier for industry to seek out these opportunities. It may be outsourced by some firms the way accounting services are, but it will be standard for all businesses.

Indeed, information barriers prevent the upcycling of many resources. Sophisticated IS requires a high level of specificity tailored to the potential uses of the resource, requiring substantial cross-sector expertise in its design for each resource. For examples, sand is not just sand to a process industry: its characteristics including grade, contaminant level, and hardness, are all relevant to its suitability to different uses. Web-based systems have not addressed (or addressed poorly): contamination, classification, distribution and timing issues (all vital to optimise re-use), or the innovation required to add value to, or transform, 'waste'.

Facilitated IS brings together producers and users of underutilised resources (including materials, water, energy, logistics, expertise, assets) with technological innovators to foster demand-led innovation (i.e., innovation that responds to the needs of the participants). Public investment in facilitation ensures all businesses, regardless of size or sector, have the opportunity to participate and benefit their bottom line whilst contributing to improving the

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35 European Resource Efficiency Platform, 2014, Manifesto and Policy Recommendations. 31 March.

36 European Resource Efficiency Platform (2014). Manifesto & Policy Recommendations. http://ec.europa.eu/environment/resource_efficiency/documents/erep_manifesto_and_policy_recommendations_31-03-2014.pdf

37 Study on the Competitiveness of European Companies & Resource Efficiency, 2011, *ECORYS* for DG Enterprise & Industry.

38 Sustainable Industry: Going for Growth, 2011, DG Enterprise & Industry

39 Based on analysis by SIC (Standard Industry Classification) code of NISP data in the UK over 7 years.

40 Murray, S. 2013. The sustainable path to profit: don't throw out the rubbish. Financial Times, Business Education Section. http://www.ft.com/cms/s/2/88e2d9f0-2f43-11e2-b88b-00144feabdc0.html#axzz3DNfZttdI accessed 15/Sept/2014

environment. Where public investment supports IS, services are generally free at the point of delivery to all comers, and engage widely, especially the difficult-to-reach SME and micro-enterprise market segments. IS activities supported by public funds are well-positioned to pursue the public good (such as natural environment which has not had a market value) and longer-term, higher-risk investments (e.g., green growth and eco-innovation). The public investment element minimises two of the largest barriers for business involvement in IS – time and money. Public investment is, however, subject to changing priorities, and short-term, sporadic investment in IS leading to sub-optimal performance; an IS business model based on public investment thus is not inherently sustainable.

To achieve mainstream transformation through IS requires engaging mainstream businesses in IS with the information needed to identify viable opportunities to move up the value chain. The NISP model of delivery (like many public-investment models) employs experts to engage with industry in the IS process: the experts provide both information and support to identify resources, assess opportunities, and complete IS transactions. To make the step-change to mainstream adoption of IS requires breaking (or dramatically reducing) the reliance on niche IS specialists – to find a way to enable all businesses managing resources (inputs or outputs) to actively engage in IS by making expert knowledge accessible to mainstream businesses.

A step-change in delivery model is needed to achieve mainstream uptake of IS across industry sectors, commercially and at scale, and enable a variety of actors (large and small consultancies, innovators, entrepreneurs and social enterprises) to successfully implement IS in a commercially sustainable way. Indeed, SHAREBOX represents a step change in the accessibility of IS delivery through improved information sharing.

State-of-the-art of ICT tools for IS facilitation: Various efforts have been undertaken to use ICT to facilitate IS since the then-ground-breaking development of Designing Industrial Ecosystems Toolkit (DIET) and Fate Assessment Screening Tool (FAST) by the US EPA in the 1990s[41]. The basic approaches, **largely research-driven rather than market-focused**, can be classified as follows:

- Waste exchanges that capture resources for others to search and identify matches;
- Databases or libraries that act as a reference for proven reuse opportunities for various materials/ wastes; and
- Process management software that actively manages the process of advancing multiple synergies opportunities simultaneously.

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Outside IS, further ICT tools are available with potentially valuable contributions –SHAREBOX is unique in incorporating their information into its solution:

- Materials databases that provide full characterisation and classifications for materials and processes, which can broaden understanding and identify potential for resources
- Regulatory data required by governments on 'waste' resources moving through the economy, with its origin and destination (in Europe: IPPC, EPR, EWC)

[41] Industrial Economics. 1998. Applying Decision Support Tools for Eco-Industrial Park Planning – A Case Study in Burlington, VT. Cambridge, MA: Industrial Economics Inc. Prepared for the US EPA.
[42] Industrial Economics. 1998. Applying Decision Support Tools for Eco-Industrial Park Planning – A Case Study in Burlington, VT. Cambridge, MA: Industrial Economics Inc. Prepared for the US EPA.

Waste exchanges have proliferated all over the world and still they remain largely unpopulated. Established in 2012, TradingWaste.com in the UK currently lists only 23 resources. The Integrated Waste Exchange (IWEX) was established by the City of Cape Town in May 2000; on 11 September 2014 (i.e. after more than 14 years), it had 100 resources listed; in contrast, the facilitated Western Cape Industrial Symbiosis Programme (WISP) identified and captured detailed information on 465 resources in less than one year – making expert-supported IS more than fifty times more effective at capturing resources. A viable waste exchange would require this level of specificity tailored to the potential uses of the resource, requiring substantial cross-sector expertise in its design for each resource, as well as a focus on those resources where market drivers create the economic incentive for businesses to prioritise time spent on seeking IS solutions. Sand is not just sand to a process industry: it has a grade, contaminant level, and hardness that are all relevant to its suitability to different uses, and its origin influences the regulatory pathway to its reuse.

Projects including the LIFE+ funded eSymbiosis and FP7 ZeroWin[43] have tried to duplicate ISL's taxonomy without practical experience to inform their development, or data derived from advancing synergies in practice. The LIFE+ project eSymbiosis is "based on the UK Industrial Symbiosis best practice"[44] although the project team were unaware of the foundation work of the US EPA. After 3 years of EU funding, eSymbiosis is populated with general test data provided by the region rather than from real companies [SHAREBOX platform will be populated with real date from the beginning – see WP 1, 3]. At the eSymbiosis final event in June 2014[45], the project leader reported that the project had failed to engage with industry or collect information from them – an experience directly contrary to this consortium's. Also after several years of funding, the European Pathway to Zero Waste (EPOW) project's online waste exchange delivered no completed transactions.[46]

This consortium has been unable to identify waste exchanges that are successful at diverting significant waste from landfill, upcycling, avoiding carbon or creating notable economic benefit. Furthermore, such systems have tended to be academic in design and have not deeply considered business drivers or real world IS practicalities such as market value of resources.

Databases/Libraries of proven IS opportunities are passive systems populated with IS ideas that may be good for replication, however, this is often from research rather than through practical data deriving from experience in delivering IS. While taxonomy, data structure and the level of information can be adequate, the requirement for initial data input and subsequent manipulation is high. A leading example, Presteo, includes several methodological tools: a data collection method, a formalization method of flows and components and a processing method to filter the results obtained, all of which require user expertise and time. Furthermore, there is limited or no ability to provide automatic matching results to the user. The IS resource databases suggest solutions linked to resource entries, but are not based on feasibility and experience. Current databases do not learn from new matches created so the library remains stagnant as real world experience advances. These databases also stop short of delivering functionality that enables users to manage identified matches.

Process management for large scale, replicable IS implementation is critical for practitioners shepherding tens of opportunities through the IS process at any given time, enabling expert intervention and coordinated effort. State of the art is ISL's SYNERGie, the only platform to be developed with practitioners based on experience, and applied in 13 countries across 5 continents. The platform currently manages resources to advance synergies, allowing users to create and manage matches between suitable resources, with the additional functionality to create and track actions. It enables the entry and searching of resource data as does a waste exchange, but further allows process management, action assigning and logging, interrogation of pre-established knowledge management, capture of results and reporting. There are a number of current limitations including adding to a library of experience and the ability to link cross border information that would enable a pan-European IS activity. Nor does it suggest potential IS solution for resources. This requires data manipulation by users which relies on their knowledge and experience to make the tacit linkages between resources.

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43 www.zerowin.eu, www.esymbiosis.gr

44 eSymbiosis. 2013. End of Action Report, Action 2: Service Architecture and Implementation. P.11.

45 Communication at eSymbiosis final event, Athens, Greece, 19-21 June 2014.

46 European Pathway to Zero Waste (EPOW), Layman's Report. UK Environment Agency March 2013.

Materials Database To date, software tools for IS lack integration with other materials selection approaches, including materials substitution. Collaborative materials information management is the next frontier for materials data, enabling the combination of dispersed materials knowledge – often relating to different fields of expertise. This direction will require significantly different framework and functionality for shared materials information.

Regulatory data is a valuable source of information currently assembled and stored for regulatory purposes but not generally accessed for productive use. State of the art is bringing this data reporting real-time (as with the Environment Agency trial in the UK) and accessing it for opportunities. [ISL is currently leading a project on this, see section 4.1 project list, "Novel materials …"]. The Integrated Pollution Prevention and Control (IPPC) data, Declaration of European Waste Catalogue (EWC) data will be ananlysed for oppotunties. The European Environment Agency (EEA) will be consulted for further relevant and available data sources.

Ground-breaking nature of the SHAREBOX platfrom: In order to overcome the limitations in current IS delivery and ICT tools for IS facilitation, a number of **key innovations** are being brought to the SHAREBOX development that will yield a world first ICT platform technology that holds the potential to be a major catalyst for increased IS among process industries and for enabling cross-sectoral interactions for a major impact within the process industry.

a) ICT-enabled next generation IS Delivery

As IS seeks to create new collaborations among economic players with the goal of exchanging information, raw materials and waste directly among area businesses, and to step up the potential pooling of services and infrastructure among neighbouring companies, the SHAREBOX project will benefit from and build on the expertise of the world leader in IS delivery, partner ISL, who has devised the National Industrial Symbiosis Programme (NISP®), and has gathered data from over 20,000 companies and identified over 10,000 potential synergies, which will populate the SHAREBOX Platform library. During the project, we will develop a 'next-generation SYNERGie®' integrated module of the SHAREBOX Platform. SYNERGie® currently offers database functionality, project management, and reporting tools for companies participating in industrial symbiosis programs, enables organisations in an industrial symbiosis to record and analyse resource data (such as waste materials, by-products, processing capacity, waste heat, water, logistics, and expertise), identifying links between participating companies. During the project, we will upgrade the tool to make it a smarter, intelligence-based module of the SHAREBOX platform towards the next generation in IS delivery. The current impact reporting capability of SYNERGie (carbon, landfill, water, hazardous waste, virgin materials, cost savings, revenue generation) will be carried over to SHAREBOX.

b) New approaches that perform cost-saving optimisation of energy and resources supply and demand

We aim to develop tools **to implement the optimum IS (i.e.** no waste to be disposed outside of the symbiotic network, no primary inputs to be purchased from external suppliers and no net energy import from outside the network**) as a pre-design model for future symbiotic networks (SNs) and use it as the theoretical optimum to which the existing SNs should reach.** As **the input-output (IO) approach** dissects complicated systems of physical flows and effectively traces the path of flows[47], it allows the modelling of the relationship between the waste emissions, primary input use, and final demand levels of main products. The main advantage of the IO modelling is its ability to measure the impacts of any factors, e.g. demand shocks, technology change, substitutability levels, at the level of whole network. Once the complete mathematical tools will be tested in real cases, the collected data will re-feed the IS design section to optimize the future symbiotic cooperation. In addition, IO modelling allows us to compute the environmental savings in terms of potential GHG reduction, landfill reduction, etc.

Converging interests of companies usually render the negotiation phase difficult to resolve. Although we can consider the diversity of company cultures as a primary barrier to cooperation (due to the different nature of sectors traditionally not used to work together), there are several important factors influencing the negotiation phase. These factors can be ranked as the geographic proximity between manufacturers, the direction of monetary flows (sometimes waste-receiving company is paid by the waste producer or vice-versa), presence of any environmental regulations or incentives, bargaining power of the supplier or buyer, and possibility of cooperating with another waste-producing/waste-receiving company on more favourable terms. Such factors strongly influence the pay-offs and render

[47] Bailey, R., Allen, J.K., Bras, B., 2004. Applying ecological input-output flow analysis to material flows in industrial systems part I: Tracing flows. Journal of Industrial Ecology 8 (1-2), 45-68.

the benefit-sharing more complex. We will tackle this problem through the **construction of benefit-sharing schemes for involved actors taking into account the role of aforementioned factors. A game theory (GT) approach** would be a particularly efficient method to compute the pay-offs that drive the decisions-to-cooperate of companies. Theoretical pay-off matrices created by GT approach can serve as benefit-sharing schemes from which companies can select the most appropriate according to their situation.

Although, the design is one of the key factors to success, deviations from an optimum IS are inevitable due to companies´ behavioural moves during negotiations over waste valorisation which represent economic trade-offs. The final physical flows, therefore, will vary according to games played by companies in the negotiation phase driven by their economic interests. Hence, we aim at adopting agent-based modelling (ABM) to reach **the third aim of this proposal: embedding company behaviour into the operations of IS**. ABM is an appropriate methodology to address such a problem as it considers the heterogeneity of the actors (i.e., agents) and proposes a self-organizing system[48]. Using ABM for SN design allows us to address the behavioural change by learning and interaction, and reveal the optimal design under different individual strategies and cooperation modes. We will study ABM based on opensource software (for example, ABM Repast) in order to **keep affordability and long term consistency**.

c) Data mining techniques for further optimisation of IS based on historical big data collected from SHAREBOX

The selection of an appropriate model for large-scale data analysis is critical. Obtaining useful information from large amounts of data requires scalable analysis algorithms to produce timely results[49]. The use of Data Mining techniques for the data analytics allows the discovering of patterns in large sets of data through machine learning, statistics or artificial intelligence (AI). This need is particularly relevant for modelling stochastic behaviours that can be difficult to explain in terms of deterministic relationships. There are many data mining techniques, according to different issues like describing the data (*descriptive methods*), discovering associations among data (*associative methods*), predicting quantitative data (*predictive methods*), and predicting qualitative data (*discriminant methods*).

One of the key innovation features of the SHAREBOX project will be the **new knowledge and new patterns discovery from large data** generated in the IS parks. These new patterns and knowledge will complement other mathematical tools (GT, ABM) used in the project aiming at getting further optimisation of IS. As the range of data mining tools is very large, it is not easy to known in advance which methods will be the most suitable ones. Anyway, from a priori analysis, it seems that Multivariate Analysis techniques, in order to search association relationships, and Artificial Neural Networks (ANNs[50]), to predict quantitative data, are suitable tools to be used. Multivariate Statistical Process Control (MSPC)[51] techniques are especially suitable for smart controlling in order to reduce product variability and increase quality. MSPC analysis comprises different techniques. All of them have proven to be robust enough for describing and predicting on the basis of very complex and noisy data sets like spectroscopic information although they are mainly limited to linear functional relationships because the most of them are based on applying linear algebra for solving statistical problems. ANNs are also powerful tools, especially for modelling non-linear behaviour of data. They have a great flexibility as well as the capacity of resembling human biological neural networks.

Notwithstanding, another data mining techniques could be considered, analysed and tested according to the available data.

As long as daily situations in an IS are solved, a great amount of experience is being produced. In most IS parks, this experiential knowledge is not used. In the SHAREBOX system, **an intelligent experience-based reasoning scheme will be used: case-based reasoning**. In this reasoning approach, the new situations/problems are solved from previous similar situations/problems solved in the past. These paradigm makes use of the historical data and experience gathered within the IS parks. This approach can learn the new solved situations, making the system more competent along time.

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[48] Macal, C.M., North, M.J. 2010. Tutorial on agent-based modelling and simulation. Journal of Simulation, 151-162.
[49] D. Talia, Clouds for scalable big data analytics, Computer 46 (2013) 98 – 101
[50] "Artificial Intelligence in Process Engineering", Ed. by Michael Mavrovouniotis. Academic Press, 1990.
[51] "Multivariate Statistical Process Control, Process Monitoring Methods and Applications. Series: Advances in Industrial Control", Ge, Zhiqiang, Song, Zhihuan, Springer, 2013

d) A flexible and secure platform that allows for secure sharing for information about resources and energy exchanges among industries in a SN and for generating operative actions

The IS methodologies, mathematical and data mining tools will be integrated into an ICT platform that will offer high security through its APIs for secure data exchange. A major feature will be the data encryption algorithms that will be used to protect the data. If the data is exchanged and even processed in encrypted format, the risk of disclosures decreases. Good encryption procedures cannot be outflanked even by the software programmers. Encryption can be used to protect data "at rest", such as data stored in files, in databases and in storage devices, and it is also commonly used for data being transferred via networks (e.g. the Internet, e-commerce), mobile telephones, wireless microphones, wireless intercom systems, Bluetooth devices and bank automatic teller machines.

SHAREBOX offers an ICT-based working environment to address the Big Data issues associated to the industrial symbiosis encouragement, implementation, management and improvement tasks.

Indeed SHAREBOX will be strengthened by-design with powerful but easy-to-use mathematical tools such as Agent Based Modelling and Game-Theory-based procedures -for assisting the achievement of optimal win-win deals- and, Intelligent Decision Support algorithms -for helping decision makers to handle conflictive, unexpected, complex or urgent issues- and Data Mining tools for unveiling hidden but relevant trends, behaviours and relationships.

Furthermore, SHAREBOX will also address the problem of acquiring and storing the data from the intrinsic plurality of data sources associated to the industrial symbiosis performance: The implementation of the SHAREBOX Platform in each industrial symbiosis network will comprehend the installation of an ad hoc communication network -or the adaptation of the existing ones- aimed at retrieving real-world data -like the ones provided by SCADA-based industrial control systems or remote real-time database servers.

SHAREBOX includes by-design a maintenance philosophy: By applying the Data Mining tools to the historically stored information, the System will be capable to suggest new trends or relationships suitable for finding new optimal deals and upgrade the decision support algorithms.

The Platform will not replace human decisions. The networked companies -normally by means of contractual agreements assisted by SHAREBOX- will arrange in advance which decisions can be automatically generated by the Platform and which cannot. In particular, legal constraints will limit the scope of the automated decisions. In any case, the decisions will be always considered as information that can be further -automatically or not- converted in operational actions to be undertaken by the involved companies. Moreover, the historical track of operational data will help to -even passively- detect deviations from the agreed behaviour as well as artefacts in the execution.

In brief, SHAREBOX, which will be deployed via a supercomputing cloud, will be the first really *smart* tool for fostering the industrial symbiosis paradigm in line with the Circular Economy and will bring about a step change in enabling resource sharing transactions for a major impact within the process industry.

Innovation potential

Environmental problems related to the consumption of natural resources and to the increase of pollution and waste have challenged traditional business models prompting a growing attention toward more sustainable ones. Growth and environmental impact must be decoupled, that is, economic growth must not automatically imply increased energy and resource use[52].

Industrial symbiosis (IS) has been proven to deliver wide-ranging benefits, from resource efficiency to eco-innovation. IS results in innovation, often times bringing new research and development to market in record time because of the demand-pull of the programme on eco-innovation. The OECD describes IS as “systemic innovation vital for future green growth.”[53]

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52 Rademaekers, K., Zaki, S.S. & Smith, M. (2011). Sustainable Industry: Going for Growth & Resource Efficiency, European Commission Enterprise and Industry, Rotterdam, The Nederlands. Available: http://ec.europa.eu/enterprise/policies/sustainable-business/files/brochure_sustainable_industry_150711_en.pdf.

53 OECD (Organization for Economic Co-operation and Development). 2010. Project on Green Growth and Eco-Innovation. www.oecd.org/dataoecd/43/48/45169190.pdf. Accessed 17 November 2010.

Indeed, based on NISP data, 20% of successful IS opportunities (called synergies) involve innovation (either new application of existing knowledge, or new research and development).

However, unlocking the value embedded in under-utilized industrial resources can be challenging, especially as it involves more than merely brokering a link between two or more companies[54]. **Extensive knowledge of different industrial processes or manufacturing techniques, combined with an understanding of the latest processing technologies and innovations, is essential to identify immediate, medium, and long-term opportunities for resource recovery and reuse[54].** To date there is no holistic tool available to IS programme developers and to the industry for enabling secure and systematic knowledge/data exchange and processing of this data for enabling operative actions that will be a facilitator for increased IS resource efficiency and turning waste into a resource. Addressing this challenge well brings to the fore the building blocks for increased eco-innovation.

By channelling new knowledge into a powerful, easy to use secure Platform that will be integrated into existing industrial information systems that will essentially bring an understanding of the flows of materials and energy in a new or existing Symbiotic Network to identify opportunities to **reuse these resources,** and to provide tools to optimize resource efficiency in product processing/manufacturing, through the SHAREBOX platform, we will address climate change by bringing together traditionally separate industries, value chains and organisations, with the aims of improving cross industry resource efficiency and sustainability and of turning waste into resource.

SHAREBOX is the first platform of its kind and our comprehensive literature review and our review of commercially available tools confirms its market uniqueness. A patent review has confirmed our freedom to operate and there are no Intellectual Property Rights that could be blockers for our developments and their commercial exploitation To this end, it can be concluded that neither the developments that will be performed during the SHAREBOX Project, not the future European and global commercial use of the outputs will be blocked by any existing commercial tools or technologies and no existing IPRs exist that could prevent the development or risk the business opportunity once we go to market. It is expected that the Platform itself will bring commercial benefits to its developers and deployers, as well as to its end-users, whereby it will serve as a tool for the optimisation of energy and resources supply and demand leading to resource sharing and exchanges for reduced costs, increased sustainability and green growth.

[54] http://2014.globeseries.com/industrial-symbiosis-new-driver-green-growth/

2. Impact

2.1. Expected impacts

Contribution to the expected impacts set out in the work programme, under the relevant topic

SHAREBOX contributes positively to each of the expected impacts of topic ***SPIRE-06-2015: Energy and resource management systems for improved efficiency in the process industries***.

IS activity, through the demand-led innovation, often leads to the redesign of processes, procurement standards, and products. *"Industrial symbiosis engages diverse organisations in a network to foster eco-innovation and long-term culture change. Creating and sharing knowledge through the network yields mutually profitable transactions for novel sourcing of required inputs, value-added destinations for non-product outputs, and improved business and technical processes."*[55] Indeed, IS is as simple and as complex as the circular economy[56] in action: simple because successful IS keeps resources circulating in the economy rather than going to disposal; complex in that the product, process, technology and procurement changes necessary are not straightforward. IS addresses information market failure by connecting traditionally separate industries, thus enabling them to divert wasted by-products and resources into productive and value-added uses elsewhere in the economy. As has been discussed, SHAREBOX holistic energy and resource management system will enable mainstream uptake of industrial symbiosis (IS) by lowering barriers to its adoption through the use of the latest advancements in secure ICT technology and numerical tools.

There is a growing market for IS across Europe, supported by the recent high-level European Resource Efficiency Platform (EREP) recommendation[57] that EU and Member States should foster IS by promoting a pan-European network of IS initiatives. A DG Environment study indicates that pan-European IS would generate €3Billion in additional sales and cost savings for Europe, alongside concomitant environmental benefits[58]. A report by Ellen MacArthur Foundation based on McKinsey & Co analysis estimates an annual net material cost savings opportunity of up to €487 Billion for a subset of EU manufacturing sectors[59]. SHAREBOX aims to tap into this market to generate revenue through the deployment of the platform as well as to contribute to growth and jobs by enabling IS adoption in the European Process industries and cross-sectorally.

If we consider the successful UK National Industrial Symbiosis Programme (NISP), "*nearly 20% of NISP synergies involved some form of innovation, many bringing new research and development to market in record time because of the demand-pull of the programme on eco-innovation*"[15]. *The* Process Industries are intrinsically linked to our quality of life and standard of living as we rely on the process industries to supply materials for many of the products we use in our everyday lives, and the ~6.8 million European workers who depend on the sector for their livelihoods. The Process Industries also have a big role to play in protecting our environment through more sustainable processes. The ability to be able to decouple the growth of the process industries from the environment would be of great benefit to us all. IS can play a major role in this decoupling in the future. By leveraging the SPIRE network, which brings together cement, ceramics, chemicals, engineering, minerals and ores, non-ferrous metals, steel and water sectors, several being world leading sectors operating from Europe, to ensure the deployment of the SHAREBOX platform as a facilitator for future industrial symbiosis between different sectors, the impacts for improve the eco-innovation capacity of the European Process industry is tremendous.

If we consider the specific expected impacts of the call: "Holistic energy and resource management systems should facilitate significant gains in sustainable processing with regard to several parameters (resource efficiency, energy efficiency and the emission performance)." The effectiveness of IS in delivering Europe's resource efficiency and

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[55] Lombardi DR and PT Laybourn. 2012. Redefining Industrial Symbiosis: Crossing Academic-Practitioner Boundaries. *Journal of Industrial Ecology* 16(1): 28-37. Both authors, at ISL, are closely involved with the IMATAIS proposal.
[56] UNEP, 2010), *ABC of SCP: Clarifying concepts on sustainable consumption and production*
[57] European Resource Efficiency Platform, 2014, Manifesto and Policy recommendations. 31 March.
[58] Economic Analysis of Resource Efficiency Policies: Final Report, 2011, COWI for DG Environment.
[59] Ellen MacArthur Foundation. 2010. Towards the Circular Economy, Vol 1: Economic and business rationale for an accelerated transition.

climate change mitigation objectives has been evidenced by research undertaken for DG Environment that identified NISP as the most effective amongst 120 policies and programmes in 23 countries globally[4].

"Optimisation of interdependencies and the identification of technology components allowing for a breakthrough regarding a cost effective reduction in the use of resources, which overcomes the difficult cross-company collaboration." We overcome the difficult cross-company collaboration via a secure platform that encrypts and anonymises company data- this will represent a step change for information sharing among enterprises.

"Optimisation of energy and resources supply and demand in selected areas should reduce the overall costs in energy intensive industries by at least 15%, by taking into consideration both economic and sustainability factors." Our LCA studies in WP6 will confirm the energy saving obtained in the industries involved in the pilot implementation of the platform.

The impacts of SHAREBOX can be estimated based on extrapolation from state of the art delivery today. NISP England model of IS has been recognised as world-leading IS delivery based on public investment. Over 7 years, the programme engaged with over 10,000 companies to deliver the impact shown in Table 3 from an investment of €42M (the third column accounts for the continuing effects of synergies, capped at maximum 5 years). If the SHAREBOX only achieves a level of market penetration across Europe comparable to NISP in England, based on a GDP proxy, exploitation of the pan-European SHAREBOX impact over 5 years would deliver a minimum of the results shown in Table 3.

Table 3- Potential impacts from applying NISP across Europe

	Total in year benefits	Total benefits if synergies last 5 years
Reduction of waste generated (tonnes)	56 M	281 M
Virgin material use avoided (tonnes)	76 M	381 M
CO2-eq avoided (tonnes)	50 M	252 M
Cost savings (€)	1.41 B	7.06 B
Increased sales (€)	1.50 B	7.48 B
Jobs created	63,000	~315,000

WP 5 and 6, are divided into two phases, by the customer (end-user) segment:

- Phase 1 – Customer is client actively delivering industrial symbiosis (Niche IS market)
- Phase 2 – Customer is client actively delivering resource efficiency, circular economy, materials security (Broader market)
- Phase 3 – Customer is Industry client actively involved in production (Mainstream industry)

Across the phases, a potential of over 1500 companies will be engaged with piloting SHAREBOX with existing partner networks. In Phase 1, piloting with consortium partners who are actively IPs in their respective countries: In Phase 2, partners engage directly with mainstream industry process industries with SHAREBOX – we are targeting 4 companies within the partnership, and have mitigated the risk by engaging now with companies to gauge interest in participation 3 years' hence. Further companies will be engaged through ISL's NISP network.

Clients will be divided in to two phases.
Phase 1 – Clients actively managing infrastructure through the management of industrial parks (IP level)
Phase 2 – Clients who are actively involved in production (company or sector level)
Operating across both these phases will be International Synergies Limited who actively deliver industrial symbiosis across a membership of 10,000 companies. The early involvement of ISL through both phases will ensure early take up of the solution.

<u>Phase 1 – Clients actively managing infrastructure through the management of industrial parks (IP level)</u>

Phase 1 will take place with organisations that are already managing industrial parks and will test the ability for SHAREBOX to impact resource efficiency when implemented at park management level. The parks have been chosen as they represent SPIRE industry sectors (chemical, engineering, minerals, non-ferrous metals, cement and ceramics) with a large number of SMEs. Over 1,500 companies will be represented through the four IPs partnering in the consortium. In addition, ISL will engage with companies based on the parks from the project outset to recruit them as

users of the software once available. Prior to the software being available, information from the companies will be collated and synergy opportunities will be identified for implementation.

We expect a wide range of synergies to be identified during this phase including direct company to company synergies as well as larger scale, infrastructure projects that require the direct involvement of the IP management organisation. Company to company synergies will likely occur at a faster rate than the larger scale synergies.

As there will be management and assistance of synergies identified in this phase, the anticipated impact of pilot activities in Phase 1 is estimated from NISP results produced from selected projects implemented in regions across the globe but taking in to account the reduced staffing and the extra functionality of the software. We have also had regard for SPIRE industry sectors.

Table 4 - Phase 1 impacts are modelled on a typical one-year pilot (England)

Impact	Phase 1
Waste avoided (tonnes)	137 k
Virgin resources saved (tonnes)	593 k
CO_2 saved (tonnes)	227 k
Cost savings (£)	8.5 M
Additional sales (£)	15 M

These **Phase one** impacts relate to clients actively managing infrastructure through the management of industrial parks (IP level). They are based on pilot scale, one year regional programmes delivered in four countries (Brazil, China, England and South Africa). While each country had different programmes, the overall impact was used to scale the results, taking in to account the length of the programmes and scale of the programmes. While the results for England have been included in Table 4, the table below represents the total and average impacts from the remaining three countries.

Table 5- Impacts from pilot programmes in Brazil, China and South Africa

Impact	Total from Brazil, China and South Africa	Average per programme
Waste avoided (tonnes)	1.68 million	560,000
Virgin material saved (tonnes)	1.68 million	560,000
Carbon Dioxide savings (tonnes)	198,000	99,200
Cost savings (€)	16.4 million	5.48 million
Increased sales (€)	14.3 million	4.8 million
Jobs created or safeguarded	81	

Within phase 1, four industrial parks will be partnering in the consortium and we will be identifying / progressing a mix of short term, smaller scale and longer term, infrastructure scale projects within the IPs. By scaling the forecast impacts within the range achieved through the pilots (and England), we feel confident that the results, coupled with International Synergies' experience make the impacts sensible and achievable.

Four industrial areas – Eskisehir, Turkey; Nules, Castellon, Spain; Chemie-Cluster, Bayern, Germany; and Bristol Ports, UK – will be piloting SHAREBOX during this phase, in addition to ISL.

- ISL will apply SHAREBOX to ongoing projects with organisations for whom large scale regional infrastructure projects are core business. These include companies in the construction, engineering and steel manufacturing industries. In addition, ISL will work with the management companies of the IPs to introduce the principles of IS at a larger scale within the IPs.
- Eskisehir, Turkey, represents engineering, manufacturing, construction and food. There are therefore multiple manufacturing sites and diverse waste streams with whom to pilot SHAREBOX. The client will have verifiable baseline data relating to resource efficiency against which the 'added value' of using SHAREBOX can be benchmarked.

- Nules, Spain (ITC, KEROS, KERAFRIT) represents the ceramics cluster, raw materials suppliers and frit tile manufacturers, with over 650 companies. Also present are other companies not directly related to the ceramic cluster, thereby providing excellent opportunities for cross industry sector synergies (ISSA, GUZMAN).
- Chemie-Cluster, Bayern, Germany has 260 companies from the chemicals sector.
- Bristol Ports, UK is a large port area that has previously engaged with ISL through the facilitated industrial symbiosis programme, NISP. During that initial engagement, over 10,000 tonnes of CO2 was saved through the realisation of synergies.

Expected short term impacts from trials conducted during this phase have been based on pilot scale, one year regional programmes delivered in four countries – Brazil, China, England and South Africa. The impacts vary depending on the type of project being implemented, the resource allocated to the pilot and the focus of the key performance indicators for each project.

Phase 2 – Clients who are actively involved in production (company level)

Phase 2 testing will take place at a company level and will be used to determine the potential for the SHAREBOX to deliver a step change in IS delivery within companies without the need for practitioner support. SHAREBOX will be implemented at four companies from within the consortium – Keros ceramic, Kernfit, ISSA-Sogefi Group and Grupo Guzman – each representing SPIRE industries, but will also be marketed by ISL to their existing client base. An example of the results obtained by ISL, through NISP, is given in table below.

Table 6- Phase 2 impacts modelled on past NISP experience with SPIRE process industries such As steel

Impact	Steel (1 company)
Waste avoided (tonnes)	12 k
Virgin resources saved (tonnes)	6 k
CO_2 saved (tonnes)	9 k
Cost savings (£)	601 k
Additional sales (£)	42 k

Phase two impacts are based on clients who are actively involved in production (company level). We have chosen SPIRE sector companies and for this phase we looked at the impacts achieved within the sectors. Two companies were chosen from each of the cement, non-ferrous metals and minerals sectors to base the project impacts on. Companies were located in England and South Africa. The results obtained from the companies are shown in the table below:

Table 7- Impacts from SPIRE sector companies

Impact	Cement (8 synergies)	Non-ferrous metals (10 synergies)	Minerals (2 synergies)
Waste avoided (tonnes)	2.2 million	200,000	250,000
Virgin material saved (tonnes)	2.2 million	2.15 million	>250,000
Carbon Dioxide savings (tonnes)	400,000	490,000	
Cost savings (€)	3.75 million	7.4 million	2.9 million
Increased sales (€)	1.7 million	39.9 million	7.4 million
Jobs created or safeguarded		34	

The impacts quoted for SHAREBOX take into account short term (during project) and medium term (3 to 5 years) impacts. Again, the impacts have regard for the number of synergies we envisage being completed within the life of the project, with the understanding that infrastructure projects take longer to plan and implement.

Based on State of the Art, it is expected that SHAREBOX will deliver greater savings pro rata than those quoted above. However, it must be recognised that each trial will have limited duration while the results quoted above are from longer time periods. It is expected that the piloting activity in the short term (duration of SHAREBOX funding) and medium term (if the pilot activities continue to deliver in the longer run, as per NISP experience) will deliver the following impact from the pilot activities themselves (this does not include impacts deriving from further exploitation) in Table below.

Table 8- Forecast Phase 1-2 impacts from SHAREBOX pilot activities

SHAREBOX IMPACTS	Short term (during project)	Medium term (3 to 5 years)
Waste avoided (tonnes)	500k	3 M
Virgin material use avoided (tonnes)	750k M	4 M
CO_2 saved (tonnes)	500k	3 M
Cost savings (£)	3 M	10 M
Additional sales (£)	3 M	10 M
Jobs created	20	200

The piloting activities detailed above are scheduled for delivery by all consortium partners in various countries. The estimated impact is derived above based on experience in the UK and elsewhere, from both mature and nascent activities.

Contribution to other environmental and socially important impacts

European firms of all sizes are actively pursuing resource and energy efficiency[60] but this topic is increasingly pressing as the European industrial sector shrinks, and energy costs and raw material demand increase[61] with increasing short-term volatility[62] in price and availability. Industry faces sustainability and inclusive growth challenges.

Industrial ecology can be a means of achieving decoupling between economic growth and environmental impact within the industry sector. A sub-field of industrial ecology is industrial symbiosis which focuses on material, energy and water exchanges between traditionally separate actors in order to achieve economic and environmental benefits. By integrating firms with each other sub-optimization can be avoided and even greater reduction in environmental impact can be achieved in total while also creating a business value and social benefits, or **so called triple bottom line benefits**[63].

As such, implementation of IS will not only occur at economic level but also environmental and social sustainability indicators will be assessed. These will include the associated energy, landfill, and GHG emission reduction impacts as well as the socio-economic impacts such as employment creation, etc. The use of IO modelling allows us to compute the indirect and induced effects of implementing IS on the above-mentioned indicators tracing the flow changes in the upstream and downstream processes of the involved production chains. Hence, we achieve a complete life cycle assessment of the IS.

NISP model of industrial symbiosis has been recognised for its successful delivery of resource efficiency, carbon reduction, eco-innovation, decoupling economic growth from resource use and supporting sustainable resource management and job creation (see Table 9, below. Column 4 presents the €/tonne if the entire investment had been only in the impact listed in that column). In relation to the powerful environmental and societal impacts that can result from industrial symbiosis, returning again to the UK NISP Programme, which serves as an independently audited example, at its peak in was actively working with an extensive 15,000 businesses, the majority of which were SMEs or micro. NISP successfully identified and closed out thousands of synergies, **collectively cutting CO_2 emissions by 42**

[60] Eurobarometer 2012. SMEs, Resource Efficiency and the Green Economy. March.
[61] DG Enterprise & Industry. 2013. EU Industrial Structure Report: Competing in Global Value Chains. © European Union.
[62] Dobbs, R, J.Oppenheim, F. Thompson, M. Brinkman and M. Zornes. 2011, Resource Revolution: Meeting the World's Energy, Materials, Food and Water Needs. November. © McKinsey & Company.
[63] Centre of Excellence in Cleaner Production (2007). Regional resource synergies for sustainable development in heavy industrial areas, Curtin University of Technology, Perth. Available: http://seg.curtin.edu.au/local/docs/publications/2007/arcbulletin1.pdf.

million tonnes and redirecting over 48 million tonnes of 'wasted resources' from landfill. It also was responsible for over 10,000 jobs[15].

Table 9- Impact of NISP in England, 2005-2013. *All metrics independently verified.

METRICS* (April 2005 – March 2013)	Total in-year benefits	Total benefits assuming synergies continue for 5 years	Value for Money Lifetime Spend (per tonne)
Landfill diversion	9 million tonnes	45 million tonnes	€0.130
CO_2 reduction	8 million tonnes	39 million tonnes	€0.150
Virgin material savings	12 million tonnes	58 million tonnes	€0.100
Hazardous waste eliminated	0.4 million tonnes	2 million tonnes	€2.740
Water savings	14 million tonnes	71 million tonnes	€0.080
Cost savings	€243 million	€1.21 billion	€0.005
Additional sales	€234 million	€1.17 billion	€0.005
Jobs	10,000+		

NISP England has operated under an external verification scheme whereby companies involved in completed synergies were contacted and synergies assessed to verify delivered outputs. ISL's standard methodology produces reporting for the metrics listed in the table above, using government-compliant calculations. ISL thus has experience operating to verification schemes and the project will be able to build on this. As part of the pilot phase, SHAREBOX partners will liaise with their local ETV bodies to identify innovative environmental technologies suitable for IS implementation based on local resources and data.

Barriers/obstacles or framework conditions that may determine whether and to what extent the expected impacts will be achieved

The market context (prices, regulations, security of supply) will determine the priority resources and sectors for commercial IS. In part this failure of existing online tools is because the market drivers aren't creating incentives for all waste to be diverted. For example, carbon reduction in Europe is not yet well-supported by the market, despite certain taxes and the Emissions Trading Scheme (ETS). At both the Global Green Growth Forum (2013) and the OECD Conference on Green Growth (2013), policy stakeholders and business leaders alike called for "getting the prices right" to incentivise resource efficient behaviour – this includes carbon, but also landfill, water, and other raw materials. In November 2013, Peter Vis (Head of Cabinet, European Commission for Climate Action) observed that the current market price for carbon, about €5/tonne, is substantially lower than expected at this time – and not sufficient to incentivise the desired behaviour in reducing emissions[64]. SHAREBOX deliverables includes policy seminars to inform the European Commission where its policies are creating barriers to the circular economy.

Industrial symbiosis (IS) is a priority for Europe, gaining substantial prominence in the last 10 years as a means to address the inefficient use of resources in business processes. Since 2009, facilitated IS has been included in policies, recommendations and case studies from the European Commission and its Directorate Generals, as a means to address the market failures that hinder Europe's transition to a sustainable economy, including (Asterisk* indicates those reports citing ISL's facilitated IS model, NISP): DG Environment, Best Practice under the EU Waste Framework Directive (2009)*; DG Environment, Roadmap to a Resource Efficient Europe (2011)*; DG Environment, Economic Analysis of Resource Efficiency Policies (2011)*; DG Enterprise & Industry, Sustainable Industry-Going for Growth & Resource Efficiency (2011)*; DG Regional & Urban Policy, Connecting Smart and Sustainable Growth through Smart Specialization (2012)*; DG Enterprise & Industry, Communication on Green Entrepreneurship (2013)*; DG Environment – Eco-Innovation Observatory Annual Report (2013); DG Research & Innovation, A Short Guide to Assessing Environmental Impacts of Research and Innovation Policy (2014)*; European Resource Efficiency Platform, Manifesto & Policy Recommendations (2014).

[64] Peter Vis, Head of Cabinet of Connie Hedegaard, European Commission for Climate Action. Climate KIC Making Transitions Happen Open Platform Event. November 2013.

The variables that must be considered as affecting the extent of the impacts can be summarised as:

- Business context: The drivers that create the business context (barriers and enablers) for IS include: legislation, policy, price, supply, tax (which can include but will not be limited to carbon, landfill and packaging), geography and business perception. At a high level, regulations across Europe are common but each nation state has different policies through which the regulations are implemented or supported. This can be a barrier to the uptake of IS across Europe. However, the direction of movement is to control resource efficiency and waste management through ever aligning implementation (taxes, landfill bans and standards as examples). We therefore believe that any potential negative impact will be reduced or mitigated through the life of the project.
- On the policy side, ISL are part of a FP7 consortia specifically looking at barriers to the uptake of resource efficiency measures by industry. Any relevant findings from POLFREE will be taken into consideration if they assist the exploitation and dissemination activities of IMATAIS.
- Standards as barriers: Procurement often involves the need to meet standards and can involve rigorous, expensive testing requirements. This could be a barrier to the uptake of IS. By focussing on the most economically viable resource streams as identified in WP 1, and the inclusion of Granta MI database which includes modules on standards and alternative materials that meet the substitution requirement these barriers are mitigated.
- Availability and quality of publicly available regulatory data from European Member States: IPPC and EPR data is often aggregated which would be useful for sector analysis but not necessarily for individual company/resource selection.

2.2. Measures to maximise impact

2.2.1. Dissemination and exploitation of results

We have drawn up plans for the dissemination and exploitation of the project results, and in particular the development of a robust business model to further develop and commercially exploit the SHAREBOX Platform as a secure integrated platform of facilitating and optimizing resource sharing/exchange among multiple production units of a single company or multiple companies on a single industrial production site **to obtain improved efficiency in the process industries**. Moreover, SHAREBOX will be offered and communicated as a valuable tool for developing further site integration advantages to a broader public of industrial site operators. Thus, clear guidelines will show how SHAREBOX can generate value added for different types of industrial areas by providing cost advantages, improving resource efficiency, reducing the environmental footprint, intensifying the integration of tenants (and thus their local commitment to the site), generating a positive influence on the innovative eco-system of an industry park and supporting site promotion activities. For example, in addition to the practical showcases by CCB, NEPIC, ECI and Ceramics cluster, a wider stakeholder network of industry park operators will be involved by CCB's Competence Center for Industrial Site Development, covering industrial park owners, facility managing companies, public authorities (running industrial sites) and other potential users and multipliers of the SHAREBOX methodology. By individual discussions, site visits and working meetings, existing initiatives in the field of industrial symbiosis and resource efficiency will be checked in order to optimize their compatibility with SHAREBOX (as part of the pilot implementation work). The implementation of SHAREBOX as a business model (WP7) shall not only focus on the market placement of a competitive IT-solution, but rather on an integrative system with a maximized connectivity to individual resource management initiatives. This way, SHAREBOX will contribute to the development of EU-wide shared standards in industrial symbiosis. A learning network of SHAREBOX-users and other initiators of resource management initiatives will be coordinated also beyond the timeframe of this project; its output will be distributed online, by brochures and at public events. Our initial exploitation and business plans, as well as our dissemination plans are discussed below. These will be built on through WP7.

Initial exploitation and business plans

In keeping with the expected scope of SPIRE-06 projects, we have drawn up an "*outline of the initial exploitation and business plans, which will be developed further in the proposal project*".

In terms of the proposed exploitation routes, the SHAREBOX consortium will **capitalise on all potential exploitation pathways** to maximise the impact of this project:

a) Using the results for developing, creating and marketing new products, processes, and services.

In H2020 it is essential that the public resources and efforts used in research are converted into socio-economic benefits to the EU[65]. This will only be truly achieved with enduring impact if we hit on a **sustainable business model and individual business plans for all the exploitable outputs of the SHAREBOX Project.**

The core exploitable output of the project is the secure cloud SHAREBOX platform for the flexible management of shared process resources that will provide plant operations and production managers with the robust and reliable information that they need in real-time in order to effectively and confidently share resources (plant, energy, water, residues, and recycled materials) with other companies in an optimum symbiotic eco-system. The project will take the platform to TRL6, whereby at the end of the project, the integrated unique SHAREBOX ICT platform, through which enterprises will securely exchange the data via APIs, will have been pilot tested at demonstration sites, using industrial partners' real production units. Our immediate post-project exploitation activities will centre on productising the beta version of the platform (which will be led by IRIS from a software development perspective and ISL will contribute their expertise in relation to IS delivery) to Minimum Viable Product for commercial release within 18 months of completion of the project. This productisation work will be done in close collaboration with the industry park operators (CCB, EIS, NEPIC and AICE), and the industry partners from the process industry and other sectors (CCB, KEROS, KERAFRIT, GUZMAN, and ISSA) so that the product is deployed in their facilities to ensure that they quickly benefit as early adopters of the platform. During the 18 month pre-launch phase, we will also implement additional industry case studies to serve as powerful endorsers for gaining market confidence.

Business Model for the SHAREBOX Platform: The SHAREBOX supercomputing cloud-enabled platform for industry resource and energy management will be commercialized through a Software-as-a-Service (SaaS) business model to exploit the platform commercially (under the SHAREBOX® brand, which is currently being trademarked). The platform will be deployed via a supercomputer cloud (such as Amazon EC2, Cloud Spectator, or others). This has the advantage of offering affordable, high performance computing to end-users for high processing power and speeds. The revenue generation will based on a subscription model- monthly or annual payments. A pricing structure will be developed that will be based on an initial registration fee (for access to the subscription-platform and initial training-on-site or via online tutorials) and payment packages for monthly payment plans, or annual plans (that will offer a 1 or 2 month discount). Flexible packages will be devised tailored to user segments (these packages will be drawn up during *Subtask 7.3.3 Business and exploitation plans*).

As SHAREBOX is a cross-sector platform technology, and target users of the platform are as follows (user segments will be further studied during the project):

- Customer is client actively delivering industrial symbiosis (Niche IS market)
- Customer is client actively delivering resource efficiency, circular economy, materials security (Broader market)
- Customer is Industry client actively involved in production (Mainstream industry)

The Sharebox Internet site (i.e. the doorway to the subscription platform) could potentially host some 'freemium' content and starter tools as a hook for securing paying clients. A user community (whereby individuals can register and manage their own user profiles, and interact with other users through the user forum) could also be created on the platform. They could enjoy the 'freemium' content (IS literature, news and updates, benefits of IS, starter training course, information about events, etc.) and starter tools. This would contribute to generating traffic to the site, which is important for gaining new clients, and would also contribute to our ambition through SHAREBOX to become the 'European grown' cornerstone of the European and Global IS movement. IRIS will exploit the SaaS business model in partnership with ISL- royalties on subscriptions will be paid to the partners who have provided the tools for the platform so that they benefit from the commercial opportunity. IRIS has already beta launched an internally developed business management SaaS Platform and they are experienced in the development of cloud-based business models. ISL can provide the commercial route to the market.

65 https://www.iprhelpdesk.eu/sites/default/files/newsdocuments/IP%20Management%20in%20Horizon%202020_the%20proposal%20stage_0.pdf

Business model for the SYNERGIE application: ISL will commercialise the SYNERGIE 2.0 applications, as a tool for identifying synergies among companies. IRIS will act as a technology partner for ISL by supporting and maintaining their SYNERGIE software application.

During the project in *Subtask 7.3.3 Business and exploitation plans*, the business models for exploiting the SHAREBOX platform and the SYNERGIE app will be further developed based on the above outline approaches. Formal business plans will be elaborated, which will detail how each of these project outputs will be used in commercial exploitation activities for business growth and competitiveness.

b) Using the results for further research and spin-off projects

Our ambition is for the SHAREBOX project to be a motor for a sustained applied research and development programme in Europe in the field of Industrial Symbiosis. To this end, an IS Research Roadmap will be developed over the course of the project. The Roadmap will take into consideration the further collaborative research spin off projects in the area of IS and resource management and the circular economy. The IS Roadmap will also engage with the authors of other Roadmaps, such as the SPIRE Roadmap, CEPI Roadmap (Forestry Sector); ECs Roadmap to a Resource Efficient Europe, the strategic research intentions of SUSCHEM, etc. to ensure we contribute to a European Knowledge Area for IS. All partners will be involved in the drawing up of the Research Roadmap in keeping with their fields of specialisation and research interests.

c) Using the results for standardisation activities

The results will also be exploited for standardisation. To this end, through DECHEMA and SPIRE, as well as through ISL and EUR-ISA we will establish dialogue with standards bodies early on in the project to pave the way forward for standard setting. The publications that will result from the project, around the non-confidential aspects of the results of the validation of the field work performed in this project will be of high importance, as they will be peer-reviewed. This will serve as a powerful independent approval of our results towards standards bodies and policy stakeholders. The project will contribute positively to the wider opportunities to be gained by incorporating industrial symbiosis into mainstream policy and identifying robust and sustainable funding instruments to further support industrial symbiosis activity in the country.

d) Use of the results for education and training

It is considered that there is very high potential to use the knowledge generated over the course of the project as content for formal third level education modules for skills generation in the area of Industrial Symbiosis. This exploitation route will be exploited by the participating Universities and Institutes (UT, UNIVLEEDS, USTRATH ZWAH, UPC, etc.). ISL and DECHEMA will develop professional training programmes for upskilling in industry so that companies are equipped with the skills required to engage in industrial symbiosis; this could be another source of revenue for ISL.

Outline Dissemination plan

Our dissemination activities during the SHAREBOX project will aim to **diffuse the non-confidential results that are generated during the project to as wide and relevant an audience as possible** in order to heighten awareness about the results, to contribute to knowledge building and upskilling in Europe, as well as to ensure that as many relevant actors and stakeholders as possible have access to these results.

As has been discussed, this is an industry driven Research and Innovation Action that has a vision **to develop the world's first platform for the integrated and holistic management of shared resources among industries, clusters, and eco-parks**. As we have a clear intention to use the results of this project for post-project commercial activities, and the project will also involves demonstration sites involving process industries (and industries from other sectors) whereby at times sensitive information regarding their processes maybe be involved, a **robust confidentiality policy will underpin all dissemination activities**. The consortium experience in collaborative projects points towards the default position that all information is confidential unless otherwise agreed as non-confidential. This is the clearest starting point. However, that said, on the one hand, the **academic participants in the consortium intend to publish results**, which is considered a very positive success indicators by all the consortium, not only for diffusing our results among the academics and research communities, but also serving as powerful peer reviewed documents towards the industry, standards bodies and policy stakeholders. On the other hand, the industry

partners and market actors also intend **to showcase and exhibit results at tradeshows, exhibitions and through other industrial channels** (technical/sectoral magazines, etc.).

As such, in order to ensure that dissemination activities are embedded in a robust Knowledge Management approach, the Project IPR Manager (Prof. Miquel Sanchez Marre, UPC) will work closely with the Communication and Dissemination Manager (Dr. Renata Korcher, DECHEMA) to manage the identification of non-confidential publishable/discloseable information that can be available for our dissemination activities. At this stage, a preliminary dissemination plan (Table below) has been drawn up for **disseminating the non-confidential or protected results of the project to our different target audiences of the project**.

Table 10- Outline Dissemination Plan

Target audience	How project results will be presented	Where projects results will be presented	Performance Indicators	Expected Impact
Industry: Industrial Parks; Process industries; Other sectors; IS and environmental engineering companies	Presentations, including physical demonstrations where appropriate	Tradeshows, exhibitions, workshops, seminars, own SHAREBOX events (trainings, etc.) CEVISAMA trade fair; ExpoQuimia; ChemSpec Europe, etc.	> 8 events reaching > 2000 companies	- Generate industry awareness about the strategic benefits of resourced sharing and IS - Stimulate interest in the SHAREBOX project and its outputs in order to generate market pull to support our business and exploitation plans
	Project Videos	Project website, You tube, websites of networks and platforms, e.g. "Ceramic observatory", process-worldwide.com, etc.	> 500 views	
	Documented IS implementation case studies (one for each site)	Technical magazines (e.g. LOCIMAP online magazine, etc.), newsletters, downloadable from project website.	> 300 downloads and 6 articles	
Research Community (scientists and early stage researches) & IS networks	Peer reviewed publications	Peer-reviewed journals (& green open access- see below): ISI-journal; Sustainable Chemical Processes J…	> 10 Peer-reviewed publications	- Contribute to strengthening the European Research Area (ERA) in Industrial Symbiosis - Attract early stage researchers to IS - Prevent brain drain by positioning EU as a scientific hub for IS
	Presentation of papers/posters at conferences	QUALICER congress. Annual "SYMBIOSIS" Conference; ACHEMA, etc.	> 10 Conference papers/posters	
	Presentation of project results at IS network events and through their channels	European Chemical Regions Network, EUR-ISA, SPIRE, SUSCHEM events and channels, among others.	> 5 Presentations	
Policy makers	Position papers, pictograms, peer reviewed publications, presentations	Face-to-Facemeetings,Policy round tables, Policy events	> 3 meetings/events	- Facilitate our outreach and engagement
General public	Press releases, Project Video, Pictograms	Mass media, You tube, Social media channels, Project Website	> 10 press releases, >500 downloads of videos/literature	- An informed public who are aware of industry sustainability efforts and their benefits.

During the project, the consortium, through *Task 7.2 Networking and dissemination of the project results* and under the leadership of the Communication and Dissemination Manager, will build on the above outline plan. Our project dissemination activities will link with the broader dissemination strategies of the SPIRE network, SusChem, European Industrial Symbiosis Nerwork (EUR-ISA)- which has a Europe-wide membership and is led by partner ISL- European Chemical Regions Network (ECRN) etc., which will contribute to efficient coordination and dissemination of the project outputs across the relevant European and international communities. Moreover, the dissemination strategy will also ensure it supports our efforts in relation to the **future exploitation of the project results**.

An important dissemination platform for the project will be the SHAREBOX website, which will be built during Task 7.4 and which will represent a 'next generation' H2020 project 'website' in that it will serve as an enduring internet site for

promoting the project and housing all the downloadable public deliverables and outputs of the project, and it will also house our 'SHAREBOX Green Channel' for open access to the peer-reviewed publications of the project. It will be widely disseminated across Europe and worldwide to build up usership and registered members from the IS community towards becoming a global hub for IS.

How the proposed measures will help to achieve the expected impacts of the project

Industrial parks and other integrated production sites represent perfect eco-systems for optimizing competitiveness of European industry by minimizing the use of resources, creating competitive advantage for the European Union as an eco-efficient production site and maximizing the innovative output of companies (in particular SME) by a deeper integration of production processes. Connecting European industry parks by bringing together aspects of resource management, cluster management and site development is an ongoing task for all industrial stakeholders, especially regarding the development of joint standards and performance indicators. While cluster management networks and industrial regions within the EU are already well connected by a broad range of initiatives, platforms for resource efficiency still lack a joint standard and practical links to other supportive structures. As CCB's Site Development Program brings together a network of European partner clusters (e.g. as part of the ESCP-initiative "Wiintech"), major players of European chemical regions (chairing the Project Council of the European Chemical Regions Network ECRN) and industry park operators, these links will become the anchor points of a learning network around the SHAREBOX approach. It will be used to share experiences among different industrial clusters, further improve SHAREBOX and adapt it to the needs of specific sites and support industrial regions in attracting foreign direct investment. This way, SHAREBOX can become a vital element of a European site development strategy.

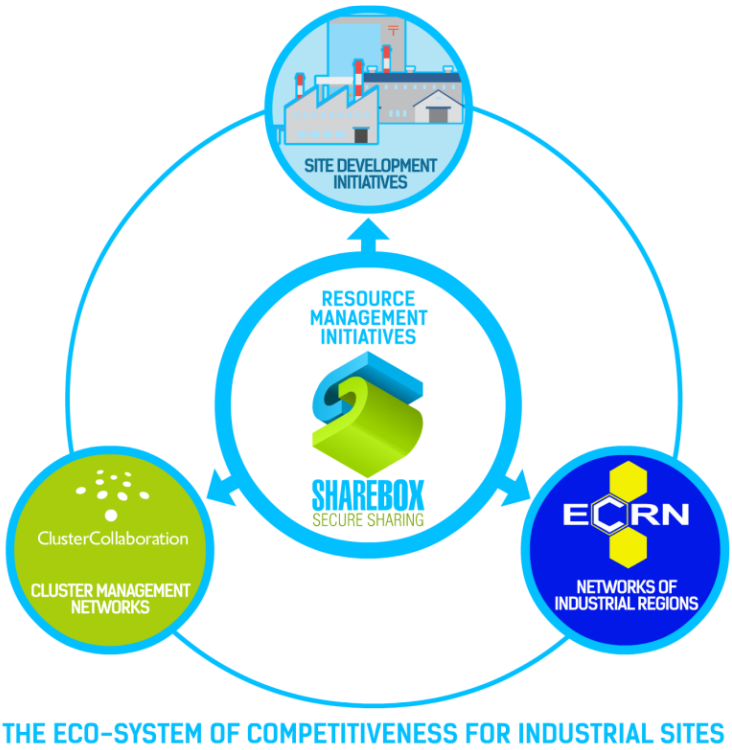

Figure 3- How SHAREBOX can be linked to a European eco-system of improving competitiveness of industrial sites

How the participants will manage the research data generated and/or collected during the project

The SHAREBOX project will not participate in the Pilot Open Access Research Data scheme, and all the research data generated by the project will be used for the purposes of the development of the SHAREBOX platform and the other project outputs.

Strategy for knowledge management and protection

To pave the way forward for exploitation and dissemination of the project results, the consortium will implement an integral strategy for the effective **management of this knowledge** *(Fig. 4)*.

Background IPR and access rights: **'Background'**[66] means any data, know-how or information-whatever its form or nature (tangible or intangible), including any rights such as intellectual property rights –that is a) held by the

66 H2020 Model Grant Agreement: http://ec.europa.eu/research/participants/data/ref/h2020/grants_manual/amga/h2020-amga_en.pdf#page=233

beneficiaries before they acceded to the Agreement, and b) needed to implement the action or exploit the results. In relation to access rights to Background, the consortium will comply with the H2020 default rules[66], whereby on the one hand, the partners/beneficiaries will provide each other access — on a royalty-free basis — to Background needed to implement their own tasks under the action/project. On the other hand the partners will give each other access — under fair and reasonable conditions — to background needed for exploiting their own results. The consortium will declare an up to date status of Background before the Grant Agreement (GA) is signed, and at proposal stage it has been confirmed that all partners have the access they need to each other's background.

Results definition, protection, ownership and access rights

'**Results**'[66] means any (tangible or intangible) output of the action such as data, knowledge or information — whatever its form or nature, whether it can be protected or not — that is generated in the action/project, as well as any rights attached to it, including intellectual property rights. Through *Task 7.1: Knowledge Management & IPR Protection,* the Results being outputted from the project will be continually assessed in line with the status of developments, whereby **novelty, patentability and protection** will be assessed. To this end, in order to **verify novelty** and to ensure that the Results do not trespass into the owned knowledge space of a third party, up-to-date patent reviews will be performed periodically. In relation to protection of the results, the partners will aim to protect all results that can be commercially or industrially exploited either through Patents (for inventions), Trademark (brand, etc.), Industrial design, Copyright (for the SHAREBOX Platform), Trade-secret, or Confidentiality, as appropriate (i.e. the form which offers the most adequate and effective protection for each type of result). As a result, over the course of the project, a clear definition of the protected IPR will be completed, whereby a verifiable list of all IPRs that have been applied for or registered (Trademark™, any European patent applications, etc.) will have been drawn up, along with a description of all the results that may have commercial or industrial applications. **Results shall be owned by the participant generating them**[66]. To avoid doubt or confusion over ownership, a protocol of procedures will be defined that each partner will follow to clearly show the development of the generation of knowledge or Results. In cases where a Result is generated jointly (i.e. where the separate parts of some result cannot be attributed to different participants), it will be **jointly owned**, with the allocation and the terms of exercising this joint ownership to be included in a Joint Ownership Agreement that will be drawn up.

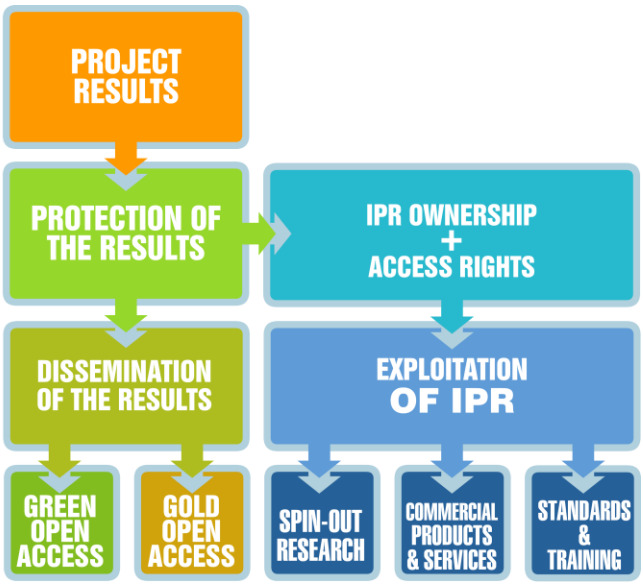

Figure 4- Integrated Knowledge Management approach

In relation to **access rights to Results**, and again in keeping with the H2020 default rules, the partners will give each other access — on a royalty-free basis — to results needed for implementing their own tasks under the action[66]. The partners will give each other — under fair and reasonable conditions — **access to results needed for exploiting their own results**. 'Fair and reasonable conditions'[66]**Error! Bookmark not defined.** means appropriate conditions, including possible financial terms or royalty-free conditions, which will be agreed in Exploitation Agreement(s) taking into account the specific circumstances of the request for access, for example the actual or potential value of the results or background to which access is requested and/or the scope, duration or other characteristics of the exploitation envisaged. However, an overriding principal will be that the **participating industry partners are provided with the access that they need to the results to exploit them in keeping with the needs and strategy of their business.**

Knowledge management activities will be led by the IPR Manager, who has significant experience in knowledge protection and ownership and access rights management in collaborative research project. This bring invaluable expertise to the project, as well as the support and expertise of the UPC highly qualified legal team. Over the course of the project, the consortium will draw on other external assistance and support, as required, from such sources as the **IPR-Helpdesk** (http://www.ipr-helpdesk.org), **Enterprise Europe Network** (http://www.enterprise-europe-network.ec.europa.eu/index_en.htm), among other sources in the partner countries. An experienced patent attorney

will also be approached for IPR protection implementation (and this has been budgeted).

Open access to peer-reviewed scientific publications

The two ways authors can provide open access are (1) by self-archiving their journal articles in an open access repository, also known as 'green' open access, or (2) by publishing in an open access journal, known as 'gold' open access.[67] It is envisaged that green open access will be the preferred option in the SHAREBOX project, whereby the partners will publish in peer-reviewed scientific publications and then self-archive a version of the article for free public use in their institutional repository, in a central repository, or on some other open access website. The project website (or Internet Portal) will house a 'SHAREBOX Green Channel' for open access to the peer-reviewed publications of the project. In relation to expected publications from the project, these include and are not limited to review papers and papers on the results of the pilot validations, etc.

2.2.2. Communication activities

Communication activities are considered to be an important aspect of this project, to ensure that we effectively promote our project, its ambitions, its progress and results, as well as its benefits and impacts to the stakeholders that we want to each. Importantly, one of the key's to successfully gaining impact **is our ability to secure stakeholder 'buy-in',** i.e. the cooperation or positive participation of our stakeholders.

Without doubt we want to secure the 'buy-in' of the European Process Industry and Industrial and EcoParks and we aim to generate awareness and interest in the project and the strategic benefits of IS. We are very interested in a two-way dialogue with industrial parks and the industry ensuring that we can involve them in the innovation cycle of the project to pick up their requirements, guidance, and feedback to ensure we are industry and market focussed at mainstream level. An important opportunity for the project will be linking with the broader dissemination/communication strategy of the SPIRE programme, which will in turn contribute to efficient coordination and dissemination of SPIRE project outputs across the programme to facilitate the market uptake of innovative solutions, whereby early awareness of key innovation themes emerging from the projects to the entire SPIRE stakeholder community as well as sectors in close relationship to the process industries (i.e. end users) is desirable, as is a communication campaign to all process industries that considers how they can adopt a synergic approach for understanding how IS can enable '*significant gains in sustainable processing with regard to several parameters (resource efficiency, energy efficiency and the emission performance)*', thereby leading to new sustainable growth opportunities, competitiveness and growth. To this end, the planned communication activities will be best used to promote and pave the way forward for the future industry exploitation of the project results.

In addition, there are enormous medium to long terms impacts to be leveraged by successfully initiating two-way dialogue with policy-makers and standards bodies to promote the SHAREBOX platform, as well as gain their interest and support towards the SHAREBOX platform and its tools becoming an industry standard or a recommended approach for industrial IS implementation. Moreover, it is important that citizens are aware of our research and its potential societal and environmental impacts, especially as we are benefitting from tax-payers contributions. This will contribute to ensuring that **citizens** are informed and feel like they are getting value for money and to promote the efforts of the European Process Industry to improve its competitiveness and sustainability to protect our planet, quality of life, as well as to lead to economic wealth and much needed job creation for European citizens.

If tackled well through a professional communication campaign (which will be led by DECHEMA, who will benefit from the experienced graphic design and marketing team in IRIS), SHAREBOX will be successful in outreaching to and mobilising of industry and policymakers behind our project and its outputs at all stages of the work plan, as well as keep the citizen informed. To this end, through Task 7.4, we will develop a communication plan, which will consist of the following steps:

1. **Identify and list all possible stakeholders**. In addition to the main stakeholder groups discussed above (industry, policy makers, citizens), the target audiences of our dissemination activities will also be analysed and any other stakeholders that might be considered important will be included as targets of our communication campaign.

[67] Jeffery, Keith G. 2006. "Open Access: An Introduction". Ercim News, 64, January 2006.

2. **Organise our stakeholder.** This will involve clustering stakeholders where relevant and establishing our different categories of audience around which we can build targeted messages and suitable communication activities for effectively reaching each.
3. **Branding.** The principles behind the SHAREBOX brand, which is visually identified through its logos, and carrying a powerful tag line, will be further developed at proposal stage.
4. **Development of project communication materials and tools.** The SHAREBOX project website, leaflet, posters, pictograms, social media sites, videos and virals will be carefully designed for use in supporting our communication campaign.
5. **Design and implement a communication campaign.** We will ensure regular communication with our stakeholders through creative campaigns that will involve press releases, digital campaigns through the social media, periodic newsletters (at least 1 per year), video releases, blog posts, poster campaigns at tradeshows and exhibitions, face-to-face meetings and round tables with policy makers, meetings and events with industry, etc.

Finally, in Task 7.4 we will build on the Dissemination Table (Table 10) at the beginning of the project and we will adapt it for planning communication activities for promoting the project, based on a process of analysing: Why – the purpose of the communication action; What will be communicated – the message; To whom – the audience; How – the method; When – the timing.

3. Implementation

3.1. Work plan

Brief presentation of the overall structure of the work plan

The project objectives have been used as the starting point for planning the work that needs to be implemented within the project. All the actions that are required to meet the objectives of this research innovation action, have been collectively analysed by the interdisciplinary consortium. It was agreed how best to distribute the workload among the various experts in keeping with their relevant fields of knowledge, capacity, infrastructure, resources and in the case of the industry end-users, and their process activity. The timings of the work, the logistics, as well as the interdependencies between the different tasks has been given due consideration to ensure a coherent work flow. This bottom-up planning process has greatly facilitated the task of arriving at a **clear, comprehensive and realistic work plan** that the consortium is confident will serve as a solid platform to ensure that the project's S&T and innovation goals are fulfilled.

The project is divided into 8 **work-packages** (WP), each with its own **tasks, deliverables and associated milestones** in order to facilitate implementation and monitoring of progress. First of all, the **end-user requirements and specifications (WP1)** will feed in to guide the case study definition and the expectations and specifications for the SHAREBOX platform that will feed into the development of the IS methodologies (WP2) and Intelligent Decision Support Tools (WP3) that will be coded and integrated into the overall Platform in WP4. The levels of pilot trials will be implemented- at Industrial Park Level (WP5) followed by connecting industries to the platform for live trials with real process conditions. Validation work will be performed to confirm the impacts via LCA assessments. This validation work will verify the platform at TRL6. The effective implementation of an Innovation Cycle **(WP7)** throughout the project will ensure the wider industry, societal, market and regulatory requirements are taken into account and fulfilled and that indeed the derived industry, competitive, environmental and societal benefits are communicated to the target stakeholders. In parallel, the post-project exploitation and dissemination activities will be planned for an effective post-project phase that will accomplish the full scaling up of the three case studies and the further improvement and market deployment of the SHAREBOX Platform in the European Process industries with 18 months of project completion, which will be highly scalable in the future. **WP8**, Project Management, will ensure the project is delivered on time, on budget, and to expected quality/scope.

The Gantt Chart shows the timings and the interdependencies between the different WPs and tasks. During the planning phase, the different WP and task leaders have collaborated together to ensure that each leader has planned his/her work in the context of its necessary interaction and potential dependency on or feed in to other WPs and tasks.

A key consideration has been to ensure that the tools and methodologies that need to be implemented into the SHAREBOX Platform are ready in a timely fashion so that coding can begin to ensure an alpha version of the Platform is ready by Month 12 so that piloting with the Parks can commence (and the progression to improved versions of the Platform will continue in parallel with the field work).

There is good collaboration between the different experts in the different WPs to ensure seamless transfer of knowledge between the disciplines as well as to ensure developers from a previous task are also stakeholders in the tasks that their work is feeding into. Logistical matters have been factored in, such as the time needed for ordering parts/components, for shipping of prototypes from development to test sites, installation times, the 'what if' scenarios of parts not arriving, of installation work throwing up challenges, seasonal factors, etc.

As this is a highly experienced development team, and in particular with many members of the consortium experienced in applied R&D under tight timeframes involving integration and demo work at industrial sites, the work planning has benefited from this 'on the ground' pragmatic experience, especially in terms of consideration of interdependencies and timings. The potential risks have been identified and contingency plans proposed. Importantly, the collective planning of the work plan has achieved the 'buy-in' of all the project developers and participants and an early understanding of the importance of a team effort, team planning and active communication among all WP and task leaders. Moreover, it is felt that the one-page Gantt itself (which also includes all the deliverables and milestones) will be a very effective tool for high level control and obtaining a helicopter overview as part of effective project management during the project.

Finally, the project work plan mobilises a strong collaboration between the academics (UT, UPC, UNIVLEEDS, USTRATH, ZHAW), the SMEs (IRIS, ISL), the park operators and innovation actors (CCB, EIS, NEPIC, DECHEMA, AICE) and the industries (KEROS, KERAFRIT, ISSA, GUZMAN) along the project. The academic partners are leading the foundational and theoretical work of the project, and will bring their knowledge and methodologies in energy scheduling and conversion (UNIVLEEDS, USTRATH), In-put-Output modelling, Game Theory, Agent Based Modelling (UT), purification and waste treatment technologies and Life Cycle Assessment (ZHAW), knowledge engineering and machine learning (UPC) towards developing the tools and technologies for the holistic SHAREBOX platform, as well as in assisting in IS implementation in the industry and providing scientific validation to demonstrate the success of the SHAREBOX platform in facilitating and optimising resource sharing. The SMEs (IRIS and ISL) will work closely with the academic partners in integrating the tools into the SHAREBOX platform (IRIS) and for improving the state-of-the-art in IS delivery (ISL). The park operators and innovation actors (CCB, EIS, NEPIC, DECHEMA, AICE) will work closely with the SMEs and academics in sharing their knowledge of their industry members in the early stages of the work plan in relation to the practical and regulatory challenges and constraints that will need to be fulfilled in order to have an exploitable and useful SHAREBOX platform, which they will validate from the perspective of its use as a tool for resource management at the park operation level. The industries (KEROS, KERAFRIT, ISSA, GUZMAN) will provide inputs, requirements and specifications from the company level to guide the development of the platform as a tool through which businesses will share data via secure APIs in relation to their process waste streams, raw materials requirements, production schedules, etc. and they will validate the platform using this real process conditions. This is a highly relevant consortium that mobilises a meaningful collaboration across all levels of the work plan.

Gantt chart showing the timing of the different WPs and their components

No	Workpackages and Tasks	Del. & Milestones	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
1	**Process Industry Requirements & Specifications**	**ML1**						#																																											
1.1	In-depth case study analysis of IS on Process industries: success factors	D1.1, D1.2 D1.3, D1.4 D1.5					■																																											
1.2	Industry engagement and End-user requirements						■	■																																										
1.3	Analysis with the "Quality Function Deployment (QFD)" method and derived system specifications							■	■																																									
2	**Industrial Symbiosis Methodologies**	**ML2**						#																																											
2.1	Taxonomies and Standards	D2.1, D2.2						■																																											
2.2	Self-sustaining database for tailored purification and waste treatment technologies	D2.3, D2.4												■																																				
2.3	Utilities and energy enabling technologies and system interactions	D2.5, D2.6												■			■									■						■																		
2.4	Engagement Methodology	D2.7										■																																							
3	**Development of intelligent decision support tools**	**ML3**								#																																									
3.1	Potential shared-resource supply-demand matching for pre-designing IS networks	D3.1, D3.2										■		■																																				
3.2	Computing pay-offs of cooperation strategies	D3.3, D3.4																		■		■																												
3.3	Designing optimum IS networks	D3.5, D3.6																														■■																		
3.4	Designing complementary decision support tools for IS networks	D3.7, D3.8																					■			■																								
3.5	Designing an intelligent module for improving the usage of the SHAREBOX system	D3.9, D3.10																																	■			■												
4	**SHAREBOX Software & Platform Architecture**	**ML4, ML5**												#												#						#															#			
4.1	Platform architecture and programming work organization	D4.1, D4.2 D4.3, D4.4 D4.5																																																
4.2	Development of the Platform: Controller (core routines), Back-end and Front-end																																																	
4.3	Development of the Platform Graphic User Interface													■																		■																		
4.4	Databases access implementation													■																																				
4.5	Virtual simulator testing and optimisation work																■																																	
4.6	Upgrading of SYNERGIES software to V2.0													■																																				
5	**Platform set up and testing in Industrial Parks**	**ML6**																								#																								
5.1	Partner engagement and opportunities identification by means of SYNERGie 1.0	D5.1, D5.2 D5.3, D5.4						■												■																														
5.2	Onsite SHAREBOX Platform Alpha Release set up and pilot trials																			■																														
5.3	SHAREBOX Platform Alpha Release feedback																									■																								
6	**Field trials with real industrial conditions**	**ML7, ML8**																											#																		#			
6.1	Definition of impact indicators and validation methodology	D6.1, D6.2 D6.3, D6.4																									■																							
6.2	Set up of communications network with participating companies																																																	
6.3	Field trials with the SHAREBOX Platform																																														■			
6.4	Verification of impacts and recommendations for further improvements																																														■			■
7	**Innovation impacts**																																																	
7.1	Knowledge Management and IP Protection	D7.1, D7.2 D7.3, D7.4 D7.5, D7.6 D7.7, D7.8																																																
7.2	Networking and Dissemination of the project results																																																	
7.3	Business and exploitation plans																									■																								■
7.4	Communication activities to support stakeholder engagement				■			■																		■■																								■
7.5	Training and skills building																																																	■
7.6	Standards setting																																																	■
8	**Project Management**																																																	
8.1 Management of consortium resources; 8.2 Implementation of project management procedures; 8.3 Interface and reporting the EC		D8.1	●♀ ■		▫			●♀ ▫			▫			●♀ ▫			▫			●♀ ▫			▫			●♀ ▫														▫			●♀ ▫			▫			●♀ ▫

■ Deliverable / # Milestones ●Steering Committee Meetings / ▫ IS Implementation Team Meeting / ♀ Innovation Committee Meeting/

3.2. Management structures and procedures

Organisational structure and decision-making

The project management will be performed by the project Coordinator IRIS based on their proven project management procedures (table 11 below) which adhere to their ISO 9001:2008 certified processes and UNE-EN-ISO 166002: Management Systems for R&D&I. In terms of the **organisational structure** (Figure 5), this consists of the **Project Coordinator** (the interface with the European Commission) who is supported by the strategic project management and decision making body- the **Project Steering Committee**. The executive management bodies are also envisaged: a **Technology Development Team** and a **IS Implementation Team**. The **roles and decision making capacity of each** is discussed below

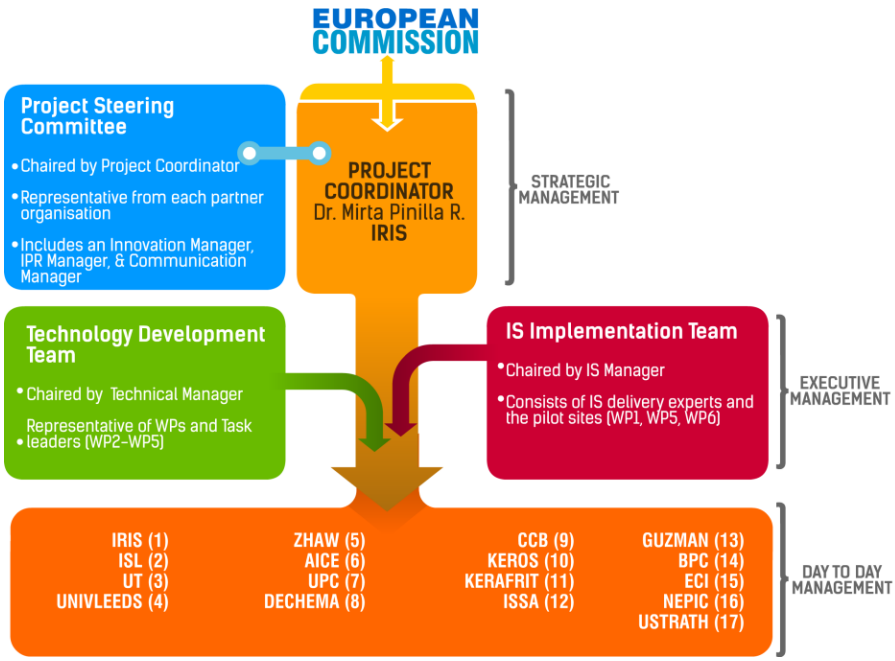

Figure 5- Overview of the SHAREBOX Management Structure

Project Coordinator

The Project Coordinator (**Dr. Mirta Pinilla Rodriguez**) is responsible for the day to day running of the project, and for ensuring that the work programme adheres to its schedule, fulfils its objectives, and the project meets the needs and expectations of all the partners. She will be active in steering the evolution of the project by closely monitoring the work performed by each partner. She will assume responsibility for ensuring that **the project management procedures are being correctly implemented** and that all deadlines and obligations are met. She will also act as the communication and administrative interface with the EC, to carry out the reporting and overall administrative management of the project. As this is also a leadership role, it requires a profile that is sufficiently experienced, qualified and capacitated not only to supervise the execution of work programme and delegation of the work packages (WPs), but also one that can foment team-building and is a strong communicator, who can motivate the team and encourage proactive participation, facilitate collective problem solving and the taking of timely corrective actions.

At company-level, IRIS has the demonstrable experience, available technical and administrative resources, linguistic and communications skills to make them a suitable candidate to successfully perform project management coordination tasks. In terms of the competence of Dr. Pinilla to successfully execute this role, she has solid project management experience, both at national and EC level. In FP7, she has effectively managed 2 Research for SME-AGs projects to conclusion, OPTIMALT (GA No. 218460-2) and COMPO-BALL (GA No. 243625), which were both completed at the beginning of 2013. With her team, she is also finalising 2 Research for SMEs projects: SMARTPADDY (GA No. 285861) and THIME (GA No. 315665) and the ERANET-MANUNET "MARE" project. As a result she is very familiar with all the modalities and administrative procedures involved in the management of EC research projects, from the negotiation process, preparing the Consortium Agreement, to the monitoring work, fulfilling reporting requirements, the use of tools for communication with the EC, etc. She understands the needs and preoccupations of industry and SMEs, and has been involved in defining specifications and delivering on research for numerous novel technologies in environmental and water management. Prior to this, she is experienced in the

management of academia collaborative projects; she has managed and participated in several scientific projects both at national and EU level. She has shown herself to be a motivational leader, a fair and intelligent facilitator, with a sound understanding not only of optics and smart systems but also of process industries.

Project Steering Committee: The Project Steering Committee (PSC) is the strategic management and decision making body within the consortium. It consists of the Project Coordinator and a key representative of each of the consortium partners/beneficiaries. The PSC will meet every 6 months and will be chaired by the Project Coordinator. The PSC will closely monitor and control the work performed, following the principles of **"quality, time, cost"**. The Project Milestones (see Table in Part A) will be used as a valuable tool for determining whether or not the project is on schedule. The PSC will also manage the implementation of the innovation cycle, and the feedback linkages to the research. It will evaluate the knowledge as it is generated in line with its market impacts. It will also ensure that the project results are **effectively protected and exploited to the full for future industrial growth and competitiveness**. Key strategic issues, such as IPR protection mechanisms, IPR sharing and routes to exploitation will be discussed extensively during meetings of this Committee. It will also ensure that the project Business and Exploitation Plans are further planned and developed based on the draft plans provided in Section 2.2 and that a professional and engaging communication campaign is implemented. In terms of **decision making mechanisms**, when strategic decisions need to be taken, such as action to solve conflicts, action to mitigate critical risks to project implementation, action to manage changes in the work plan, all matters with regard to IPR and exploitation, etc., it will be the responsibility of the Project Coordinator to seek consensus with the Project Steering Committee, whereby each beneficiary will be **assigned one vote**. Decisions will be taken by simple majority. In the event of a tie situation, the Coordinator will take the final decision. Each representative of the PSC will be responsible for mobilising and coordinating the active and coherent participation of the other personnel from their organisation who are involved in the project team, and they will also ensure that reporting and cost statements from their organisation are submitted on time and correctly. Importantly, the following managerial positions will ensure a decentralised, expertise-centred approach to how core aspects of the project are run:

- ***Innovation Manager*- Mr. Alejandro Rosales (IRIS).** Responsible for leading the partners in effective management of the innovation cycle of the project (this role is discussed under 'Innovation Management'). He will act as a driver for the partners in defining the post-project strategy for exploiting the project results and in driving technology transfer. Alejandro has been the Chair of the Innovation Unit in IRIS for the past 2 years and is experienced in Innovation Management.
- ***IPR Manager*- Prof. Miquel Sanchez-Marre (UPC).** Responsible for knowledge management within the consortium. He will lead the consortium in protecting the project results where appropriate, and in maintaining confidentiality of the results until the IP is protected to safeguard the innovation impacts. Miquel has vast experience in relation to knowledge management, patent applications, open access publications, etc.
- ***Communication & Dissemination Manager*- Dr. Renata Körfer (DECHEMA).** Responsible for leading the consortium in further planning and implementation of the project Communication and Dissemination activities, and ensuring the project reaches out to different audiences in the most effective way. Renata has extensive expertise in outreach and engagement with policy makers and stakeholders, and in the dissemination of research results to EU Process Industry.

Technology Development Team: This is an executive body for managing the multidisciplinary technology developments within the project. This team, which will consist of the WP and Task leaders from WP2-WP5, will be led by the qualified **Technical Manager**, Dr. Devrim Yazan (UT), who will ensure an integrated approach to the research and development work. This Team will meet at least every 3 months under the chair of the Technical Manager (the Project Coordinator will also be present). This meetings (which may be virtual, as appropriate) will review the progress of the developments, and will plan upcoming activities. The Technical Manager together with the team and the Project Coordinator will continually monitor for technical, operational or logistical risks or problems that could jeopardise the timely delivery of the technological outputs of the project so that appropriate mitigation and contingency strategies can be planned as appropriate. Any technology development matters of **strategic importance that require decisions to be taken** will be communicated to the PSC via the Project Coordinator so that a decision can be made.

IS Implementation Team: The IS Implementation Team will be led by the **IS Manager**, Rachel Lombardi (ISL) and will ensure a coordinated an expert-led approach to the pilot implementations in real industrial settings. Rachel has longstanding expertise in IS and is a very well-respected professional within the IS networks in Europe and

internationally. She comes with the weight of expertise of ISL's 10 years in the practical implementation of IS globally. Other qualified IS experts from the consortium will be included in this team (from UT, UNIVLEEDS, USTRATH…) and they will cooperate with the pilot implementation sites (parks and companies) to ensure a standardised approach and that best practice is followed in how the pilots are implemented and validated. The IS Team will also be a valuable source of knowledge for the development teams.

Management Procedures

The SHAREBOX project (like any other project) consists of a series of inter-related activities that will require a set of resources (time, money, people, infrastructure, materials, consumables, communication, quality, risk, etc.) to meet pre-defined objectives. Robust management procedures (Table 11)- adhering to ISO 9001:2008 and UNE-EN-ISO 166002: Management Systems for R&D&I- will serve as a tool for ensuring the project is successfully managed. These procedures are supported by certified processes, methodologies and documentation in IRIS, and will be available to partners through the Project Management Procedures Handbook that all partners will be fully briefed on at Project launch (D8.1).

Table 11-Summary of project management procedures

Type	Description of Procedure
Work plan Review	The workplan will be reviewed on a monthly basis (via monthly Technical Reports submitted by the leaders of each of the active WP and Task). A high level one page planner tool will be used to review the schedule, progress and spend. The critical path of the project will be evaluated during the meetings of the PSC, whereby the Project Milestones (see Project Milestones Table in Part A) will be used as a valuable tool for determining whether or not the project is on schedule. Any potential adjustments to the workplan will be discussed and proposed to the PSC. Risks will be flagged and managed under the Risk Management Procedure.
Issues Management	Issues can be raised through verbal or written means, but eventually an issue must be formally documented using an Issue Submission Form, which is then entered into the Issues Log. The PC will assign issues to an appropriate responsible from the consortium (depending on nature of the issue) for resolution. Impact to the project in terms of budget, schedule, and scope will be estimated. The resolution or course of action will be documented on the Issue Submission Form. If resolving an issue involves changing the scope of the project, the Issue will be closed and the Scope Change Management procedure will be used instead to manage the resolution.
Scope Management	While scope changes are considered unlikely, should a scope change be requested (only to be recommended in extraordinary circumstances and maybe arising through the resolution of an important issue or a high-level risk) a Scope Change Request form will be completed and brought before the PSC for discussion. If deemed vital, the request will be entered into the Scope Change Log and will be assigned to the Technology Manager for investigation. If the scope change involves an addition in scope that will not significantly impact on project cost, effort, and duration and the project will still be completed within the agreed upon cost, effort, and duration, the PSC may approve the Scope Change Request. However, if a scope change involves a shift in the scope of the project, the EC will be informed.
Risk	Discussed below under '*Critical risks and mitigation measures*'.
Conflict of interest Management	The project ethos is based on common understanding and vision, transparency, open dialogue and active participation, which will mitigate against conflicts. Project participants will be encouraged to raise any issues of concern or worry during meetings, or in private with the PC. The PC will also actively monitor for conflicts, to ensure early identification, an analysis of their stage and the implementation of the appropriate resolution process (see Table 12). The PC is responsible for ensuring that all conflicts are dealt with as soon as they arise in order to minimise their negative effect and progression to the next stage. Conflicts that cannot be resolved between the parties concerned will be brought before the PSC for a resolution or decision.

Communication Management	An **Internal Communication Plan** is in place to guide the internal communication process, which includes: **Formal project meetings**: Agendas for all on-site and virtual meetings will be required. Discussions, decisions and actions will be recorded in minutes, which will be circulated to all partners, edited and agreed. The PC will use the 'Kick-off' (KO) meeting to establish positive working practices and to set the tone for all future conduct. Thereafter the PSC will meet every 6 months to review the general progress of the work and to discuss all strategic matters that are common to all project beneficiaries. Extraordinary meetings of the PSC will be convoked in the event of identified risks, conflicts, problems, etc. The Executive Teams will meet every 3 months (teleconferencing facilities can be used as appropriate) to enable the WP and Task leaders to review together the technical issues of the project. The technical progress and results of each of the active tasks will be presented, along with the actions required for the upcoming 3 month work period. Active working groups may meet amongst themselves more regularly if required, and especially during periods of particularly high task activity or prior to major project milestones. Virtual tools will be used in as much as possible for these extra meetings. **Day-to-day communication**: The project intranet, e-mail, Skype, and videoconferencing tools will be used for regular and agile communication. The partners will assume the responsibility to report to the PC on all significant actions/conclusions and immediately should any problems arise.
Quality Management	The Quality Function Deployment (QFD) method will be employed to transform user demands into design quality, to deploy the functions forming quality, and to deploy methods for achieving the design quality into subsystems and component parts, and ultimately to specific elements of the manufacturing process. The QDF will be implemented at the start of the project (through WP1) and will continually review the project management structure and plans, the specific quality objectives and quality indicators for each development stage of the project, the quality assurance and control activities (what is controlled, who controls, when), the details of personnel with respective responsibilities and assignments for each partners, the information exchange and dissemination, control of information exchange and reporting, and the documentation management and control including identification of critical points and risks assessment. In combination, the 'V- Cycle' will be implemented, which fulfils the criteria of the Demming Cycle for continuous improvement and client satisfaction based on the previously programmed validation and verification work.

Table 12- Conflict stages and resolution strategy

Stage No.	Conflict stage	Resolution strategy
1	Potential for conflict exists	Open dialogue and conciliation between the parties concerned
2	Latent conflict- where a situation could easily spill over into conflict	Open dialogue and conciliation between the parties concerned
3	Open conflict	a) Negotiation first among PSC (whereby decision-making procedures for PSC apply); b) Mediation next; c) Arbitration as a last resort
4	Aftermath conflict	a) Negotiation first among PSC (whereby decision-making procedures for PSC apply); b) Mediation next; c) Arbitration as a last resort

Appropriateness to the complexity and scale of the project

Our organisational/management structure and decision-making procedures are appropriate in view of the nature of the action (Research and Innovation action), its size and geographic coverage, its ambition and expected innovation impacts. In short, they will ensure:

1. Effective management of the day-to-day running of the project and its resources, and appropriate interfacing with the EC (Project Coordinator);
2. Strategic management and decision making (Project Steering Committee);
3. Integrated management of the overall technological development and it potential risks (Technology Development Team) and continuous engagement with the IS Manager as well as the industries for an industry/market driven development.
4. Holistic coordination, planning and execution of the IS implementation pilots in the industry (IS Implementation Team). Active link between the development work and the pilot sites to ensure the Process and the other cross-sectorial industries are engaged in a meaningful innovation cycle.
5. Decentralised management through the appointment of additional qualified managers to ensure we deliver high impact: Innovation Manager (towards market-oriented outputs and planning to realise technology transfer

opportunities); IPR Manager (results protection for gaining competitive advantage); Communication & Dissemination manager (stakeholder engagement and wide communication/dissemination of project's results).

Dedicated committees, teams and managers with key roles and responsibilities will allow for the effective management of this highly diverse and multidisciplinary project. It will ensure that the necessary skills-sets are mobilised at the different levels of leadership, as opposed to a centralised, one-dimensional management approach, which would be highly unlikely to deliver the necessary versatility. As such we have created an integral approach to the management of the project, giving all participants have a stake in management, to ensure that the project achieves its development objectives, activates a dynamic innovation cycle, manages the knowledge and IP generated, as well as the other innovation-related activities required to deliver market exploitable results and to map out the route for post-project use of the project results for impact.

Innovation management

As a key objective of publicly-funded research is that it should lead to the exploitation of results, which goes one step further than the mere production and dissemination of new scientific knowledge[68], an active **innovation cycle** is a vital component for ensuring success in market-oriented research projects.

Our two Executive Teams (Technology Development Team and IS Implementation Team), and each its own experienced manager, will regularly engage to ensure that the worlds of technology development and IS pilot implementations in the industry are systematically connected in an Innovation Cycle, whereby the technology developers benefit from feedback from the industry and guidance from the IS experts, and the pilot industries and parks can influence on the development work. This entire process needs to be well managed, through a robust innovation strategy, whereby the incorporation of feedback and the exploration of market opportunities should start from the very beginning of the innovation cycle and not long after the beginning of development activities[68]. In the case of our SHAREBOX Innovation Cycle, it will commence at the start of the project with the QFD process that will translate the "voice of the client" (i.e. KEROS, KERAFRIT, ISSA, GUZMAN, CCB, ESI, ITC, NEPIC -and DECHEMA as the wider voice of the European Process Industry), into the "voice of the engineer/developer" to ensure that the system specifications for the platform (the "House of Quality" from the QFD process) are robust and market oriented. From there we will mobilise a **continuous iterative process** whereby the development work, which is ongoing throughout the project, will be closely linked to the industry pilot implementation sites. This will ensure that the Process Industries are involved in a true co-creation process. At a holistic level, a series of feedback loops will incorporate feedback from designers, engineers, IS experts and end-users, as well as, incorporate feedback and explore market opportunities in parallel with other key steps of the innovation cycle (e.g. research, prototyping, industrial demonstration etc.). Our "House of Quality" from the QFD process will provide a solid benchmark against which the outputs from the development work can be evaluated. As SHAREBOX is a highly multidisciplinary development, the outputs of which will be trialled via practical industry case studies, these trials will commence (in M13) well before the end of innovation cycle to enable user feedback to play a prominent role in the innovation cycle and allows for further advancement of the project outputs (through optimisation work that will be performed as necessary during the pilot implementations).

As the successful management of innovation requires an understanding of both the market and the technical aspects, a competent **Innovation Manager** who is capable of fomenting an **integrated team approach** will be appointed to mobilise the involvement of this **multidisciplinary team**. Mr. Alejandro Rosales, a business focussed professional who has solid experience chairing the Innovation Unit in IRIS, is competent in leading teams to ensure "no thinking in terms of silos", and is savvy enough to engage with external actors and the diverse stakeholder groups as required to ensure "openness for ideas from outside"[68]. Moreover, the Project Steering Committee (PSC) brings together the necessary multidisciplinarity combining the freest thinkers from the development team, the market and business-focussed skills sets from the technological/supply SMEs, and the senior actors from the end-user industries who will bring all the inputs of early adopters of first clients. Bi-annual meetings will allow the PSC to **effectively manage the innovation cycle**, the core elements of which have been embedded as tasks into the Innovation Impacts WP (WP7)

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[68] Innovation. How to convert research into commercial success story? Part 3: Innovation Management for Practitioners. European Commission. Directorate-General for Research and Innovation, 2013. Nanosciences, Nanotechnologies, Materials and New Production Technologies (NMP)

to facilitate their formal management and execution. Combining all these skill sets with knowledge management, IPR and patent expertise, ensures that the SHAREBOX project has mobilised a solid and experienced approach to innovation management.

Critical risks and mitigation measures

Given that an inherent characteristic of research and innovation projects is technological and scale-up uncertainty, which may result in project failure, and time and resources spent without any tangible return, these risks will be carefully managed by Project Coordinator, with the support of the other Managers in the consortium and by instilling a risk management and 'red flag raising' culture across the entire team. Risks are not only technology-related, and also include: market related risks, technology related risks, environment related risks, and organisation related risks[69]. The process of risk identification and management will be based upon the principles and guidelines of the FERMA (ISO 31000 related) Standard[70]. Some initial risks have been identified already and are listed in the Critical Risks Table in Part A with the assistance of the Risk Exposure Level assessment matrix (Table 13).

Table 13- Risk Exposure Level assessment matrix

		LIKELIHOOD		
		Low (L)	Medium (M)	High (H)
IMPACT	Negligible (N)	Low	Low	Medium
	Marginal (MA)	Low	Medium	Medium
	Critical (C)	Medium	Medium	High
	Uncontrollable (U)	Medium	High	high

During the project the management of these (and other risk as identified) will be based on a list of co-ordinated activities embracing the 7Rs and 4Ts of (hazard) risk management: Recognition or identification of risks; Ranking through evaluation of risks; Responding to significant risks through consideration of: Tolerance of risk; Treatment of risk; Transfer of risk; Termination of risk; Resourcing controls; Reaction planning; Reporting and monitoring risk performance; Reviewing of risk management framework. Classification and importance of each risk is determined by two factors: likelihood and impact, which together determine the exposure level as presented in the matrix below. A continual process of monitoring during the management of this project, will ensure that critical risks are quickly identified and managed so that they do not jeopardise the future of the project.

3.3. Consortium as a whole

Description of the consortium

The SHAREBOX consortium is made up of 17 partners from 6 European countries (Spain, United Kingdom, Netherlands, Germany, Switzerland, Turkey) and collectively represents a highly focused and synergistic balance of research and industry fields, expertise, skills, resources, and European reach, to constitute a complete value chain of actors that together are **capable of achieving the project objectives.** The call topic requires that "projects should enable the implementation of a broad variety of technologies, encompassing a wide range of disciplines, such as fundamental science, and plant engineering and management. This requires a multidisciplinary team. Without a pan European approach, the critical mass of expertise required for the multidisciplinary research and development team could not be gathered. Single technique specialists cannot, by definition, resolve multidisciplinary and multifaceted problems.

The consortium leverages a powerful **industry-academic** partnership.

The project is **industry driven**- under the leadership of the experienced SME, IRIS (SPIRE members and the coordinator of a successful SPIRE-01 project called PROPAT)- to ensure the research and innovation work and the resulting technological outputs are relevant for market needs. The **strong industrial and commercial involvement** through the individual companies on the one hand (IRIS, ISL, KEROS, KERAFRIT, ISSA, GUZMAN- numerous of which are SMEs), and via the Industrial Parks and Clusters (CCB, NEPIC and ECI) and the sectorial research

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69 Kasap et. al. (2007). Risk Management in R&D Projects. Management of Engineering and Technology, Portland, US. 5-9 Aug 2007. Print ISBN: 978-1-8908-4315-1

70 Federation of European Risk Management Associations (www.ferma.eu)

association (ITC-AICE), who provide a pathway to thousands of companies representing SPIRE and cross-sectors. This industry participation spans the entire value chain necessary to succeed in delivering impact and exploitation pathways for the projects results: from IS delivery specialists (ISL), to technology and ITCs (IRIS) to and the process industries and cross sectors (KEROS, KERAFRIT, ITC-AICE, CCB, ISSA, GUZMAN, NEPIC, ECI). The presence of a highly qualified and experienced **market actor** (DECHEMA, who are also SPIRE members) with over 5500 members- scientists, engineers, companies, organisations and institutes- provides an impressive pathway to the process industry in Europe. Moreover, there is strong industrialisation capacity post-project to further develop the project results and with a clear route to market (IRIS, ISL), as well as to adopt and further roll out the platform for major impact in terms of the circular economy, competitiveness and growth.

The industrial consortium has secured a very solid participation of **scientific/academic actors** (UT, UNIVLEEDS, UPC, ZHAW, USTRATH), who will contribute their knowledge and methodologies towards developing the tools and technologies for the holistic SHAREBOX platform, and in assisting in IS implementation in the industry and providing scientific validation to demonstrate the success of the SHAREBOX platform in facilitating and optimising resource sharing and IS in the participating Process industries and Parks. These organisations have a strong applied approach to their research and the majority of them are members of the SPIRE Association (UNIVLEEDS, ZWAH, AICE-ITC). They are committed to working closely with the European Process Industry for realising the goals of the SPIRE research roadmap. In order to evidence how each partner is relevant for the project and to **show the complementarity between the participants**, the partners are clustered according to their positioning on the value chain. The broad role of each cluster of partners in contributing to the project objectives, and, indeed, in contributing to the expected impacts of the project are illustrated in Figure 6, testament to how they are suited **and are committed to the tasks assigned to them.**

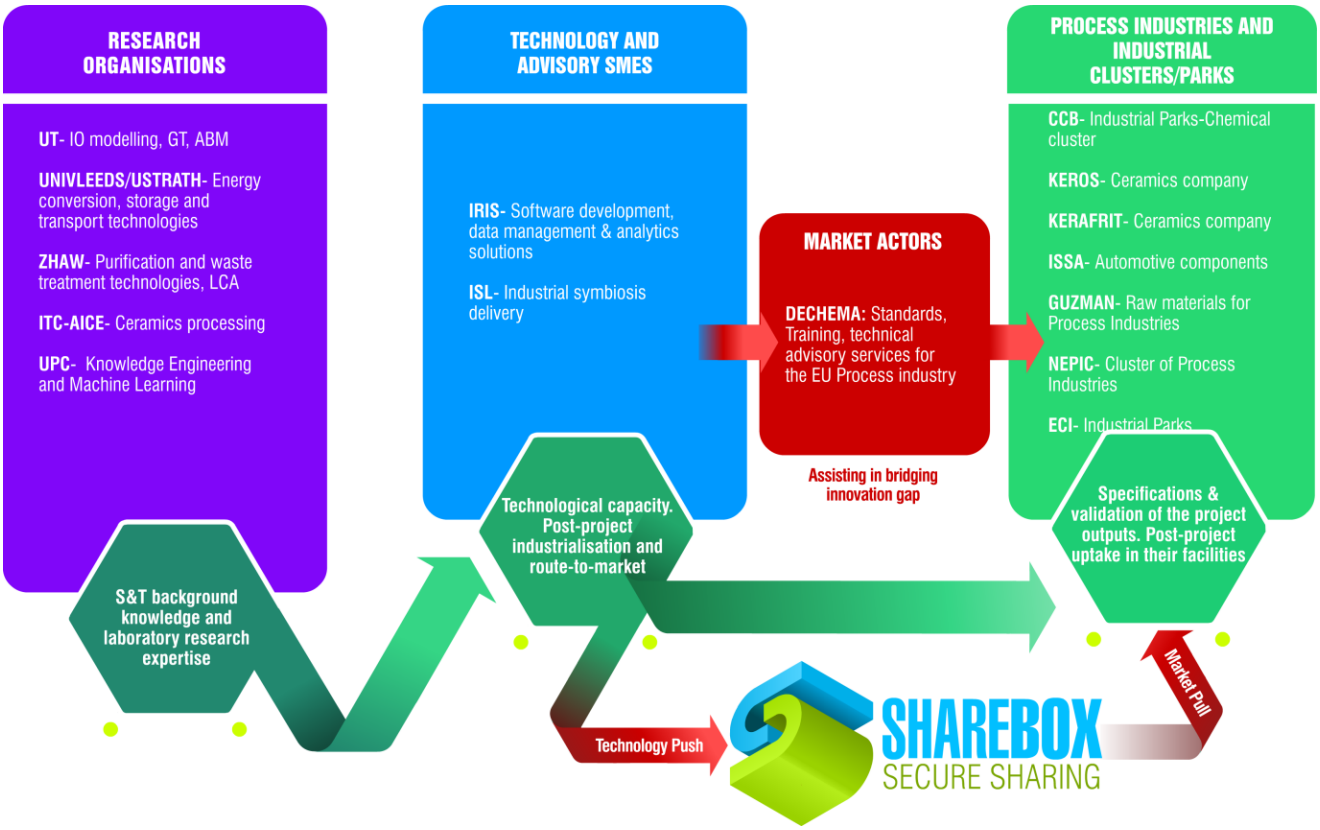

Figure 6- Overview of the consortium and how each contributes to SHAREBOX

In terms of the relevance and capacity of the consortium to implement the work plan, on the one hand, there is sufficient research and technological background and expertise in the fields of: **Industrial symbiosis** (ISL), **energy scheduling and conversion** (UNIVLEEDS and USTRATH), **In-put-Output modelling, Game Theory, Agent Based Modelling** (UT), **purification and waste treatment technologies and Life Cycle Assessment** (ZHAW), **knowledge engineering and machine learning** (UPC), **ICT and software platform development** (IRIS); **Ceramics processing**

(KEROS, KERAFRIT) and raw materials (ISSA); **Chemical processing** (CCB); **Cross sector industry** (GUZMAN); **Park Operators and clusters** (CCB, NEPIC, EIS, AICE-ITC) and outreach to the wider **EU Process industry** (DECHEMA).

There is no conflictive overlap among the consortium and each participating organisation has a clearly defined role in the project, which will mitigate the risk of conflicts arising during the project. Indeed, our consortium covering the whole value chain, brings a series of key strengths due to some key features of its participants: **Chemie Cluster Bayern**- linking over 260 Bavarian chemical companies**; ISL-** World Leaders in Industrial Symbiosis; Initiators of the award winning National Industrial Symbiosis Programme; **DECHEMA**- Outstanding positioning within the European Process industry; **IRIS-** SME leader of a SPIRE-1 project and active in developing monitoring, control and decision support platforms solutions for the industry, and who have just recently beta launched a SaaS cloud based platform for SME operations management that is generating very positive independent reviews; **UNIVLEEDS-** Participants in F3 Factory project, which was the Project that led to the creation of the SPIRE PPP; **KEROS, KERAFRIT, ISSA, GUZMAN**- cross sector companies who currently have no symbiosis amongst themselves, but who have come together to establish symbiosis and to widen the network during the project under the management of **ITC-AICE**; A core group of Suschem members who lobbied for the SPIRE programme creation; to mention but a few statements that reinforce the outstanding quality and relevance of our consortium.

In short, SHAREBOX unites sufficient reputable actors from the scientific fields, as well as from the industry value chain, in order to be able to realistically implement the ambitious work plan, to protect the knowledge and plan for its use and communication to a wide audience of different stakeholders, so that the true impacts of the results can be felt:
The SHAREBOX Platform will bring a world-first secure and flexible resource and energy management platform to the industry that will be based on the latest developments in secure ICT tools and artificial intelligence offering an holistic approach to energy and resource management systems in process industries and overcoming the difficult cross-sectorial resource exchanges for improved efficiencies.

The industrial/commercial involvement in the project to ensure exploitation of the results

There is **strong industry leadership and involvement** in this consortium. Each industry participant has clear expectations and commitments towards commercially using the results of this project in their businesses, which is manifested through the initial exploitation and business plans presented in Section 2.2.

While the involvement of industry in R&D consortia increases the success rate of market-oriented exploitation, **vertical integration (i.e. including the whole value chain)** in R&D projects is a success factor for market-oriented exploitation[68]. To this end, SHAREBOX includes a technological SME with research performing capacity who can contribute to **technology push** (IRIS), who are not only capacitated to contribute to the R&D effort, especially in the context of **scale-up, integration and demonstration of the results during the project,** but who can absorb the knowledge generated to protect it, in order to further develop and **industrialise the outputs post-project towards commercialisation.** The inclusion of an SME dedicated to **industrial symbiosis delivery** (ISL), will ensure an exploitation pathway for the platform to serve the industry demand that will be stimulated as a result of **the market pull** generated by the participating process and cross sectorial industries and parks (KEROS, KERAFRIT, ITC-AICE, CCB, ISSA, GUZMAN, NEPIC, ECI), who during the project will contribute end-user industry knowledge and experience to assist in mainstreaming the results of the project and in validating them in the field. Importantly, the presence of large enterprises, such as KEROS, KERAFRIT, ISSA, GUZMAN, can make a difference in successful market oriented exploitation as long as the research outcomes are in line with their 'expectations'[68]. SHAREBOX has secured the interest and involvement of these industries at the early stages of the preparation of this bid, and they all understand its scope and have been very committed in collaborating with the rest of the partners to contribute to the definition of the Platform specifications, as well as participation as demonstration sites whereby SHAREBOX will be used in their plants and will help them to establish symbiosis amongst themselves. Their engagement during the proposal preparation stage has helped to shape the scope and ambition of this bid.

Finally, the involvement of a non-profit making society for chemical engineering and biotechnology (DECHEMA) who is dedicated to **transferring research results to industry**, as well as **providing training and educational programmes**, is an invaluable market actor to complete the consortium by contributing to close the innovation gap

that frequently prevents industry adoption the results and outputs of R&D, by facilitating the advice and training needed for the successful implementation of new technologies such as SHAREBOX in the European process industry.

Other countries: Switzerland is now a non-associated third country participant. Our Swiss partner, ZHAW, the Zurich University of Applied Sciences, has been included in this consortium to their excellent expertise in purification and waste treatment technology and LCA. Funding will be provided from Switzerland. Erni Peter, director at the Eurosearch Head Office in Bern, Switzerland said that "In the event that Switzerland will be treated as a third country in calls for Horizon 2020 project proposals, funds for contributions to the EU framework programme will instead be paid directly to researchers in Switzerland, by the State Secretariat for Education, Research and Innovation of Switzerland.".

3.4. Resources to be committed

Table 3.4b: 'Other direct cost' items (travel, equipment, other goods and services, large research infrastructure)

1 IRIS	Cost (€)	Justification
Travel	27600	Travel costs to project meetings (13 meeting in 3 years, 2 people travelling- ~1.2K€/meeting x13=). Travel of engineers to industry sites for installation and validation work (2 engineers x 3 trips x~2K/trip- €12K)
Equipment		
Other goods and services	82000	Cost of IPR protection and publication fees for open access journals; exhibition space and subscription to 2 tradeshows (€12 K); Leaflets, posters, other dissemination materials (€10K); Supercomputing cloud hosting and servers (60 K)
Total	109600	

2 ISL	Cost (€)	Justification
Travel	66180	Travel costs to project meetings: 9 General Steering Committee meetings x ~1 K€/meeting = 7.2K; 5 technical meetings- €4 K; travel to demo sites- €4K; travel to dissemination/training events- €3 K, plus see additional travel costs detailed below associated with marketing and trial of system in Turkey, Germany and UK
Equipment		
Other goods and services	12000	Cost of IPR protection, organistaion/subscription dissemination events, etc. Computing.
Total	78180	

3 UT	Cost (€)	Justification
Travel	18200	Travel costs to project meetings: 9 General Steering Committee meetings x ~1 K€/meeting = 7.2K; 5 technical meetings- €4 K; travel to demo sites- €4K; travel to dissemination/training events- €3 K)
Equipment	8000	ABM software
Other goods and services	8000	Cost of IPR protection and publication fees for open access journals + subscription to conferences and events (8K);
Total	34200	

4 UNIVLEEDS	Cost (€)	Justification
Travel	17547	Travel costs to General Steering Committee meetings and technical meetings, demo sites, dissemination/training events
Equipment	3000	Computer equipment
Other goods and services	3000	Consumables
Total	23547	

5 ZHAW	Cost (€)	Justification
Travel	18200	Travel costs to project meetings: 9 General Steering Committee meetings x ~1 K€/meeting = 7.2K; 5 technical meetings- €4 K; travel to demo sites- €4K; travel to dissemination/training events- €3 K)
Equipment		
Other goods and services	8000	Open access publications. Dissemination materials. Organisation of training session and workshop. Subscription to events
Total	26200	

6 ITC	Cost (€)	Justification
Travel	18200	Travel costs to project meetings: 9 General Steering Committee meetings x ~1 K€/meeting = 7.2K; 5 technical meetings- €4 K; travel to demo sites- €4K; travel to dissemination/training events- €3 K)
Equipment		
Other goods and services	8000	Open access publications. Dissemination materials. Organisation of training session and workshop. Subscription to events
Total	26200	

7 UPC	Cost (€)	Justification
Travel	18200	Travel costs to project meetings: 9 General Steering Committee meetings x ~1 K€/meeting = 7.2K; 5 technical meetings- €4 K; travel to demo sites- €4K; travel to dissemination/training events- €3 K)
Equipment	2000	Computing
Other goods and services	8000	Cost of IPR protection, open access publishing, organistaion/subscription dissemination events, etc.
Total	**28200**	

8 DECHEMA	Cost (€)	Justification
Travel	20000	Travel costs to project meetings (13 meeting in 3 years, 1000€ per meeting), conferences, workshops, training sessions, participation in the relevant standardization technical committees of an existing European standards
Equipment		
Other goods and services	15000	Cost of IPR protection and publication fees for open access journal; consumables and supplies; licence to acces the standard sources; organisation of training session and project workshop
Total	**35000**	

9 CCB	Cost (€)	Justification
Travel	18200	Travel costs to project meetings: 9 General Steering Committee meetings x ~1 K€/meeting = 7.2K; 5 technical meetings- €4 K; travel to demo sites- €4K; travel to dissemination/training events- €3 K)
Equipment		
Other goods and services	8000	Dissemination materials. Organisation of training session and workshop. Subscription to evenrs
Total	**26200**	

10 KEROS	Cost (€)	Justification
Travel	18200	Travel costs to project meetings: 9 General Steering Committee meetings x ~1 K€/meeting = 7.2K; 5 technical meetings- €4
Equipment		
Other goods and services	12000	Dissemination materials. Organisation of training session and workshop. Subscription to events. Consumables (4K)
Total	**30200**	

11 KERAFRIT	Cost (€)	Justification
Travel	17000	Travel costs (meetings, conferences, workshops, mobility between demo sites, traning events)
Equipment		
Other goods and services	19000	Dissemination materials. Subscription to events. (8k) Flow meter, gas analyzer for obtaining data (8K) Consumables (3K)
Total	**36000**	

12 ISSA	Cost (€)	Justification
Travel	16000	Travel to project meetings. Travel between demo sites. Travel to dissemination events/trainings
Equipment		
Other goods and services	15000	Dissemination materials. Subscription to events. IPR and legal costs. Consumables (6K)
Total	**31000**	

13 GUZMAN	Cost (€)	Justification
Travel	6800	Travel costs to project meetings
Equipment		
Other goods and services	8000	Dissemination materials. Organisation of training session and workshop. Subscription to events
Total	**14800**	

14 BPC	Cost (€)	Justification
Travel		
Equipment		
Other goods and services		
Total	0	

15 ECI	Cost (€)	Justification
Travel	18200	Travel costs to project meetings: 9 General Steering Committee meetings x ~1 K€/meeting = 7.2K; 5 technical meetings- €4 K; travel to demo sites- €4K; travel to dissemination/training events- €3 K)
Equipment		
Other goods and services	8000	Dissemination materials. Organisation of training session and workshop. Subscription to events
Total	**26200**	

16 NEPIC	Cost (€)	Justification
Travel	6000	Travel costs to project meetings and events
Equipment		
Other goods and services		
Total	6000	

17 USTRATH	Cost (€)	Justification
Travel	17829	Travel costs to project meetings: 9 General Steering Committee meetings x ~1 K€/meeting = 7.2K; 5 technical meetings- €4 K; travel to demo sites- €4K; travel to dissemination/training events- €3 K)
Equipment	3100	Computer equipment
Other goods and services	23500	Cost of IPR protection, organistaion/subscription dissemination events, etc. (8k); Consumables (15.5K).
Total	44429	

4. Members of the consortium

4.1. Participants

Participant no.	1	Name (short name)/ Country	Innovacio i Recerca Industrial i Sostenible (IRIS) / Spain

Description of the legal entity

IRIS was established in Barcelona in 2007 as an advanced engineering and R&D company, specialising in the field of process engineering. IRIS has expertise in the following fields of engineering: Production Processes (with a strong specialisation in novel technologies); Smart Systems; and Optical Systems, combined with a knowledge and research capacity in environmental science, food technology and materials science. IRIS applies its knowledge and expertise to develop solutions for industry predominantly in the fields of advanced process monitoring and control, as well as production data management, for the food, agriculture, chemical and pharmaceutical industries, among others. The company has a policy of continuous investment for growth, and currently employs 48 staff based in 2 sites (Barcelona and Dublin), which are equipped with engineering workshops (3), analytical and wet chemistry laboratory, optics labs (2), and an electronics/telecommunications lab. IRIS is **ISO 9001:2008 certified** and also is certified under **UNE-EN-ISO 166002: Management Systems R&D&i**.

Main tasks that will be performed

IRIS will act as the Project Coordinators. They have extensive expertise in the management of industry-driven projects (B2B projects and FP7 R&D projects). Their project management procedures are ISO and UNE certified, and they have an internal PMO that has the necessary resources and skills to ensure smooth administrative and financial management of the project. In addition, IRIS will also lead WP1, whereby they will lead the consortium in the implementation of the QDF tool (and Kano methodology) for engaging with the Process Industries to map out their requirements that will feed into system specifications for the Platform, and for further definition and planning of the Demo trials. They will also lead WP4 whereby the will build the Platform: designing the architecture, coding the tools, establishing the API with the supercomputing cloud provider), building the databases, the software, etc. They will also build a virtual simulator for testing and troubleshooting the platform. In addition, they will deploy iterative versions of the platform among the demo sites and they will set up the necessary communications in the industries. IRIS will also contribute their skills in knowledge/IPR management, as well as in commercial branding, reaching industry, the development of SaaS products and business models. They will also contribute to knowledge transfer and training activities and gearing up for the post-project phase. They will lead the development of the business plans, and they will be the chairs of the Innovation Committee.

How the legal entity profile matches the tasks in the proposal

The Smart Systems group has expertise in the development of software, web and mobile applications, as well as cloud-based systems. The group unites expertise in sensor readings, process logic and control, and modelling, coupled with knowledge predictive and adaptive learning capable of describing and analysing the given situation and taking smart actions based on the available data. The Group is currently working on a number of projects related to Wireless Sensor Networks where the above mentioned is applied. Specialised staff (software and telecommunication engineers with programming experience) that work to the Agile methodology and philosophy (scrum and kanban) for developing complex applications that include hardware interfaces, SQL and No-SQL databases (MySQL, PostgresDB, Couchbase

among others) and advanced and user-friendly graphic user interfaces not only for PCs (using Windows Presentation Foundation) but also for smartphones (Android, iOS) or web-based access (HTML 5). Software engineers skilled in a wide variety of languages and operating systems as well as low-level programming (microcontrollers). Strong background in data processing and data mining, chemometrics, and mathematical and predictive modelling in the context of advanced monitoring in industrial and agricultural applications as part of collaborative projects as well as B2B projects with industry. This team of developers includes software engineers, modellers, programmers, back-end and front-end web and mobile platform developers, as well as graphical designers. They have worked on the development of predictive models, decision support systems, data-based driven web portals, Apps, and cloud-based applications for numerous B2B and European projects. The team also has expertise in Software As A Service (SaaS) new product development, deployment and support. Strong culture in Innovation Management- since 2012 IRIS has operated an Innovation Committee and Innovation Unit for monetizing the company's knowledge and development capacity towards marketable products & services.

In terms of project management expertise, IRIS is a highly experienced coordinator of over 35 projects during FP7, as well as leading 2 MANUNET projects, not to mention the management of B2B contracts with industry. Internally, IRIS uses agile KANBAN methodologies to manage the work of the engineering team. IRIS are also coordinators of a project that was funded under H2020 SPIRE-1-Integrated Process Control (ProPAT- Grant Agreement no. 637232.

CV of the persons who will be primarily responsible for carrying out the work

DR. MIRTA RODRIGUEZ PINILLA. (Female) Doctor of Science and Technology from the Helsinki University of Technology in 2004 (Ph-D thesis- "Signatures of Superfluidity in Atomic Fermi Gases"). Degree in Physics from the Universidad Complutense de Madrid. Masters in Energy and Sustainable Development from the Universitat Politècnica de Catalunya. Prior to joining IRIS she held different research positions CSIC (National Research Council of Spain), Institute of Photonic Sciences in Barcelona, University of Oxford and the Nanoscience Centre in Jyväskylä, Finland. Expertise in the management and development of scientific projects in theoretical quantum physics. Author of over 23 scientific papers in peer reviewed Journals (Science, Nature Physics, Nature Photonics, and Physical Review Letters). Co-supervised a doctoral thesis. Member of the ANEP, National Evaluation Agency and has participated in several evaluation panels for the Ministry of Science and Innovation of Spain.

MSc. ALEJANDRO ROSALES (male) has a M. Sc. in Physics from the Universidad de La Habana (1986). His background comprises theoretical and technical skills in several scientific and technological areas: optics, lasers, high vacuum, colorimetry, UV-VIS-NIR-MIR spectroscopy, chemometrics and design of measurement instruments. He has extensive practical experience in programming, electronics and mechanical CAD design as well as R&D methodology and management. His career has included positions as researcher applications engineer, design engineer, business developer and R&D manager. In 2000s, he worked in the FTIR spectroscopy field applied to quality control in the agro-food (wine) industry, whereby he developed two new measurement instruments for wine analysis. His career as industrial physicist has been focused in applying his wide expertise to provide solutions for industrial problems. Since 2011, he has been the Technical Manager in IRIS, leading a staff of 20 engineers and researchers involved in R&D projects aimed at developing novel high-end solutions for industry. Currently, he is the Science & Technology Manager in IRIS.

SERGI CASBAS (male). Scrum Master. Computer Science Engineer by Open University of Catalonia and Certified Scrum Master by Scrum Alliance. Scrum Master with numerous scrum teams in diverse areas (Game Developing, IT department, R&D) applying agile techniques adapted for each team and project. Large experience in programming management tools, accountability applications, statistical and data collection systems on many programming languages like C, C++, VisualBasic, PHP, ASP, Java and .NET. Wide knowledge on database design and Database Management Systems (MySQL, Postgres and MSSQLServer). The following software products have been developed by Sergi as Scrum Master: Phonefocus – phone bill platform; ZITRO (recreational machines factory) machines software.

DR. ANDRAS GEFFERTH (male). Modeller. Ph.D. in Computer Science (2005) and M.Sc. in Computer Science (1998) from Budapest University of Technology and Economics (BUTE). Besides computer science his background includes mathematical modelling and stochastic processes. Prior to IRIS, he has industry experience at Morgan Stanley as a quantitative researcher working on pricing and risk models for various financial derivatives. Research fellow at BUTE participating in several R&D projects in the fields of video transmission, data network traffic identification, data network traffic modelling and analysis. Teaching and supervision of M.Sc. students, giving lectures and supervising laboratory work. Expertise in various programming and scripting languages-C++, Matlab, Tcsh, Awk and Python. Co-author of several peer reviewed publications.

NORMAN MORELL ASENSIO. (Male). Computer Engineer from Polytechnic University of Catalonia (2012). For his

thesis he developed a model-view-controller PHP (OOP) framework for a website and Intranet for an NGO. He began his career in a home automation company (Windows network administrator and Access/Visual Basic developer) and in a computer academy teaching about web knowledge. Afterwards he joined the Rationalization and Methods Department of a financial entity where he analysed and upgraded all the business processes of their office network in order to automate then. In the same entity he was promoted to the e-business department where he spent 10 year as a Java developer-analyst and also managed development projects. Prior to joining IRIS he was working in a software consulting company developing a project to improve data-mining's parallel task executions throughout a Marklogic (NoSQL platform for Big Data applications) and Storm**.**

FERNANDO ROMAN GARCIA. (Male) Wicket Specialist. Telecommunications Degree from Polytechnic University of Catalonia (expected). Experienced in programming microcontrollers using C such as Arduino and in web development, above all in backend using Servlets in Apache Tomcat or frameworks such as Apache Wicket. Experience in XNA Framework for Game Developing using DirectX. Familiar with object oriented programming languages such as Java and C#.

MIROSLAV SPAJIC (male). Senior Programmer. B.Sc. in Programming, Uni. Belgrade. M.Sc. Programming, Megatrend University, School for Computer Sciences. Senior Programmer in IRIS since March 2009. Main fields of interest are Programming, DB administration, SAP, Networking and multimedia.

IVAN PAQUICO RODRIGUEZ (male). App Developer. Telecommunications Degree in UPC (Barcelona) and a Joint Degree in Multimedia Development in LUAS (Finland). An official iOS and Android Developer since 2010, creating apps for iPhone: e.g. 'BeThere' developed by iOS SDK from Apple and using Couchbase technology for the database. Experience in different programming languages such as C, C++, Objective C or Java, 8 bits embedded and real time systems. Expertise in multimedia methodologies ranging from Virtual Reality programming techniques to 3D/2D modelling software such as OpenGL, Quest3D, Autodesk Maya, Autodesk 3DS Max or Adobe Photoshop.

PAU PUIGDOLLERS DE BALLE. (male). BSc in Environmental Science (Research Project: "Estimating patterns of leaf herbivory in NE Queensland Rainforests,Australia. Might global warming increase leaf herbivory in tropical rainforests?", 2009) from the Universitat Autònoma de Barcelona. MSc in Terrestrial Ecology (Research Thesis: "Effects of biochar from fast pyrolysis on soil water availability and basal respiration", 2010) from the same university and MSc in Renewable Energy Engineering and Management from the Universitat de Barcelona. Prior to joining IRIS he held positions as technical sales person on Servicios Normativos consultancy group, working with environmental and quality projects, and quality field inspector on waste treatment plants for Ingeniería IDOM International, performing quality audits and reports. He has also held different research positions at the CSIRO (Commonwealth Scientific and Industrial Research Organization), in Australia and in the CREAF (Centre de recerca ecològica i aplicacions forestals) in Barcelona. He has also experience as environmental consultant at Eco Intelligent Growth, working with renewable energies, life cycle assessment, C2C certifications and green building.

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. Data processing analysis solutions for the industry (http://iris.cat/wp-content/uploads/2014/02/IRIS-DataProcessingAnalysis-2.pdf), whereby IRIS has delivered numerous B2B services and product developments to industry.
2. Software-As-A-Service product: developed and commercialized by IRIS www.sapenta.com.
3. Development of a commercial system for real-time monitoring of moisture, CO2, temperature and filling level of grains in industrial silos. It comprises a network of multi-sensor probes and a data management subsystem. B2B project (private contract research).
4. System for real-time on-line multipoint monitoring the olive oil extraction (centrifuge) in order to optimize the yield and reducing wastes. B2B project (private contract research).
5. A platform for in-field multi-point monitoring of salinity in rice paddies in order to optimize the water consumption (AgriBox, in IRIS products line)

A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. In H2020 under SPIRE-1, IRIS is currently leading the development of a Global System Control Platform for the integration of PAT analysers and for the data management for PAT implementation in the process industries: PROPAT (Robust and affordable process control technologies for improving standards and optimising industrial operations). GA-637232.

IRIS has performed roles of development of data mining tools in the following FP7 project:

2. STARTEC. Project full name: Decision Support Tools to ensure safe, tasty and nutritious Advanced Ready-To-Eat

foods for healthy and vulnerable Consumers. Grant Agreement no: 289262.

IRIS has worked on the data processing and chemometrics software in the following FP7 projects, among others:

3. Project Acronym: OPTICLEAN Project full name: Effective and reliable optical system for cleaning validation in pharmaceuticals manufacture Grant Agreement no: 285836
4. PUREFORMULA- Project full name: Development of a novel control technology for ensuring high quality Powdered Infant Formula; Grant Agreement no: 605134; 2 years
5. CRYSTALVIS- Project full name: A process analytical technology for characterizing the physical properties of crystals; Grant Agreement no: 605814; 2 years

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Electronics workshops, sprint room for scrum sprints, server room and IT infrastructure.

Participant No.	2	**Name (short name)/ Country**	International Synergies Ltd. (ISL)/United Kingdom

Description of the legal entity

International Synergies Limited (ISL) is the world's leading expert on industrial symbiosis, recognised by the UN, UNEP, EU, OECD, China, UK, Canada, Mexico through its award-winning flagship programme and international brand NISP®. The European Commission cites IS as an innovative means to achieve resource efficiency; calling for Europe-wide implementation of IS as part of its 'Roadmap to a Resource Efficient Europe' and the high-level 'European Resource Efficiency Platform' (EREP).

ISL provides expertise and facilitation to private companies, and capacity building and training to enable partners to implement IS projects at local, regional or national level. It has worked in 20 countries on five continents since 2005. ISL also offers a state-of-the-art resource management database to support resource efficiency at all levels (site, company, supply chain, local/national network). ISL is a partner of the European Climate Knowledge and Innovation Community (Climate KIC), and has successfully delivered projects through LIFE+, ERDF, FP7, SAVE II, Intelligent Energy Europe and SWITCH ASIA. ISL has ISO 9000 and ISO 14001 certifications.

Main tasks that will be performed

Exploiting their vast experience in IS, they will lead the IS methodologies in WP2 and they will also lead the setting up of the sites in WP5. They will also identify the non-technical barriers in WP1. In WP4 they will engage with IRIS for the upgrading of their SYNERGie® software that facilitates the capture of data, information and knowledge around synergies, and has an integrated document management system for linking relevant documents to synergies and searching for documents previously added to the system. They will also have a key role to play in WP6, whereby they will mobilise their experience for measuring the impact of the symbiosis to validate the SHAREBOX platform. They will also play an active role in dissemination, training, communication and planning for the post-project phase. Overall contribution from ISL builds on 10 years' experience with the successful marketing and delivery of industrial symbiosis tools and techniques across Europe and around the world. This experience includes measurement of impact with strong business engagement, grounded in the needs of practical industrial symbiosis implementation.

How your organisation is suited to these tasks

ISL has developed and implemented the world's leading industrial symbiosis model (NISP), based on demand-led engagement with business and other organisations. This model uses information provided by the companies as well as available public data. In addition, ISL has pioneered practitioner informed ICT tools such as SYNERGie®, which is currently being used in 9 countries.

ISL has over 1000 industrial symbiosis case studies/output reports from 20 countries. These provide comprehensive, externally verified data on successful synergies including:-

- economic successes (cost savings, revenue generated, investment attracted)
- environmental aspects (avoided CO2 emissions, landfill/waste, water and resource use)
- social benefits (jobs created and safeguarded)

The benefits calculation uses a before-and-after methodology. In IMATAIS, this methodology will be strengthened with the Life cycle analysis LCA performed by ENEA and COWI.

ISL has extensive business networks across the world, particularly Europe, together with institutional (WB/UNIDO/UNEP/ISIE/IFC etc.) and press relationships that will be of great assistance in disseminating the outcomes of IMATAIS leading to extended IS uptake and market exploitation.

CV of the persons who will be primarily responsible for carrying out the work

PETER LAYBOURN (Male) is the founder and chief executive of ISL and the creator of the National Industrial Symbiosis

Programme – NISP. NISP was the first national industrial symbiosis programme in the world, has won several major awards and continues to be world leading at delivering economic, environmental and social benefits. Using the industrial symbiosis approach, the NISP model is now being exported to many countries across five continents.
Peter graduated in Economics and Ecology from the University of Southampton. Peter spent 18 years in the aerospace manufacturing industry as Contracts and Commercial Manager at GEC Marconi, Smiths Industries and Meggitt Aviation and Marine. After completing a second degree in Marine Environmental Management, Peter took on several roles including Environmental Co-ordinator for a local authority. Whilst working as a sustainability consultant for Shell Exploration and Production, Peter was inspired to create NISP and had his own industrial symbiosis column in the Environmentalist and recently contributed a forum piece and co-authored an article for the Journal of Industrial Ecology's Special Issue on industrial symbiosis. He launched the industrial symbiosis Public Private Partnership (PPP) at the Global Green Growth Forum (3GF) in 2013 and will be updating status at 3GF in October 2014. Peter is the current Sustainability Leader of the Year awarded by edie.net and has been involved in thought leadership sessions for many corporates including Solvay and Hewlett Packard. During 2014 he has been an expert for the EU on eco-innovation/eco-industrial parks and the circular economy, delivering 3 presentations at Green Week.
DR. RACHEL LOMBARDI (female) is Director of Business Development for ISL, representing the world's leading deliverer of industrial symbiosis around the world. Rachel has been involved in sustainability and business issues for almost 25 years, largely from an industrial ecology perspective, with experience spanning both academia (Yale University, USA and University of Birmingham, UK) and business (IBM, McKinsey & Co., and ISL). Her work has influenced practitioners and policy makers internationally through bespoke training, participation in regional and international workshops, workshop and conference organisation and facilitation, traditional academic journals and conferences, and course development at the undergraduate and post-graduate levels. She holds degrees in mathematics and physics with honours, a MSc. in Electrical Engineering, and a PhD in molecular electronics. Rachel is an invited expert on systemic eco-innovation by The European Commission DG Research and Innovation, and invited expert and facilitator at the Global Green Growth Forum (2013, 2014).
Dr. Adrian Murphy is the Innovation and Technical Director for ISL, overseeing the company's delivery of a comprehensive range of business resource efficiency services. Adrian has a wealth of experience in resource efficiency delivery, including managing and delivering the National Industrial Symbiosis Programme (NISP). Adrian has 16 years' experience with LIFE funded projects as both an assessor and fund manager. Adrian is responsible for developing innovative and bespoke resource efficiency programmes from clients, and has expert knowledge in programme development, business systems, knowledge management and coordination of technology innovation teams. Adrian's breadth of skills and experience provide him with expert knowledge in the application of resource efficiency within diverse business environments including aerospace, chemistry, construction, engineering, environmental management, research and technology sectors. He has a degree in Chemistry and a PhD in the environmental arena, and has worked in industry throughout his career specialising in industrial symbiosis and resource efficiency.
MR. JAMES WOODCOCK (Male), International Manager, ISL James is a Project Manager with extensive experience both in the UK and internationally, working with funders and delivery partners. To date he has managed over 15 international projects for the company, including those funded through LIFE+ and Switch Asia. He has extensive environmental and industrial knowledge gained from over 20 years in roles that cross industry boundaries. His responsibilities include project management and delivery to pre-defined targets, training of new and complex concepts to various audiences (including non-English speaking audiences working with translators), and support of project delivery to in-country teams. James also works to identify and progress new industrial symbiosis opportunities throughout the world, recognising potential business partners and producing bids. He works to develop new products and services for the company relating to the establishment and delivery of industrial symbiosis programmes, including the development of SYNERGie, ISL's proprietary system for managing data and synergies. Having worked in the water industry for 15 years with a wide range of customer focused roles that could often require balancing conflicting requirements, revenue raising and tackling pollution incidents. James was also responsible for granting and enforcing trade effluent consents to industry. James holds a degree in Applied Science with honours.

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. Lombardi DR and PT Laybourn. (2012). Re-defining industrial symbiosis: crossing academic-practitioner and disciplinary boundaries. Journal of Industrial Ecology Special Issue on Industrial Symbiosis 16 (1): 28-37.
2. PT Laybourn and Lombardi DR (2012) Industrial Symbiosis in European Policy. Journal of Industrial Ecology Special Issue on Industrial Symbiosis 16 (1): 28-37.
3. Peter Laybourn and Maggie Morrissey, 2009 NISP The Pathway To A Low Carbon Sustainable Economy ISBN: 978

0 9564235 0 4
4. Chertow MR and DR Lombardi (2005). Quantifying economic and environmental benefits of co-located firms. Environmental Science & Technology, 39 (17) 6535-6541.
5. Lombardi DR and PT Laybourn (2007). Report on 3rd annual International Industrial Symbiosis Research Symposium, Yale University Press, USA.

A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. ISL is the only SME to have coordinated and hosted a track at the Global Green Growth Forum (3GF), leading on industrial symbiosis at 3GF 2013. ISL will present an update at 3GF 2014, 3GF 2015 & 2016 provide an excellent dissemination platform for IMATAIS
2. ISL devised and hosted the world's first International Working Conference on Applied Industrial Symbiosis (IWCAIS) launched by Commissioner Potočnik in Birmingham in 2012.
3. ISL hosted in Brussels "Friends of Industrial Symbiosis" for DGs Enterprise, Environment, Regions, Innovation and Climate (2012) which attracted senior Commission officials and MEPs
4. ISL founded the European Industrial Symbiosis Association (EUR-ISA) in 2013 which was launched in Brussels by Commissioner Potočnik
5. ISL hosted the 3rd International Industrial Symbiosis Research Symposium in 2006 and ISL hosts the industrial symbiosis channel on the 2 Degrees business social networking site– established 2014

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

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Participant No.	3	**Name (short name)/ Country**	UNIVERSITEIT TWENTE (UT)/Netherlands

Description of the legal entity

The University of Twente (UT) is where talent can best realize its full potential. Students and staff are the key. Together, 3,300 scientists and professionals carry out groundbreaking research, bring about socially relevant innovation, and provide inspiring teaching for more than 9,000 students. To us, entrepreneurship comes as second nature. The campus is home to around 100 businesses, including student-run businesses. The University of Twente has also generated more than 700 successful spin-off companies. The university's business park, Kennispark Twente, encourages and assists entrepreneurs to start new companies. The UT is one of the three technical universities in the Netherlands. UT-IEBIS

IEBIS – Department of Industrial Engineering and Business Information Systems-, which comprises about 30 scientists (including PhD assistants) of which about 50 % is working on logistics and supply chain management. The group holds a sound track record on research in fields such as transport and logistics and production and maintenance planning and control. Models and methods from Operations Research form the methodological basis of the group. In particular, one may think of combinatorial optimization for location planning and physical distribution planning, stochastic network models for production and logistics, queuing models, multi-agent technology to support distributed decision-making, and discrete event simulation to evaluate various policy alternatives. The group is involved in various projects aiming at more sustainable transport, logistics, and production and works closely with private industrial partners to achieve this.

Main tasks proposed in the project

There are physical, technological and economic conditions that favour the implementation of the industrial symbiosis. Primarily technological and physical conditions should occur to pre-design and approximate the optimal state of symbiosis from operational perspective while the economic conditions are critical to vitalize the cooperation. These conditions would make not only direct impact on the production processes involved in industrial symbiosis but also on the upstream and downstream process flows (material/energy/waste/product flows) of the whole production chains.
First, waste is not produced upon demand, but appears as a secondary output whose quantity depends on the final demand for the principal product of each supply chain. When the waste substitutes the primary input of another supply chain, then, the fit between the final demands of both production chains is critical for both the collective and individual economic benefit of the involved companies (i.e. physical condition). Second, there is also a relationship between the technological process efficiencies of both chains, which is also decisive for the waste emission and primary input use quantities, which in turn, affects the waste supply-demand matching (i.e. technological condition).
Then, to efficiently implement industrial symbiosis at the whole production chain level, we should understand the impact of final demand fit between two production chains on the operational performance of the industrial symbiosis and reveal the role of process efficiencies on waste supply-demand matching. Accordingly, we can propose dematerialization, waste-reduction, technology change, collaborative forecasting, or other sustainable production policies to reach the optimal (set

of) solution(s) to improve the cooperation. To this aim, we propose to adopt a waste input-output (WIO) model. Since WIO represents the process efficiency thanks to the observed waste and primary input coefficients and computes the total waste emissions and primary input use according to the main product final demand quantities, it is practically suitable for such an analysis.

Physical and technological conditions are not sufficient to vitalize the cooperation because not only value-added is obtained but also additional costs (e.g. recycling cost, transportation cost, production structure change cost, coordination cost) emerge from industrial symbiosis. Then, companies produce benefit- and cost-sharing strategies to maximize their economic gains formed by cooperation in industrial symbiosis. Individual needs drive the strategies but they are also influenced by the presence of competitors, environmental regulations, power asymmetry between companies, ratio of potential benefit to the gross income of companies, flexibility of the production, and the information availability about the co-operators. Hence, a game emerge between the companies, which produce different economic gains under each different strategy adoption. Game theoretical approach is suitable to integrate the company behaviour in such games defining the minimum individual needs of companies. Accordingly, we can understand the willingness to-to-cooperate and bargaining power of each company in the industrial symbiosis. Once the technological, physical, and economic conditions are set, then, we aim at adopting agent-based modelling (ABM) to propose to companies best set of solutions for maximizing both individual and collective economic gains. ABM is an appropriate methodology to address such a problem as it considers the heterogeneity of the actors (i.e., agents) and proposes a self-organizing system. Using ABM for industrial symbiosis design allows us to address the behavioural change by learning and interaction, and reveal the optimal symbiosis design under different individual strategies and cooperation modes. Hence, ABM is critical to achieve the optimal result in the negotiation phase between companies.

Finally, we propose to use input-output modelling to measure the total environmental impact of the industrial symbiosis at the whole chain level from a life cycle perspective. Input-output modelling is particularly useful for such a life-cycle-assessment as it is capable of tracing the all material/waste/energy/product flows of the involved production chains.

How your organisation is suited to these tasks

IEBIS is suited to the above-mentioned tasks as it largely uses quantitative methods, particularly ABM and Game Theory relevant to this proposal. It is in excellent cooperation with the industrial partners particularly in the Netherlands. In this proposal, it will also deal with setting the physical and technological conditions for industrial symbiosis and measuring the life-cycle impact of implementing industrial symbiosis.

CV of the persons who will be primarily responsible for carrying out the work

PROF. HENK ZIJM. (M) Henk Zijm (1952) is a full professor in Production and Supply Chain Management at the *University of Twente* (UT). Until October 1, 2014 he was also Research Director of the *Dutch Institute for Advanced Logistics* (DINALOG), the national institute responsible for the execution of the Innovation Action Plan for Logistics in the Netherlands. He is also vice-chair of the European Technology Platform for Logistics ALICE, which had delivered significant input for the Transport WP2014-2015, and is currently doing the same for the now designed Transport WP2016-2017. Both within the Netherlands and at a European level he is involved in various projects on collaboration and consolidation in logistics, on behalf of the University of Twente and on behalf of DINALOG, with the aim to reduce the ecological footprint and improve sustainability. A variety of national projects is carried out under the umbrella of DINALOG, with a focus on Supply Chain collaboration and coordination.

DR. DEVRIM MURAT YAZAN. (M) Devrim Murat Yazan graduated in Marmara University (Istanbul, Turkey) in Industrial Engineering and holds a PhD degree in Advanced Production Systems (Innovation Management Area) from Interpolytechnic School of Doctorate (Polytechnics of Bari-Turin-Milan, Italy, 2010), by the thesis entitled: 'Economic and Environmental Sustainability of Production (Supply) Chains: An Enterprise Input-Output Approach'. In CSTM, he is dealing with the sustainability of biofuels supply chains, development of local second generation biomass markets, and environmental and economic returns of alternative energy production and use. He is the author/co-author of a number of journal and conference articles and research reports in various research streams like spatial analysis for bioenergy supply chains, design and coordination of joint supply chains, transportation in re-engineered supply chains, sustainable cement, concrete, and ceramic production, industrial symbiosis networks, and cooperation and negotiation in reverse supply chains. He uses input-output modelling, life-cycle assessment and game theory methodologies in sustainable supply chain management research stream.

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. Romano, V.A., Yazan, D.M., Albino, V. 2012 How far is an input-output production system from perfect industrial symbiosis?, *20th International Input-Output Conference,* Bratislava, Slovakia, June 25-29.
2. Yazan, D.M., Clancy, J., Lovett, J.C. (2012) Supply chains, techno-economic assessment and market development for

second generation biodiesel. In: Luque R., Melero J.A. *Advances in biodiesel production. Second generation processes and technologies*, Cambridge, Woodhead Publishing.
3. Haneyah, S.W.A., Schutten J.M.J., Schuur, P.C., Zijm, W.H.M. (2013) A generic material flow control model applied in two industrial sectors. *Computers in Industry*, 64 (6), 663 - 677. ISSN 0166-3615.
4. Shanthikumar J.G., Yao, D.D., Zijm, W.H.M. (2003) Stochastic Modeling and Optimization of Manufacturing Systems and Supply Chains, Kluwer Academic Publishers, Boston.

A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. CONCOORD: Consolidation and Coordination in Urban Areas. Joint Programming Initiative Urban Europe (currently running)
2. SoCool@EU: Sustainable Organisation between Clusters Of Optimised Logistics @ Europe. FP7 project (currently running)
3. CO_3: Collaboration Concepts for Co-Modality. FP7 project (currently running)
4. Alternative energy sources in cement manufacturing: a systematic review of the body of knowledge. Network for Business Sustainability. Available at: http://nbs.net/wp-content/uploads/NBS-Systematic-Review-Cement-Manufacturing.pdf

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

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Participant No.	4	**Name (short name)/ Country**	University of Leeds (UNIVLEEDS) / UK

Description of the legal entity

University / Higher Education Institute. The University of Leeds is one of the largest universities in the UK. It is a research-intensive university, a member of the Russell Group and ranked in worlds top 100. The University has a multi-disciplinary and internationally leading energy research community and its current research council energy research portfolio is valued at in excess of £20M. In order to support this community and coordinate its activities for maximum research and educational value and impact the university has created the Energy Leeds virtual institute. As part of its educational mission, Energy Leeds has launched a new multi-disciplinary Sustainable Energy Systems master's programme addressing energy from technical, environmental, financial and political standpoints and with Professor Heyes as programme director. The MSc is taught by staff from five schools within the engineering and business faculties at the university. Students from this programme will be integrated into the research project by virtue of their individual MSc projects achieving the dual aims of providing students with real world problems and experience to enhance their learning and employability and providing additional support to the research project. This will also allow us to explore interesting and potentially valuable tangential issues that emerge but which are not on the critical path. This will be done at no cost to the project.

Main tasks you would propose to perform in the proposal

As part of WP2, working in close cooperation with USTRATH we will assess, catalogue and derive schemes for optimisation of new energy transfer, storage and conversion infrastructure. These will be required to realise the IS opportunities identified by the SHAREBOX programme. We will also identify and develop business opportunities and plans for additional actors in the form energy service companies within the industrial ecosystem.

How your organisation is suited to these tasks

As part of its research support and development mission, Energy Leeds has created the Smart Energy Systems research network. This is a network of over thirty academic and research staff from eleven schools and four faculties with expertise in many areas (energy storage, smart grids, energy policy, system modelling, low-grade waste heat conversion technologies, energy from waste etc.) relevant to the current research project. The executive team of the proposed research are all members of the network and the project will be regarded as a Smart Energy Systems Initiative so that the researchers will have access to this multi-disciplinary community for advice and support throughout the project. In additional to access to the Energy Leeds research community as detailed above the PDRA will be supported and managed by a core executive management team with diverse but complimentary skills. The team brings expertise in heat storage, transfer and transformation, chemicals processing, renewable energy, system modelling and policy analysis and waste conversion technologies to bear on the project and, as such, all the important facets of the research challenge can be addressed and supported. Details of the individual members of the team can be found below:

CV of the persons who will be primarily responsible for carrying out the work

PROFESSOR TIMOTHY COCKERILL (Male) Chair in Efficient Energy Utilisation. The Centre for Integrated Energy

Research (CIER) at the University of Leeds (UoL) was established in 2010, bringing together more than 40 academics to integrate the University's research and innovation capabilities across the full range of energy disciplines. CIER forms part of an ongoing strategic investment by the University, being located a new £12.5M Energy building that plays host to 150 staff and research students. CIER members have active involvement with all three of the recently awarded EPSRC DTC's in which the University has and interest (CCS, Bioenergy and CFD). Prof. Tim Cockerill, was recently appointed as Chair of Efficient Energy Utilisation within CIER. He also has a visiting appointment at Imperial College London. As PI or Co-I on projects supported by EPSRC, NERC, Tyndall Centre, and the European Commission, he specialised in the integrated systems modelling and techno-economic optimisation of low-carbon technologies with an emphasis on CCS and energy storage. He is an associate member of the NERC Peer Review College and contributes to the SSE Accounting for Sustainability Board. Recent projects include SUPERGEN Highly Distributed Energy Futures (EP/G031681/1), modelling the techno-economics of solids looping NG PCC Technology (EP/J020745/1), and investigation of the wider land use implications of differing renewable energy strategies using spatial modelling techniques (NE/H010521/1). A range industry and RCUK funded PhD studentships currently support work on the deployment of energy storage and the lifecycle environmental impacts of CCS

DR COSTAS VELIS, (Male), BSc MSc DIC PhD MInstP MCIWM, Lecturer in Resource Efficiency Systems. The Institute for Resilient Infrastructure (iRI) in the School of Civil Engineering conducts interdisciplinary research at the interface between the disciplines of engineering, environmental, economic, community / societal and political domains. It has a multimillion pound research portfolio including reliance on critical materials, radical approaches to new infrastructure business models, and resource recovery from waste via whole-system interventions. Dr Costas Velis is Lecturer in Resource Efficiency Systems and as part of iRI he leads an international research team working on recovering value from waste. He directs a relevant state-of-the-art laboratory, supervising 8 PhD students. With over 10 years of work and research experience in waste and resources management, he serves as Associate Editor of the journals *Waste Management & Research*, and *Critical Reviews in Environmental Science and Technology*. Dr Velis was recently honoured with the prestigious ISWA 2013 Publication Award for a paper offering novel inter-disciplinary approaches that consider the waste management systems interplay with other systems such as materials supply chain, socioeconomic and business /organisational aspects. A Chartered Waste Manager (CIWM), he is also registered with the Institute of Physics. He serves in many national and international committees and is involved in many of the initiatives addressing waste and resource-related challenges at a global level (core author member team of the UNEP Global Waste Management Outlook; coordinating member of the International Solid Waste Association Globalisation and Waste Management Task Force; scientific co-ordinator of Waste Atlas project). His areas of expertise include the production and quality management of solid recovered fuels that can replace fossil fuels; novel methods to measure their biogenic content to mitigate greenhouse gasses emissions (TSB/SBRI/971195); optimising the energy efficiency of energy from waste plants via the R1 formula; material flow analysis of mechanical processing in waste plants. He is the chief Co-I in a NERC 'Resource Recovery form Waste' catalyst call cross-disciplinary £1.2M project (NE/L014149/) advancing the multi-objective systems optimisation of value recovery from waste between multiple players (C-VORR). Prior to joining University of Leeds, he worked for two year at Imperial College London as Research Associate in Resource Recovery. He sits at the Society of Chemical Industry (SCI) Environment, Health and Safety Technical Interest Group, where chemical industry meets academia. Dr Velis is Vice Chair of the Waste to Energy Resource and Technology – UK, and he runs his own consultancy (Fuelogy), which provides specialised technical and modelling research and development solutions for advanced aspects of resource recovery and energy efficiency.

PROFESSOR DR. IR. FRANS MULLER *PHD CANTAB CENG FICHEME* ***(Male)*** Professor Muller holds a joint chair in Chemical Process Engineering in the School of Process Environmental and Material Science and the School of Chemistry. He works on the interface between chemical engineering and chemistry and his research interests lie in the dynamics of manufacturing processes and understanding of the interaction between process and equipment. His work is currently supported through a £1M strategic investment of the University of Leeds as part of the establishment of the Institute of Process Research and Development (iPRD) and directly from industry (AstraZeneca (AZ), Syngenta, Alfa Laval and Aesica). Before joining iPRD in August 2011, he spend 17 years in industry (AstraZeneca (AZ), Syngenta) as a process engineering scientist working on the design and development of industrial processes & equipment. His group includes 4 PhD students funded by Leeds and industry (AZ, Syngenta). He was the lead investigator for AZ on a large EU consortium grant (F3 Factory, 19M€ EU funding, 25 partners involved in continuous mobile modular manufacture, www.f3factory.com), representing AZ's 5 academic partners on the consortium management team and leading the AZ project team. At Syngenta he developed and in 2000 subsequently commercialised a batch processing simulation program, known commercially as DynoChem, to a SME software company (Scale-up Systems Ltd, http://www.scale-up.com). Now over

450 DynoChem licences are used on ~100 sites in 23 countries by 13 of the top 15 pharmaceutical companies. Scale-up Systems employs 20 people in 5 countries and has an annual turnover of over €2M.

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. Velis C.A. and Cooper J. Are solid recovered fuels resource-efficient? Waste Management & Research, 2013, 31(2): 113-114.
2. Velis C.A., Longhurst P.J., Drew G.H., Smith R., and Pollard S.J.T., Production and Quality Assurance of Solid Recovered Fuels Using Mechanical-Biological Treatment (MBT) of Waste: A Comprehensive Assessment. Critical Reviews in Environmental Science and Technology, 2010, 40(12): 979-1105.
3. Velis C.A., Wagland S., Longhurst P., Robson B., Sinfield K., Wise S., and Pollard S.T., Solid recovered fuel: materials flow analysis and fuel property development during the mechanical processing of biodried waste. Environmental Science & Technology, 2013, 47(6): 2957-2965.
4. Velis C.A., and Brunner, P., "Recycling and resource efficiency: It is time for a change from quantity to quality". Was. Manag. Res., 2013, 31(6): 539-540.
5. McGlashan N.R, Childs P.R.N, Heyes A.L, "Chemical Looping Combustion Using The Direct Combustion Of Liquid Metals In A Gas Turbine Based Cycle" Trans. ASME: J. Eng. Gas Turb. And Pow, 2011, Vol. 133, Issue 3.

A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. PI for UoL. Call H2020-WASTE-2014-one-stage: WASTE-4b-2014; Project: EWIT: Developing an e-waste implementation toolkit to support the recycling and the secondary raw material recovery strategies in metropolitan areas in Africa. "EU-African collaboration addressing resource recovery from e-waste by identifying optimal management practices based on material flow analysis tools." Total 1,64M Euros, University of Leeds: 183,750 Euros.
2. Chief Co-I; University of Leeds only; Funding: NERC and ESCR, £1.1M plus over £1M in-kind industrial partner contributions; Call NERC Resource Recovery from Waste – Full Phase: 'C-VORR: Complex Value Optimisation for Resource Recovery': Cross-disciplinary Innovation with a view to 2050: Developing and evaluation framework based on systems dynamics to optimise sustainable resource recovery from waste considering all waste and resource management steps and material / products life-cycle, from sustainable consumption and production, to prevention, waste collection and sorting, to recycling recovery and final disposal where needed.'

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

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Participant No.	5	**Name (short name)/ Country**	Zurich University of Applied Sciences (ZHAW)/ Switzerland

Description of the legal entity

The ZHAW is one of the leading universities of applied sciences in Switzerland. Teaching, research, continuing education, consulting and other services are both scientifically based and practice-oriented. Graduates of the ZHAW hold their own on national and international markets and in the intercultural working world. About 11'000 students are enrolled at the ZHAW in a total of 26 Bachelor's degree programmes and 13 consecutive Master's programmes. The university also offers a broad range of continuing education courses, including over 40 Master's of Advanced Studies programmes. 30 institutes at the

ZHAW network their resources to provide business and industry with customised services.

Our Institutes and centres create opportunities and entrepreneurial scope for specialised, professional expansion of research and development. Some of them are managed as profit centres and enable expertise to be developed and extended in an interdisciplinary working environment, and made available to interested partners. Institutes and Centres are led by a professor who also has teaching responsibilities – in degree programmes and in continuing education. This ensures that state of the art knowledge is continually transferred into the ZHAW's degree programmes and its continuing education activities. The research staff and assistants, supported by a state-of-the-art, comprehensive infrastructure, form the backbone of the Institutes and Centres.

Life Sciences are key fields of activity for the 21st century, demanding interdisciplinary thinking and action. In response to this challenge, the School of Life Sciences and Facility Management provides opportunities for networking – from biotechnological, chemical and food technology related issues to logistical, ecological and management aspects – and carries out complex projects using an interdisciplinary approach.

Depending on the particular challenges of a project, researchers, research assistants and students from various disciplines work together in interdisciplinary teams. The ZHAW's equipment and apparatus are state-of-the-art and allow professional processing of application oriented research and development projects. Modern laboratories as well as testing and

production facilities provide a solid basis for this.

At the interface between theory and practice, the Institutes work closely with industry and business. Expertise and infrastructure are made available in accordance to clients' needs.

Main tasks you would propose to perform in the proposal

With our knowledge in purification and waste treatment technologies we want to contribute in implementing a self-sustaining database for tailored purification and waste treatment technology. Besides product requirements defined by the process industries, efficient purification requires the detailed knowledge on the physical chemical properties of the product(s) as well as of the impurities to select the best suited purification technology, e.g. membrane filtration, depth filtration or ion exchange. On the other hand, the performance of filtration aids depends on their physical chemical properties as well, such as pore size, streaming resistance, and polarity in case of membranes or properties controlling the adsorption efficiency of filtration aids such as surface area, pore size, zeta potential. We have already a large set of physical chemical data from commercially available filtration aids together with respective processes, where these are employed. This database shell grow within the SHAREBOX project. Furthermore, we will apply our competences in LCA for detailed quantitative environmental sustainability evaluation of processes and products in analysed case studies (test sites) within the frame of the SHAREBOX.

How your organisation is suited to these tasks

The Center of Functional Materials and Nanotechnology focus on the development of new filtration processes and technologies including innovative materials such as molecularly imprinted polymers or nanofibrous membranes. In addition, we are characterizing membranes and filtration aids based on their physical chemical properties such as surface area, pore size, pH depending zeta potential or streaming potential in case of membranes to correlate their performance in a filtration or purification task with their physical chemical properties.

The ZHAW Institute of Natural Resource Science will focus on applying the Life Cycle Assessment methodology for an environmental sustainability assessment of the processes involved in SHAREBOX..

CV of the persons who will be primarily responsible for carrying out the work

DR. JÜRGEN EBERT (male). He has been working in the filtration industries for more than 15 years where he was developing custom tailored depth filtration sheets. He is also in the board of one of the biggest suppliers of filtration aids such as perlites and zeolites in Switzerland. In parallel, he is lecturing at the ZHAW filtration technology and related topics. Together with his team at ZHAW, he is also running research activities with the aim to better understand the relation between physical chemical properties of filtration aids and the performance of the filtration process. There, his key expertise is the characterization of organic (cellulose) and inorganic (diatomaceous earth/perlite/zeolite) filter aids, membranes, adsorbents (activated carbons) and ion exchange resins. The characterization includes analytical methods like “particle charge detection”, “zeta potential”, “porosimetry”, “SEM”, etc. Furthermore, the development and/or modification of tailor-made depth filter sheets and membranes depending on the chemical and physical properties of the impurities lays is in the focus of his work. He covers the whole range of solid/liquid separation in the food and beverage industry, biotech and pharmaceutical industry (product purification), as well as water and waste water treatment (waste management).

MSc. MATTHIAS STUCKI (male) received his Master of Science Degree in Environmental Sciences from ETH Zurich in 2007 and is an expert in the field of Life Cycle Assessment (LCA). Between 2008 and 2012 he worked as a project leader at ESU-services Ltd. where he conducted a range of LCA projects in the various fields such as power generation, solar energy, bioenergy, geosynthetic materials, biogenic plastics, leisure activities, consumption patterns, sanitary systems etc. He published his findings in scientific journals such as the International Journal of LCA and submitted datasets to the worldwide ecoinvent database. Affiliated with the ecoinvent Centre, he started the Life Cycle Inventory Network in India and trained Indian engineers in conducting LCA projects. Since 2012 he holds a position as a Research Associate at the Institute of Natural Resource Sciences of the Zurich University of Applied Sciences, where he is head of the LCA research group and responsible for research and teaching in the field of LCA. Currently, he is engaged in national and international projects that analyse the environmental impacts of food production (including dairy), sewage treatment, and power generation. He is a reviewer at the Journal of Industrial Ecology, the Journal of Waste Management, and the International Journal of Life Cycle Assessment, member of the editorial board of the ecoinvent database, and an individual verifier at the International EPD® System.

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. J.Ebert and C.Adlhart, Intelligent nanofibres as basis for new filter materials, Filtration & Separation, Global Guide 2012-2014, p. 181ff

2. J. Ebert, Zetapotential measurements for surface characterization of filter aids, ZHAW-Transfer 3/2009
3. R. Büttiker, J. Ebert, C.Hinderling, and C. Adlhart, Membranes for Specific Adsorption: Immobilizing Molecularly Imprinted Polymer Microspheres using Electrospun Nanofibers, CHIMIA 2011, 65, No. 3
4. Jungbluth, N., Büsser, S., Frischknecht, R., Flury, K., & Stucki, M. (2012). Feasibility of environmental product information based on life cycle thinking and recommendations for Switzerland. Journal of Cleaner Production, 28(0), 187-197
5. Faist Emmenegger, M., Stucki, M., & Hermle, S. (2012). LCA of energetic biomass utilization: Actual projects and new developments, April 23, 2012, Berne, Switzerland. The International Journal of Life Cycle Assessment, 17(9).
Patents: Filter layer for TCA removal (cork taint) – EP000001961805A1; Filter layer with a fibre matrix made of cellulose and chitin and/or at least partially deacetylated chitin - EP000001649916A1; Membrane, membrane module and process for cross-flow depth filtration - EP000001721656A1

A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. Process optimization and purification of herbal drugs
2. Removal of organic pesticides from vegetable oils
3. Development of a cleaning process of water-ethanol mixtures for re-use of extraction liquids in pharmaceutical processes
4. Characterization of activated carbons for biogas cleaning
5. Development and characterization of specific adsorbents for the removal of micropollutants in water and wastewater Hydrothermal carbonisation as an energy-efficient alternative to thermal drying of sewage sludge

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Equipment to perform filtration tests in laboratory scale or using industrial filter systems (plate and frame, alluvium, cartridge, module, membrane), a lab scale depth filter sheet production plant, and ion exchange cartridge systems.
Analytical equipment to characterize filtrate and filtration residue
Analytical equipment to characterize physical chemical properties of filtration aids and membranes such as zeta potential meter, particle charge detector, BET surface measurement, pore size analysis.

Participant No.	6	**Name (short name)/ Country**	Ceramic Industry Research Association (ITC-AICE)/Spain

Description of the legal entity

The Ceramic Industry Research Association (ITC-AICE) is a Private non-profit Research Institute recognized by the Spanish Ministry of Economy and Competitiveness as Technology Centre by Royal Decree 2093/2008. ITC-AICE is committed to providing solid support for ceramic sector in the defence and enhancement of their strategic positioning in the current global context, principally through innovation-enabling research and development actions, but also through whatever activities might serve to foster the competitiveness and growth of the sector, always based on sustainability criteria and commitment to societal well-being.
In 1969 ITC-AICE concerned an agreement with the Universitat Jaume I of Castellón in response to the needs of companies from the Spanish ceramic cluster. During its more than 40-year history, this agreement has articulated a successful university–business cooperation system that has borne its fruits, witness the significant development of the Spanish ceramic tile manufacturing industry.
The competence attained through ITC-AICE's wide-ranging research activity enables to extend its field of action to other types of processes and materials. Particularly ITC-AICE works in the field of ceramic processing, energy efficiency, and industry's environmental impact minimization (innovative waste recovery solutions and emissions reduction), as well as in the functionalisation of ceramic surfaces and in the achievement of new technical performance and aesthetic features of products related to the habitat hyper-sector and other industries, such as high-tech tools advanced ceramics, automotive or petrochemical sectors. Its activities include products and processes improvement, residues and greenhouse pollutants reduction, control systems development, energy saving plans, etc.
From the outset, as technology partner of the ceramic sector, ITC-AICE has undertaken several R&D activities in the ceramic and related industries, through projects funded by public bodies and by ceramic sector companies.

Facts & Figures (last five years):

- Number of Autonomous and National funding projects: **174**
- Number of European projects: **16**
- Number of R&D collaborative projects funding by industry: **380**
- Number of Scientific papers with impact factor: **180**

- Number of Institutional bibliography references : **54**
- IPR through Patents: **17**
- Number of awards recognizing the R&D and Technology Transfer's impact in the ceramic cluster: **5**

Main tasks you would propose to perform in the proposal

Implementation of demo tests/case study: development of the SHAREBOX platform in a Spanish industrial park, where there are several companies related to the manufacturing of ceramic tiles. (WP5, WP6)

This task will consist of:

- Identification of the inputs to each company, including water, fuels, materials, etc.
- Identification of the outputs from the companies: solid wastes, dirty water and sludges, wasted energy, etc.
- Analysis of the general management issues that could be able to share among different companies located in the same industrial area: security services, cleaning services, suppliers of some materials, etc.
- Implementation of a new strategy to optimize resources and management of wastes, favouring the industrial symbiosis by reducing the use of primary resources and enhancing waste recovery.
- Identification of new opportunities to improve the efficiency of the companies in their use of energy, raw materials and water by implementing new technologies.

How your organisation is suited to these tasks

ITC-AICE is a research centre with more than 40 years of experience in the ceramic industry, working on projects related to raw materials preparation and optimisation, atmospheric emissions, wasted water treatment, solid waste management, energy efficiency and energy recovery, among others.

The industrial park selected for implementing SHAREBOX platform is formed by different kind of companies. Some of them are related with ceramic manufacturing such as raw materials suppliers, glaze and frit manufacturers and ceramic tiles industries and are well-known enterprises since ITC-AICE have been collaborating with them in different R&D projects.

ITC-AICE broad experience in the topics covered by this project is demonstrated by the projects carried out in the last years (in the field of resource efficiency, energy and environment). In the field of energy efficiency, ITC-AICE is working with ceramics manufacturers to ascertain effective energy saving actions to be implemented in the manufacturing processes, optimising the energy consumption and enhancing overall energy efficiency. Thus ITC-AICE capabilities in the field of energy saving and energy efficiency include, among others: the Performance of energy audits, the realisation of studies of energy reuse in industrial facilities, the adaptation of heat recovery systems, the optimisation of energy consumption at industrial scale, etc…

Resource efficiency and waste is another area of expertise of ITC-AICE. In fact the production of waste is a another major environmental concern in any industrial production process, in relation to both the quantity and diversity of the arising types of waste and to the ensuing regulatory obligations. ITC-AICE therefore collaborates with companies in order to enable them to assure appropriate waste management, from the start of their production processes to the final destination. ITC-AICE know-how in the field of resource efficiency and waste include: waste recycling in the ceramic process, recycling of the ceramic process waste in the own process or alternative sectors, full life cycle analysis and cost assessments, waste minimisation plans and waste characterization.

Finally, ITC-AICE has a wide experience in wastewater treatments. Indeed, the manufacture of ceramic products requires great water consumption. ITC-AICE is working with ceramics manufacturers to ascertain effective water saving and recycling actions to be implemented in the manufacturing processes, optimising the water consumption and enhancing overall wastewater recovery, treatment and resource efficiency. In addition, ITC-AICE participates in others R&D activities aimed at identifying alternatives to the use, recycling and treatment of wastewater, in ceramics manufacturing in order to reduce the environmental impact of such industrial activities.

CV of the persons who will be primarily responsible for carrying out the work

ENCARNA BOU SOLSANA (Female) is doctor in Chemical Science from the Universitat Jaume I (Castellón, Spain). She is part of ITC team since 1991 and is currently a Senior Researcher in the area for Materials and Ceramic Technology. Encarna Bou has a strong expertise in ceramics technology, especially in the field of frits and glazes, raw materials and resource efficiency. She has leaded several R&d and technological projects as principal investigator, under private and public funding (at regional, national and European level). In addition, she is responsible for coordinating all actions related to the European association "Energy Efficiency Building Association (E2Ba)".

Encarna Bou is the author of more than 54 scientific publications in Spanish and international journals and has participated in various national and international Congresses with 42 communications. She has also co-authored 2 books in the field of

ceramic technology.
Encarna Bou will participate in the project as responsible of the cermic case-study implementation and expert in resource efficiency.

ANA MEZQUITA MARTÍ (female) has a Chemical Engineer degree from Universitat Jaume I. She has joined ITC in 1999 in the area of industrial processes. Since 2006, Ana Mezquita is the head of the Energy Unit, which aim is to support ceramic in any activity related with energy efficiency and energy savings. In addition, she has participated in R&D activities aimed at identifying alternatives to the use of natural gas, a fuel of fossil origin, in ceramics manufacturing in order to reduce the environmental impact of such industrial activities.
Ana Mezquita has collaborated in more than 35 R&d and technological projects, under both private and public funding (at regional, national and European level). She is the author of more than 20 papers in scientific and technological journals and around 20 communications to congresses, mainly in the field of ceramic technology. She has also given 9 courses about energy efficiency and environmental aspects in the ceramic manufacturing, to private companies and in national and international research centres.
Ana Mezquita will participate in the project as Energy Efficiency expert.

IRINA CELADES LÓPEZ (Female) is doctor in Chemical Science from the Universitat Jaume. She is part of ITC team since 1997 and she is the head of the Environmental Technologies Unit since 2000. She has collaborated in more than 50 R&D projects (20 as principal investigator) under private and public funding.
Irina Celades has co-authored more than 40 cientific publications, 2 books and over 55 communications to congress, in the field of Environmental technologies and Ceramic process. In addition Irina has taught in 8 courses to form ceramic companies in the field of Environment.
Irina Celades will participate to the project as expert in environmental technologies and their application to the ceramic sector.

RAÚL MOLINER SALVADOR (Male) is graduated in Chemical Engineering from Universitat Jaume I in 2002 and has a Master's Degree in Ceramic Materials Science, Technology and Applications also from Universitat Jaume I.
Raúl Moliner is part of the Environmental Technologies Unit of ITC and has participated in more than 30 R&D and technological projects developed by ITC under both private and public funding scheme. He has been granted by the IMPIVA (Valencian public organism) in 2005 to realised a 6 months research stay at the European Membrane Institute (Institut Européen des maembranes, Montpellier, France), increasing his knowledge in water treatment. In addition, he is responsible for coordinating ITC activities related to the Spanish Water Technological Platform (Plataforma tecnológica española del agua).
Raúl Moliner has also coauthored 9 scientific papers and 9 communications to congresses in the field of ceramic technology and chemical engineering.
Raúl Moliner will collaborate in the SHAREBOX Project as expert in water management and wastewater treatment.

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. Chapter 8.09. Ceramic Manufacturing Processes: Energy, Environmental, and Occupational Health Issues. In Comprehensive Materials Processing. 1st Edition.
Monfort, E.; Mezquita, A.; Vaquer, E.; Celades, I.; Sanfelix, V.; Escrig, A. Bassim, N.,
Ed, Vol. 8; Elsevier Ltd., 2014; pp 71–102.
ISBN: 9780080965321
2. Reduction of CO2-emissions in ceramic tiles manufacture by combining energy-saving measures.
Mezquita, A.; Monfort, E.;Vaquer, E.; Ferrer, S.; Pitarch, J.M.; Arnal, M.A.; Cobo, F. Cfi Ber. DKG 85, Vol. 91, Issue 5, May 2014, Pages E37-E42, 2014.
3. Case study of glazing waste valorisation.
Monfort, E.; Bou, E.; Felíu, C.; Silva, G.; Cruz, R.; Portolés, J.; Martí, V. *Cfi Ber. DKG,* 81(1-2), 33-36, 2004
4. Recycling marble working wastes in manufacturing ceramic products. Part I Waste characterization & Part II Ceramic wall tile manufacture.
García-Ten, J.; Mallol, G.; Bou, E.; Silva, G.; Fernández, J.; Molina, A.; Romera, J *Cfi Ber. DKG,* 80(9), 84-90, 2003. & *Cfi Ber. DKG,* 80(10), 30-32, 2003
5. Use of nanofiltration membrane technology for ceramic industry wastewater treatment.
Deratani, A.; Palmeri, J.; Moliner-Salvador, R.; Sánchez, E. *Bol. Soc. Esp. Ceram. Vidr.,* 51(2), 103-110, 2012.

A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. Reduction of carbon dioxide emissions in the ceramic manufacturing process (REDUCER). With the support of the

European Commission. Sustainable Industry Low Carbon Scheme, SILC I (2012-2014). Ref. Grant agreement nº: SI2.639876.
2. Zero waste in ceramic tile manufacture (Life CERAM), supported by European Comission through the Programme LIFE+, through the Programme of Environmental Policies and Governances. Ref. Grant agreement nº: LIFE12 ENV/ES/230.
3. Innovation and Ecodesign in the Ceramic Industry (SINNDESIGN). With the support of the European commission through Leonardo da Vinci 2013 (Lifelong learning Programme). Ref. Grant agreement nº: 2009-1-PT1-LEO05-03237.
4. Advanced membrane separation technologies for boron removal in wastewaters of ceramic and other manufacturing industries (NANOBORON). Funded through the European program EUREKA. Ref. Grant agreement nº: ∑! 4121
5. Ceramic waste valorisation in cement manufacturing (VALOCER). Supported by the Spanish Ministry of Industry, Energy and Tourism. Ref. Grant agreement nº: CTM2004-06619-C02-02

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

The ITC-AICE has a mobile unit that provides all equipment necessary to perform in industrial facilities several activities such as energy audits, atmospheric emission measurements to carry out research projects or consultancy activities.
In the case of atmospheric pollution studies, the equipment property of ITC-AICE, allow the study of atmospheric emissions of different nature and from different type of sources, channelled and diffuse ones, these last generated due to the transport, handling and storage of bulk raw materials.
On the other hand but not included in the lab mobile unit, ITC-AICE has in its installation two mobile plants for feasibility studies of tertiary treatment in industrial waste water, based on membrane techniques. These plants are semi-industrial and pilot scale, both possible to transport to the industrial facilities to assess the viability of the results obtained at laboratory scale.

Participant No.	7	**Name (short name)/ Country**	Universitat Politècnica de Catalunya (UPC) / Spain

Description of the legal entity

The Universitat Politècnica de Catalunya · BarcelonaTech (UPC) is a public institution dedicated to higher education and research, specialised in the fields of engineering, architecture and science. With a focus on intellectual rigour, critical thinking, a cross-disciplinary approach to knowledge, educational innovation and entrepreneurship, the UPC produces competent professionals with the skills they need to tackle present and future challenges.
The academic and research activity that goes on at UPC campuses and schools has made the University a benchmark institution. UPC is a member of European CLUSTER (Consortium Linking Universities of Science and Technology for Education and Research, http://www.cluster.org), which is a consortium of 12 elite European Universities in Science and Engineering (and architecture) with associate members from around the world. UPC is ranked in the 8th position among best European universities, and the 37th best university around the world in Computer Science subject, according to USNews ranking (http://www.usnews.com/education/best-global-universities).
The University harnesses the potential of basic and applied research, and transfers technology and knowledge to society. These actions make the UPC—in partnership with industry—an agent and driver of economic and social change. UPC has 22 Schools and Faculties spread over 7 Catalan cities, 42 Departments, 10 Research Institutes and 201 research groups. One of these research groups is the Knowledge Engineering and Machine Learning Group (KEMLG).

Main tasks you would propose to perform in the proposal

- Design and development of an appropriate intelligent recommendation system, based on data mining techniques, and/or knowledge-based reasoning, for suggesting optimal actions for plant and operation managers or recommendations for shared resources decision support complementing other mathematical/computer tools like game theory algorithms, scheduling algorithms, etc.
- Design and development of an intelligent module, based on case-based reasoning, for analysing and learning from the day-to-day operation of the SHAREBOX-ISI core module. These experiences will be used to improve the competence of the SHAREBOX-ISI system, learning to solve operational problems /suggest recommendations in a better way along time.

How your organisation is suited to these tasks

The Knowledge Engineering and Machine Learning Group (KEMLG, http://kemlg.upc.edu) belongs to the Universitat Politècnica de Catalunya · BarcelonaTech (UPC), and is the group involved in this proposal. The KEMLG group has been distinguished as an excellent and consolidated research group by the Catalan Quality Agency since 2005 (2005 SGR 0001, 2009 SGR 1365, 2013 SGR 1304). Currently, the group has 6 senior researchers (1 Full professor and 5 Associate professors, all with tenure tracks), 1 post-doctoral researcher, 6 PhD students and 1 Master's student. The main fields of interest of the group are Data Mining, Knowledge Discovery, Intelligent Decision Support Systems, Recommender

Systems, Case-Based Reasoning, Knowledge-Based Systems, Knowledge Engineering and Management, Machine Learning, Autonomous Agents, Assistive Technologies and Ambient Intelligence, Electronic Institutions, Organizational Theory and Multi-agent Systems, e-Commerce, Semantic Web and Distributed Intelligent Systems. The main aim of the KEMLG research group is the analysis, design, implementation and application of several Artificial Intelligence techniques and methodologies to support the operation or behaviour analysis of real-world complex systems or domains. The research is focused on the analysis, design, management or supervision of these domains, such as in healthcare, in environmental processes and systems, in social and internet-based systems and in the industrial and enterprise sector. Its major area of expertise regarding the objectives of the SHAREBOX proposal relays in the development of Data Mining techniques, Recommender Systems and Intelligent Decision Support Systems (IDSS) applied to Environmental Systems (IEDSS), to e-Health systems (assistive technologies) and to Industrial Systems (Textile Industry).

The group has been active in the Artificial Intelligence field since 1986. Recently, the team has been involved in EU funded projects TexWIN (Textile Work Intelligence by closed-loop of product and process quality in the Textile Industry) (CP-FP 246193-2); in project "Rapid Application of Innovation in Small Medium Enterprises (RAISME)" (FP7 2624269); in project FATE. FAll deTector for the Elderly (ICT-PSP-2011-5-297178); in project IDONTFALL. Integrated prevention and Detection sOlutioNs Tailored to the population and Risk Factors associated with FALLs (ICT-PSP-2011-5-297225), and in project SUPERHUB: SUstainable and PERsuasive Human Users moBility in future cities (ICT-2011 289067). Previous EU projects were "Development of a rapid configuration system for textile production machinery based on the physical behaviour simulation of precision textile structures (MODSIMTex)" (CP-FP 214181-2), Agentlink III (Semantic Knowledge Based Systems), ASPIC (Semantic Knowledge Based Systems) and Provenance (Grid computing). A team of the KEMLG group won the International "Hacking Bullipedia" contest in November 2013. Prof. Ulises Cortés, which is the founder of the group, was one of the coordinators of the "Agentcities" (DEP-2003-00024) finalist application to the Descartes Prize 2003 Outstanding Scientific and Technological Achievements Resulting from European Collaborative Research. In addition to European Project activity, the group participates is a wide range of national / international initiatives including: "An Intelligent Decision Support System for minimisation and forecasting of volcanic activity impact" (R31971-A), funded by CONACYT in México, Spanish and Catalan funded research projects such as in the past projects within data mining field such as "Improvement and new developments in an Intelligent Data Mining System for Knowledge Extraction and Management in Environmental Databases (GESCONDA-II)" (TIN2004-01368) and "Development of an Intelligent Data Analysis System for Knowledge Management" in Environmental Data Bases (GESCONDA, TIC2000-1011) funded by Spanish CICyT entity.

CV of the persons who will be primarily responsible for carrying out the work

DR. MIQUEL SÀNCHEZ-MARRÈ (Male) (http://www.cs.upc.edu/~miquel) received a B.Sc. in Computer Science in 1988 and a M.Sc. in Computer Science in 1991 both from the Barcelona School of Informatics. He received a Ph.D. in Computer Science in 1996 from the Universitat Politècnica de Catalunya · BarcelonaTech (UPC). He is Associate Professor in the Computer Science Department (CS) of the UPC since 1990 (tenure 1996). He was the head of the Artificial Intelligence section of CS Dept. (1997-2000) and head of the KEMLG group (2006-2011). He is the Academic Coordinator of the PhD program of Artificial Intelligence (UPC) since February 2013, and he was the Academic Coordinator of the Interuniversity Master program of Artificial Intelligence (UPC-URB-UB) since February 2010 to August 2011. He was on a sabbatical leave from UPC at Catalan Institute of Water Research (ICRA), since September 2011 to august 2012. He is a pioneer member of ACIA (Catalan Association of Artificial Intelligence) and in 1994-1998 was member of its board of directors. He is a Fellow of the International Environmental Modelling and Software Society (iEMSs) since 2005. He received an Accesit of the Oms i De Prat 1991 Prize, granted by Caixa de Manresa, in the field of Experimental and Applied Sciences for the Master's Thesis entitled, DEPUR: an Application of Knowledge-Based Systems to Wastewater Treatment Plants Diagnosis (1991). In 2003, he founded the spin-off company Sanejament Intel·ligent S.L. (SISLtech), which received the *Energy Excellence Award 2013* in the category of Entrepreneurship and Employment, awarded by the Catalan Energy Institute (ICAEN) for the business plan developed. The firm has developed an innovative and high benefit for the wastewater treatment process and has consolidated in the area of energy optimization of wastewater treatment plants, a market that has allowed implementing projects in different countries in Europe and South America. November 10th, 2013. Furthermore, the company received in 2012 the *Bioenergy Silver Award 2012*, awarded by ATEGRUS (Technical Association of Waste Management, Urban and Environment), for the project of energy recovery from industrial wastes and/or agricultural processes of anaerobic co-digestion of sewage sludge to increase biogas production and, consequently, the generation of electricity. May 23, 2012. In addition, he is a pioneer member of iEMSs (International Environmental Modelling and Software Society) and a board member of iEMSs, too. He is also a pioneer member of the MIDAS (Spanish Network on Data Mining and Learning) and ÁTICA (Spanish network on Advance and Transference of Applied Computational Intelligence), and he is the main coordinator of the ÁTICA network. He is a member of the Editorial

Board of International Journal of Applied Intelligence and of the Environmental Modelling & Software journal. He has been researching in several projects granted by the Catalonian government, the Spanish council and the European Union (EU). He has authored 153 referred journal or conference publications, being 132 at international level and 21 at a national level, and is author/editor of 10 books. He has been an invited editor for some special issue journals, reviewer for some journals and has organised some international workshops and conferences in the AI field. From 1998, he co-organized several periodic workshops on Data Mining, IEDSS, Artificial Intelligence and Environmental Systems in international conferences, such as (BESAI in ECAI, DM-TES and AITENV in iEMSs). His main research topics are case-based reasoning, machine learning, data mining, recommender systems, knowledge discovery, knowledge engineering, intelligent decision support systems, and integrated AI architectures. He has an special interest on the application of AI techniques to Environmental systems, to Industrial systems (textile industry) and to e-Health systems (Assistive technologies).

DR. KARINA GIBERT (Female) (http://www.eio.upc.edu/homepages/karina) is a researcher at the Universitat Politècnica de Catalunya · BarcelonaTech (UPC) since 1990. B.Sc. in Computer Science from the Barcelona School of Informatics in 1990. M.Sc. in Computer Science from the Barcelona School of Informatics in 1991. Ph.D. in Computer Science (in the Ph.D. Program on Technical and Informatics Applications of Statistics, Operation Research and Optimization) from the Universitat Politècnica de Catalunya (UPC) in 1995. Associate professor at the Statistics and Operations Research department at UPC from 1990. She is a lecturer for graduate students related to her research topics at Barcelona School of Informatics (FIB), and Mathematics and Statistics Faculty (FME) from 1990; also, she is an academic staff in the Ph.D. Programme of Statistics and Operations Research, from 1994, and in the Ph.D. Programme of Artificial Intelligence from 1998. From 2005, she is a lecturer in the Interuniversity Master in Artificial Intelligence, from 2005, in the Interuniversity Master in Statistics and Operations Research, from 2008, at UPC, and both with the quality award from the Spanish Government from 2003; Also in Master in Health Care Innovation, Universidad de Cádiz, from 2010. She focus her research on the design of hybrid Artificial Intelligence and Statistics methodologies in Data Mining for intelligent data analysis of very complex domains to support complex decision-making. She also works in Intelligent Decision Support Systems (IEDSS) for both managing and control as well as in the formalization of expert-knowledge for decision-support. She has long experience on medical domains, and environmental systems, in which she has active research from 1990, being nominated consultant of the Mental Health Department of the WHO (World Health Organization) from February 2008 and elected fellowship from International Environmental Modelling and Software Society in 2008. She participated in several competitive projects, funded from both Spanish government or European projects, recently in K4Care, (IST-2004-026968), which was distinguished by the EU with the award of *best e-Health project of the month* in February 2010 because of the high quality standards achieved. She has published 93 papers in research journals, from which 55 in international journals, most of them well indexed (31 in ISCI journals), 26 book chapters and more than 200 papers in Conferences and Workshops. From 2001, she co-organized several periodic workshops on Data Mining, IEDSS Artificial Intelligence and Environmental Systems in big international conferences, such as (BESAI in ECAI, DM-TES and AITENV in iEMSs). She is pioneer member of ACIA (Catalan Association of Artificial Intelligence) and SCE (Catalan Statistics Association) and ordinary member of ACM (Catalan Association of Mathematics), ESTYLF (Spanish association of Fuzzy Logics), EUSFLAT (European Association of Fuzzy Logics and Technologies), BESAI (Binding Environmental Sciences and Artificial Intelligence), iEMSs (International Environmental Modelling and Software Society) and PSICOST Scientific Association. In addition, she is a pioneer member of the MIDAS (Spanish Network on Data Mining and Learning) and ÁTICA (Spanish network on Advance and Transference of Applied Computational Intelligence).

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. A. Gatzioura & M. Sànchez-Marrè. A Case-based Recommendation Approach for Market Basket Data. *IEEE Intelligent Systems*. Accepted. September 2014. IEEE Computer Science, ISSN 1541-1672. DOI: 10.1109/MIS.2014.57
2. M. Poch, U. Cortés, J. Comas, I. Rodríguez-Roda & M. Sànchez-Marrè. *Decisions on Urban Water Systems: some Support*. Servei de Publicacions, Universitat de Girona. ISBN 978-84-8458-401-8. September 2012. http://dugi-doc.udg.edu/handle/10256/7162
3. B. Sevilla & M. Sànchez-Marrè. Case-Based Reasoning Applied to Textile Industry Processes. *20th International Conference on Case-Based Reasoning (ICCBR'2012*). Lecture Notes in Computer Science LNAI, 2012, vol. 7466, p. 428-442. September 2012. DOI: 10.1007/978-3-642-32986-9_32
4. K. Gibert, M. Sànchez-Marrè, B. Sevilla. Tools for environmental data mining and intelligent decision support. *6th International Congress on Environmental Modelling & Software (iEMSs 2012*). iEMSs'2012 Proceedings, pp. 1726-1734. Germany, July 2012. ISBN: 978-88-9035-742-8. http://www.iemss.org/society/index.php/iemss-2012-proceedings.
5. K. Gibert, M. Sànchez-Marrè & I. Rodríguez-Roda. GESCONDA: An Intelligent Data Analysis System for Knowledge

Discovery and Management in Environmental Data Bases. *Environmental Modelling & Software* 21(1):116-121. January 2006. Elsevier Science, ISSN 1364-8152.

A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. Project "Big Data and real-time decision-making applied to food and beverage online industry" (BIG FOOD). (AVANZA project, TSI-100300-2013-161). Funded by Spanish Ministry of Industry, Energy and Tourism (MINETUR) (10.000€ for KEMLG, which has been subcontracted by Supercomparador de Consumo SL and Label Grup Desarrollo de la Información SL companies). January 2014-December 2016.
2. Project "SUstainable and PERsuasive Human Users moBility in future cities (SUPERHUB)" (ICT-2011, FP7-ICT-289067). Funded by European Union (463.133 € for KEMLG). October 2011-October 2014.
3. Project "Textile Work Intelligence by closed-loop of product and process quality in the Textile Industry (TexWIN)" (FP7-NMP- 246193-2). Funded by European Union (312.125 € for UPC). March 2010-March 2013.
4. Project "Rapid Application of Innovation in Small Medium Enterprises (RAISME)" (FP7-SME-2624269). Funded by European Union (100.800 € for KEMLG). December 2010-December 2012.
5. Project "Improvement and new developments in an Intelligent Data Mining System for Knowledge Extraction and Management in Environmental Databases (GESCONDA-II)" (TIN2004-01368). Funded by Spanish Ministry of Science and Education (MECD) (36.570 € for KEMLG). December 2004-December 2007.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

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Participant No.	8	**Name (short name)/ Country**	DECHEMA - Society for Chemical Engineering and Biotechnology e.V. (DECHEMA)/ Germany

Description of the legal entity

DECHEMA - Society for Chemical Engineering and Biotechnology e.V., is an interdisciplinary scientific society with more than 5,500 personal and institutional members (including about 500 SMEs), whose activities cover wide areas relevant to the process industries. DECHEMA is dedicated to the support of R&D progress and implementation in various fields of chemical engineering, including reaction engineering, catalysis, materials, process efficiency and environmental protection. DECHEMA is board member of the European Technology Platform Sustainable Chemistry (SusChem) and hosts several secretariats of organisations in this field: European Federation of Chemical Engineering (EFCE) and its international equivalent (WCEC), the European Federation on Corrosion (EFC) and the German Catalysis Society (GECATS). DECHEMA is organiser of the International Exhibition-Congress on Chemical Engineering, Environmental Protection and Biotechnology (ACHEMA) and of around 50 conferences with over 10.000 participants per year, and offers more than 20 training courses on different topics. Co-ordination and support of DECHEMA's activities is carried out by about 190 employees.

DECHEMA hosts more than 50 websites including project websites and secure collaborative workspaces for information exchange and knowledge management of project consortia.

DECHEMA hosts two secretariats for international (ISO) and European (CEN) standardisation and hosts the German standardisation (DIN) working group on laboratory apparatus.

Main tasks that will be performed

DECHEMA will be actively involved in activities related to maximizing the innovation impacts of the project, which will be executed through WP8. Within WP8, DECHEMA is mainly responsible for standardisation gap analysis, mapping of relevant standards and guiding the consortium in preparing for regulatory compliance. DECHEMA will lead the partners, with the support CCB in designing an effective communication strategy for engaging with Process Industries, Industrial Parks and other stakeholders, as well as disseminating the results to the European process industries, Industrial Parks and stakeholder groups. DECHEMA will also heavily assist CCB in the training activities to ensure upskilling in the industry in relation to IS and the use of the SHAREBOX platform. They will also contribute to communicating the requirements of the process industries in WP1 and ensuring that the process industry is included in the innovation cycle of the project as appropriate. Dr. Renata KÖRFER will act as the Communication and Dissemination Manager of the project.

How the legal entity profile matches the tasks in the proposal

DECHEMA has more than 25 years of experience in identifying and closing technological gaps, in facilitating communication between research, industry and the public sector and in supporting the market uptake of innovation. Research coordination and management; strong network in chemical engineering and process industries; high expertise in linking industry and R&D; strong skills in international communication, dissemination and exploitation.

CV of the persons who will be primarily responsible for carrying out the work

DR. RENATA KÖRFER (Female) is head of standardization department at DECHEMA and has 15 years of professional experience working in science and research management. Her background includes extensive knowledge of assessment and use of innovative process and methods in the field of process engineering especially in Food and Chemical industry. She has more than 8 years experience in coordinating and participating in various EU projects (FP6/FP7) as well as EC Project Technical Advisor (PTA), proposal evaluator and project reviewer. She is also an executive secretary of technical committees DIN 055, CEN 332 and ISO 48.

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. Member of PPP SPIRE
2. Secretary of SusChem
3. Member of the Managing Board, task force DECHEMA/GDCh 'Process analytical technologies (PAT)'
4. Executive secretary of technical committees DIN 055, CEN 332 and ISO 48General verification protocol (produced in AdvanceETV, adapted and translated in 11 languages by the EC and used by the EU ETV Pilot Programme - http://iet.jrc.ec.europa.eu/etv/verification-bodies)
5. Guide for Proposers to the EU ETV Pilot Programme" (produced in AdvanceETV, translated by the EC in 11 languages and used for the EU ETV Pilot Programme – http://iet.jrc.ec.europa.eu/etv/technology-proposers)

A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal

EU project coordination and management of NanoBioPharmaceutics, Next-GTL, Synflow, CO2NET, E4Water, ChemWater, M4CO2, R3Water

1. AdvanceETV: "Coordination Action on Environmental Technology Verification ETV – Building a framework for international cooperation" (FP7, Grant no. 226824)
2. ChemWater:
3. E4Water: "Economically and Ecologically Efficient Water Management in the European Chemical Industry" (THEME FP7 NMP.2011.3.4-1, Grant no: 280756)
4. R3Water: "Demonstration of innovative solutions for Reuse of water, Recovery of valuable substances and Resource efficiency in urban wastewater treatment" (FP7-ENV-2013-WATER-INNO-DEMO, Grant no.619093)
5. RiSKWa: "Risk Management of Emerging Compounds and Pathogens in the Water Cycle" (BMBF – German Federal Ministry of Education and Research)

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

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Participant No.	9	**Name (short name)/ Country**	Chemie-Cluster Bayern (CCB) / Germany

Description of the legal entity

CCB is a cluster network of 270 member institutions from the chemical and process industry in Bavaria and Southern Germany, including 40 universities and 8 Fraunhofer Institutes. Member companies cover producers of basic, fine and special chemicals, coatings, plastics and new materials. The cluster furthermore represents operators of industry parks (Mainsite, InfraServ Gendorf) and leading experts in chemical engineering, such as Linde or Siemens. CCB chairs the Project Council of the European Chemical Regions Network and is awarded with the "Gold Label for Cluster Excellence". The cluster has started an international "Value Creation Partnership" (http://www.wertschoepfungspakt-chemie.de/en/wpc/) to explore additional markets for chemical companies in certain industrial value chains. CCB is running special cooperation programs with international partner sites and industry parks, also including member companies of more than 20 partner clusters in Europe (mainly UK, France, Italy, Austria, Poland and Spain).

Main tasks you would propose to perform in the proposal

-commitment to WP 1, in particular input to the case study analysis from Bavarian Industry Parks
-bringing in at least 1-2 demonstration sites for applying the toolbox, supporting and monitoring implementation (WP5/6),
-adding a management component for industry park managers, so that the toolbox could easily be included into the service offer of park managers; conducting market studies of target users, development of a business model (WP7, we would be ready to take the lead),
-commitment to dissemination activities.

How your organisation is suited to these tasks

CCB's Competence Center for Sustainable Development of Industrial Sites coordinates the work of globally leading and well experienced construction and planning companies with a focus on process industry and industrial production in fields like bio-economy, clean-tech, renewable energies or other cutting-edge technologies as well as scientific clusters. By its

"Site Development Program", the Competence Center follows the quality approach of creating "Value Optimizing Parks". This label describes the concept of an industry park, which provides optimized value creation for partners on site, park operators, municipal authorities and the social community around the park. Value added for public authorities is being created by providing the label of a Model Industry Park, which helps – in combination with a tailor-made investment promotion concept – to attract high-value production in order to leverage local value creation.

CV of the persons who will be primarily responsible for carrying out the work

MR. DANIEL A. GOTTSCHALD (male)**.** Dipl.-Kfm. Daniel A. Gottschald has studied European Business Sciences at the universities of Bamberg, Germany, and Jyväskylä, Finland. Until 2009, he was responsible for Internationalization of Northern Hesse's industry network for renewable power generation, deENet e.V.. At same time, he was working as assistant to the CEO of the regional economic development agency. Since 2009, he is managing director of Chemie-Cluster Bayern GmbH. Mr. Gottschald has coordinated various industry projects at international level; he is president of the Cluster Excellence Expert Group and chairman of the Project Council of the European Chemical Regions Network (ECRN).

MRS. IRINA NUNBERGER (female). Irina Nunberger has studied economic cybernetics in Nowosibirsk (Russia) until 1994. She worked as investment analyst in several financial institutions and as expert in the field of regional and private sector development prior to her appointment as head of CCB's international department in April 2012. As project leader, she realized a broad number of international projects (EuropeAid, World bank, UNDP among others) in Europe, Caribbean and South-East Asia regarding regional competitiveness, technology transfer, strategic development of regions and industry parks for public donor organizations.

Competence Center for Industrial Site Development, expert in industry park management

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. International Site Development Program
2. Value Creation Partnership

A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. CEOPS project (NPM3-SL-2012-309984, http://www.ceops-project.eu/)
2. Joint planning and management of a Sino-European Eco-Industry Park in Yancheng, Jiangsu (China)
3. Framework cooperation of realizing a Technology Park for Shale Gas Production in Yibin, Sichuan (China)
4. Implementation of the Value Creation Partnership in the PlasChem Park, Jubail New City, Saudi-Arabia

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

CCB's Competence Center for Industrial Site Development is located at Industrie Center Obernburg, one of Bavaria's traditional industry parks, and works in close cooperation with the site operator Mainsite in order to implement new tools and structures of integrated site management. The network of the Site Development Program brings together leading cluster networks of European industry and science, experts in industry park planning and public authorities in order to support sustainable cluster building, planning, development and operation of eco-efficient industry parks on well-developed and emerging business sites all around the world.

Participant No.	10	**Name (short name)/Country**	KEROS CERÁMCIA S.L.

Description of the legal entity

KEROS CERÁMICA S.L. is a company founded in 1994. Its business is the manufacture and commercialization of ceramic products. The company has a facility (production and offices) in Nules (Castellón), the facility occupies 22.000 sqm in an area of 80.000 sqm. The annual production capacity is 4 millions sqm. Since 2009 a filial was set up in Russe (Bulgary) (KEROS BULGARIA EAD), where it produces white body wall and floor tiles, with an annual production capacity of 2,6millionssqm. KEROS BULGARIA has a spray drier acting in some cases as waste managers. The company employs 141 workers in Spain, showing annual sales of € 23 million. It has over 1500 customers in more than 80 countries. KEROS BULGARIA has a turnover of € 13.5 million, including over 800 current clients diversified in more than 60 countries, employing 139 people.

Main tasks you would propose to perform in the proposal

KEROS will act as coordinator of the companies located in Nules industrial park. Moreover it will carry out the following key tasks:

- Identification of the inputs: including water, fuels, materials, etc. (WP1)
- Identification of the outputs: solid wastes, dirty water, and sludges, wasted energy, etc. (WP1)

- Analysis of the general management issues that could be able to share among KEROS and other companies located in the same industrial area: security services, cleaning services, suppliers of some materials, etc. (WP5)
- Identification of new opportunities to improve the efficiency of the companies in their use of energy, raw materials and water by implementing new technologies. (WP5)
- Field trials of the developed tool in industrial conditions (WP6)

How your organisation is suited to these tasks

KEROS is located in the Nules industrial park selected for implementing SHAREBOX platform in the ceramic cluster. The company is aware of reducing the environmental impact of the ceramic tile production, having carried out different projects to reach this objective.

KEROS has an Environmental and Quality department that nowadays is collecting data about the environmental impacts of its production. This department will work actively in the project, mainly in gathering information about the current situation of ceramic tile production sites with respect IS (WP1).

Currently, the company keeps track of solid and liquid wastes generated in the manufacturing process. Wastes are managed through authorized agents which take them to their corresponding treatment centres (sprays dryers, etc..).

Regular checks of gaseous emissions are carried out in order to control and not exceed the minimum values required by the environmental regulations.

KEROS has also a R&D department that will be in charge of establishing the ceramic cluster requirements for IS (WP1), gathering information (WP5) and carrying out field trials in industrial conditions (WP6).

CV of the persons who will be primarily responsible for carrying out the work

PEDRO CASTEJÓN SEGARRA (Male), DIRECTOR OF ENVIRONMENTAL AND QUALITY DEPARTMENTS. 44 years old. Degree in Business Studies (Universidad Jaime I of Castellón). During 7 years he was devoted to teaching middle and upper level students about administrative subjects. In 1998 (17 years), he joined KEROS CERÁMICA, S.A. for the Quality Area. Nowadays he is the director and responsible for the implementation of the environmental management, quality and labour risk prevention systems.

Pedro Castejón as maximum responsible of the industrial waste management of the company. He will participate in collecting and feedback the information about inputs and outputs, type, quantities and management procedure.

CARLOS GIL ALBALAT (Male). R& D DIRECTOR, 36 years old, Degree in Chemistry and diploma in advance Studies about molecular chemistry and materials, mainly ceramic materials (Universidad Jaume I, Castellón). During 4 years was working in the Solid State Chemistry Research Group of the Inorganic Chemical Department of the University Jaume I (Castellón). During that time he acquired deep knowledge about ceramic materials (influence of the raw materials in the physical-chemical properties, synthesis and development of new body, frit and glaze compositions, etc.). Moreover he also acquired knowledge about experimental and instrumental techniques and physical-chemical analysis. Later, on 2005, he joined KEROS CERÁMICA S.A. where he is working as R&D director of the company. Carlos Gil is author of more than 30 papers in scientific and technological national and International journals and in different national and International congresses. He has managed more than 10 R&D projects financed by the regional government (IMPIVA), national government (PROFIT and CDTI) and international (LIFE and FP6).

Carlos Gil as responsible of the R&D project management will be in charge of the internal coordination of the project and the administrative management.

JOAQUIN EMILIO MARTÍ SEGARA (Male), GENERAL MANAGER, 47 years old, Degree on Chemistry, Specialization in Chemical Engineering, from the University of Valencia. He was working during 5 years in the "Instituto de Tecnología Cerámica (ITC)" of Castellón. Later on he joined KEROS CERÁMICA S.A. during its foundation in 1994. Since then he developed his professional career working in the ceramic sector. Over the past 16 years he has gone through almost all technical departments of the company (Operator in the section of glazes and inks preparation, in charge thereof, Head of Production and Technical Direction) until the current position. Hence he has a wide knowledge of the ceramic product, ceramic technology and market.

J**oaquin Emilio, as general manager and person that knows perfectly the ceramic sector, will be in charge of the coordination of the industrial park of Nules (Castellón), together with the rest of the project partners of the industrial park. He also will be on charge of monitoring the project activities.**

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. GARCÍA-TEN, F.J.; QUEREDA, Mª F.; **GIL, C**.; CHUMILLAS, D.; ZAERA, V.; SEGURA, Mª C. LIFE CERAM. Zero waste in ceramic tile manufacture. 2014
2.. **GIL,C**.; SILVESTRE , D.;PIQUER, J.; GARCÍA-TEN, J.; QUEREDA, Mª F.; VICENTE, M. J. Preparation of porcelain tile granulates by more environmentally sustainable processes. Bol. Soc. Esp. Ceram. Vidr., Vol 51, 2, 67-74, Marzo-Abril 2012, 2012
3. GARCÍA-TEN, F.J.; QUEREDA, F.; VICENTE, Mª. J.; **GIL, C.;** MIRALLES, C.; FLORS, J.R.; DE LA PEÑA, A. **OBTAINMENT OF CERAMIC TILES USING SUSTAINABLE GRANULATION SYSTEMS. QUALICER 2014**

A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal

European Projects:

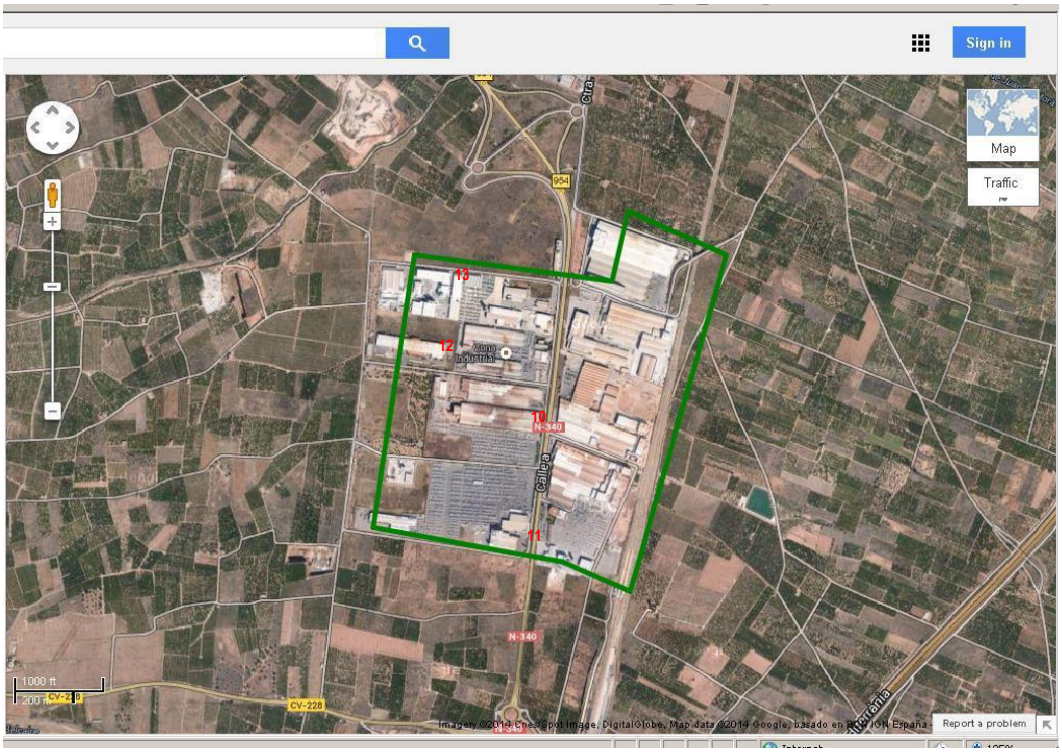
1. LIFECERAM (LIFE12 ENV/ES/230). "Zero waste in ceramic tile manufacture". http://www.lifeceram.eu/es/
2.EC project-G1RD-CT-2002-00783-MONOTONE: "Automatic Quality Control for Industrial Printing"

National Projects:

3. INIDIV/2010/020: "MICROGENERACION ELECTRICA A PARTIR DE CALOR RESIDUAL DE BAJAS TEMPERATURAS PROCEDENTES DE UN HORNO CERAMICO" (Electric microgeneration from waste heat at low temperatures coming from a ceramic kiln)
4. INIDIV/2009/145: "Investigación de variables críticas en hornos industriales para el desarrollo de un sistema de monitorización en continuo que minimice el consumo energético y en el impacto ambiental". (Research in critical variables of industrial furnaces for the development of a continuous monitoring system that minimizes energy consumption and environmental impact)

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

KEROS is located in the Nules Industrial Park (Castellón), which a photograph is included with the location of the different partners involved in the SHAREBOX project. 10-KEROS, 11-KERAFRIT, 12-ISSA, 13-GUZMAN



Environmental and R&D departments.

KEROS is aware of the environmental impact produced by industrial processes, in general, and in particular the one produced by the production of ceramic tiles. The company has spent years conducting research to improve the energy efficiency of the process. KEROS participated in the design and installation of various systems for the recirculation of combustion fumes in order to minimize the consumption of natural gas. Moreover, the company has also been working in a system of electrical microgeneration, innovative in the field of ceramic tiles production. This system produce electricity using the heat coming from the combustion fumes produced during the firing process of the ceramic tiles (in the ceramic

kilns).

Participant No.	11	Name (short name)/Country	KERAFRIT, S.A. (KERAFRIT)/Spain

Description of the legal entity

Kerafrit s.a is a chemical ceramic company with headquarters in Nules (Castellón) and was inaugurated in 1997; it is dedicated to the manufacture and commercialization of frits, enamels, colours, silk-screen printings, mediums, additives, digital inks, 3rd firing products and other ceramic materials.

In 2014 the Kerafrit commenced a new stage with a new majority shareholder (Ceramic Group). This made the company grow, providing a solid financial position, with more investments in added value products and an international expansion with better means, more technical personnel and focusing the commercial strategy in obtaining new markets.

With the help of a significant technological effort, of R&D investment and with the capture of new clients, and with the support of our new Ceramic Group we have increased our sales by 20 % compared to the previous years, reaching around of 20 M €/year.

Our working lines are focused on the creation of more added value products for our clients, and Kerafrit S.A. tries to continue growing, taking advantage of the synergy of its Ceramic Group, extending productive facilities in Spain, innovating with new products which are nowadays under investigation, with new patents and taking advantage of its growth on the export markets, as well as a great progress in our program of internationalization to be able to increase our assistance to the clients with a personalized, rapid and flexible service as well as effective and close as possible.

Main tasks you would propose to perform in the proposal

KERAFRIT will participate mainly in WP1, WP5 and WP6 carrying out the following tasks:

- Identification of the inputs: including water, fuels, materials, etc. (WP1)
- Identification of the outputs: solid wastes, dirty water, and sludges, wasted energy, etc. (WP1)
- Analysis of the general management issues that could be able to share among KERAFRIT and other companies located in the same industrial area: security services, cleaning services, suppliers of some materials, etc. (WP5)
- Identification of new opportunities to improve the efficiency of the companies in their use of energy, raw materials and water by implementing new technologies. (WP5)
- Field trials of the SHAREBOX tool in industrial conditions (WP6)

How your organisation is suited to these tasks

KERAFRIT is located in the Nules industrial park selected for implementing SHAREBOX platform in the ceramic cluster. The company is committed to reducing the environmental impact of the frit, pigment and glaze production.

KERAFRIT S.A. has a wide structure to manufacture products following the strictest procedures of quality and environmental protection. Our investments have allowed us to **extend our production capacity**, to improve our facilities of storage (warehouse), as well as to improve our facilities of **environmental protection**, with a new station of purification and new filters to reduce the emission to the atmosphere in lower levels than the allowed ones for the in force legislation. Kerafrit S.A. has the **ISO-9001:2008** Certification, and also the **ISO-14001:2004** for a suitable environmental management. Kerafrit takes care of the environment because of the good use of the existing technologies in order to minimize the emissions, water consumption and for a better **environmental protection**.

KERAFRIT has an Environmental Department that nowadays is collecting data about the environmental impacts use of raw materials, electricity, gas, water for its production and emissions of CO_2. This department will work actively in the project, mainly in gathering information about the current situation of ceramic tile production sites with respect to IS (WP1).

KERAFRIT has also a R&D department that will be helping KEROS to establish the ceramic cluster requirements for IS (WP1), gathering information (WP5) and carrying out field trials in industrial conditions (WP6).

CV of the persons who will be primarily responsible for carrying out the work

Rosario Flors Rogla (Female) → **R&D and Quality Control Laboratory Manager** -
Chemistry Degree
-Expertise in research and development of ceramic frits, special enamels and nanotechnology
-Expertise en FRX
Languages: English (medium)

Carol González Tardif de Petiville (Female) → **R&D Technician** -
Master in Science and Technology of Ceramics
-Master in Environmental Science and Enterprise

-Chemical Degree
-Expertise in inkjet development (nanotechnology)
Languages: English, French

Sonia Hernandez Gozalbo (Female) → **Quality & Environmental Manager**
-Master in Environmental Science and Risk Prevention
-R&D management course
-Chemical Degree
-Expertise in ISO-standards, Legislation, Environmental tasks and emissions.
Languages: English, French (medium)

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

Our installation measures the air emissions of their focus with the periodicity established in the law. Some ashes recollected in the filters are reintroduced to the kiln, so we can reduce wastes. We have a defined reduction plan for wastes that we have incorporated in our internal environmental objectives. The water has a treatment system to recover in our process and we have no waste.

A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal

"**Nanoglaze",** is the project which has granted us the "Alfa de Oro 2009" award. It's a product fabricated on the basis of two patents developed by the ICMM research team.
This technological success is the result of previous studies carried out by both the ICMM and ICV teams on the basis of:
•the so-called percolative materials
•the fact that the nanometals are embedded in sepiolite fibers
•the dewetting process of the metallic nanoparticles which takes place
•during the fast firing of the ceramic glaze
•the scale-up process which was performed at both laboratories.
Coordinator of the project: Fernando Lucas (Kerafrit, S.A.)

"**Kershield",** is another Project developed with the collaboration of the Technical Ceramic Institute, is a project that allows producing ceramic pieces with a significant reduction to the X radiations. This has been obtained by the development of a new ceramic paste with a high percentage of PbO, Bismuth and Barium. This product can be used in Hospitals, Radiological Rooms, Protection zones, etc. This Project is in process of being patented and and it wil be presented in the Alfa of Gold in Cevisama 2015. Coordinator of the project: Fernando Lucas (Kerafrit, S.A.)

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

KERAFRIT is located in the Nules Industrial Park (Castellón), 11-KERAFRIT in the photograph included in KEROS partner profile.
4 KILNS
4 INKJET MILLS
R&D LABORATORY
TECHNICAL ASSISTANCE LABORATORY
INDUSTRIAL PLOTTER
4 ENAMELS MILLS
3 MEDIUM MIXERS
1 RINAJET (SECADERO DE ESMALTES)
FRX
DILATOMETER
PARTICLE SIZE DISTRIBUTION EQUIPMENT
SURFACE TENSION EQUIPMENT

Participant No.	12	**Name (short name)/ Country**	Ibérica de Suspensiones S.L (ISSA) /Spain

Description of the legal entity

Ibérica de Suspensiones S.L (ISSA) produces helicoidally suspensions for the automotive industry; it belongs to the metallurgic sector.

Founded in 1989 and headquartered in Alsasua (Spain) ISSA is a joint venture between Sogefi group and NHK Spring. The company offers products ranging from the design to the manufacturing and supply of suspension coil springs and stabilizer bars made both with cold and hot forming technologies. It operates in business segments including the supply of OEM parts for passenger cars and light commercial vehicles. ISSA has 212 employees in the two manufacturing sites of Alsasua and Nules, and in 2012 its turnover was 49.2 Million €.

The plant in Nules occupies 11310 sqm in an area of 55000 sqm, and automotive suspensions spring are manufactured. In this plant ISSA has highly automated means to manufacture all kinds of coil springs.

Main tasks you would propose to perform in the proposal

ISSA will mainly carry out the following tasks:

- Identification of the inputs: including water, fuels, materials, etc. (WP1)
- Identification of the outputs: solid wastes, dirty water, and sludges, wasted energy, etc. (WP1)
- Field trials of the tool developed in industrial conditions (WP6)

Share the results of the project between the metallurgic sector. (WP7)

How your organisation is suited to these tasks

ISSA is located in the Nules industrial park selected for implementing SHAREBOX platform in the ceramic cluster. Even ISSA is a company that does not belong to the ceramic cluster, it is typical that in the industrial areas were ceramic product producers are located, other companies are also set up there. Because of that, it is considered that the participation of ISSA in the project is important in order to demonstrate that there is also possibility of industrial symbiosis between companies of different sectors. ISSA is in metallurgic sector. ISSA is aware of reducing the environmental impact of its production, having carried out different projects to reach this objective.

CV of the persons who will be primarily responsible for carrying out the work

ISABEL SIERRA (Female). Industrial Engineer, Environmental Speciality. Master in labours risks prevention (3 specialities: industrial safety, industrial hygiene and ergonomic & psychosocial). She is working in ISSA since September 2003 in the Quality Department, being responsible of security and environmental issues

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. PRTR (European Pollutants Release and Transfer Register)

A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. ISO 14.001
2. Internal projects related to the reduction of wastes and depuration of wastewater using physico-chemical treatments.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

ISSA is located in the Nules Industrial Park (Castellón): 12-ISSA in the photograph included in KEROS partner profile.

Water waste treatment plant

Participant No.	13	**Name (short name)/ Country**	GUZMAN GLOBAL S.L. (GUZMAN)/Spain

Description of the legal entity

GUZMAN GLOBAL is a leading raw materials distributor company based in Spain which belongs to Grupo GUZMAN, a family-owned business. Guzmán was founded in 1939 and our core business is and has always been the distribution of raw materials, from minerals to rubber, polymers, chemicals, raw coffee beans, nuts and dried fruits amongst others.

But we are not only raw materials suppliers since we also cover essential aspects for our customers as quality control, technical assistance, research & development & innovation as well as logistics services according to their needs.

Our mission is to provide products and services to our customers with high standards of excellence, dynamism and innovation.

MINERALS BUSINESS UNIT. The business unit of **GUZMAN GLOBAL** specializing in industrial minerals primarily covers the ceramic industry, but is also present in sectors as diverse as glass, plastics, refractory, paints, etc. The production plant (I) located in **Nules** (Castellón) covers 55,000 m^2 of facilities and has a total capacity of milling and processing for 185,000 tons/year of zirconium sand, nepheline syenite, feldspar sodium, alumina and spodumene among other products. The production plant (II) located in the **Port of Castellón** has warehouses covering 22,000 m^2 for optimising logistical needs, mixtures of clays and capacity for bulk load. We have a laboratory with the latest technical means for quality control, formulation and product development.

Main tasks you would propose to perform in the proposal

GUZMAN will participate in WP1, WP5 and WP6 carrying out the following tasks:

- Identification of the inputs: including water, fuels, materials, etc. (WP1)
- Identification of the outputs: solid wastes, dirty water, and sludges, wasted energy, etc. (WP1)
- Analysis of the general management issues that could be shared with the other companies located in the same industrial area: security services, cleaning services, suppliers of some materials, etc. (WP5)
- Identification of new opportunities to improve the efficiency of the companies in their use of energy, raw materials and water by implementing new technologies. (WP5)
- Field trials of the tool developed in industrial conditions (WP6)

How your organisation is suited to these tasks

GUZMAN GLOBAL forms part of the industrial park selected for implementing SHAREBOX platform and has been collaborating with ITC in different R&D projects.

The company has developed some projects related to raw materials alternatives for the tile manufacturing companies in order to get energy benefits and environmental advantages.

The company is actually collecting data of inputs and outputs and has the capacity to determine which ones can be shared with other companies.

CV of the persons who will be primarily responsible for carrying out the work

Santiago Domínguez Perelló (Male)**: Head of Industrial Operations. Minerals B.U.**

- Industrial Engineering from Universidad Politécnica of Valencia in 2001.
- Master Degree in in Engineering Maintenance from Universidad Politécnica of Valencia in 2009-2010.
- Degree in Organization and Engineering of Production and Management of Industrial Plants from Universidad Politécnica of Cataluña in 2013.
- Expertise in the development and management of Maintenance, Industrial Operations and Production Projects since 2002.
- Languages: English

Raul Carregui Balaguer (Male)**- Industrial Technician. Minerals B.U.**

- Technician of Management IT Systems from Universitat Jaume I in 2002
- Expertise in general computer maintenance
- Since 2005 he has developed performance management of the function module production in current ERP (enterprise resource planning), being part of the computing tasks. He has also carried out tasks as support center for help at startup and commissioning of the two ERP implanted along these years "Oracle e -business suite" and " Microsoft Dynamics Navision " in accounting, billing , purchasing, payments, production, inventory and ordering modules.
- Languages: English

Ana Orenga Castelló (female)**: R& D and Quality Control Technician. Minerals B.U.**

- Chemistry Degree from Universitat Jaume I in 1997.
- Expertise in research and development of raw materials, ceramic applications and milling technologies since 2002.
- Expertise in Quality ISO-standards and management of Quality System since 2005.
- Languages: English

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

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A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal

Development of additives for energy improvement in porcelain tile firing. Project developed in collaboration with the Technical Ceramic Institute.

In this study a material has been developed that can be added to porcelain tile compositions to increase the sintering rate, thus reducing energy consumption and CO2 emissions. Lithium oxide was selected as the base constituent of the additive, which is a vigorous flux that does not display the disadvantage of increasing the tendency to deform by pyroplasticity. The use of these fluxing additives allows:

- A considerable reduction in the peak firing temperature (about 40ºC with an addition of 5%).
- The duration of the firing cycle to be reduced.

- Energy consumption and CO2 emissions to be reduced.

Use of wollastonite in wall tile manufacture. Environmental Implications and product quality. Project developed in collaboration with the Technical Ceramic Institute.
The use of wollastonite instead of calcite in frits and ceramic bodies provides different properties and advantages both environmental implications and product quality.
These advantages may be particularly important for frits and glazes industries, also for tile manufacturing industries, especially those affected by the Kyoto protocol.
From the study conducted, it has been able to confirm the following improvements:

ENERGETIC IMPROVEMENTS

- Reduction of the firing cycle in tile manufacture (17% reduction) and thus a reduction in gas consumption.
- Decrease of energy to melt raw materials in frit manufacture (5,5% maximum).
- Decrease of energy to fire tiles in an industrial kiln (6,8% maximum).
- Decrease CO_2 emissions.

PRODUCT QUALITY IMPROVEMENTS

- Decrease of infused content in frits produced with wollastonite, improving the appearance of the glazes developed from it.
- Decrease of "foams" in the production process of melting frits, achieving more homogeneous compositions.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

GUZMAN is located in the Nules Industrial Park (Castellón): 13-GUZMAN in the photograph included in KEROS partner profile.
6 continuous industrial dry ball mills
1 impact mill
R& D Department
Technical assistance laboratory
Two particle distribution equipments
FRX Chemical analysis equipment

Participant No.	14	**Name (short name)/ Country**	Bristol Port Company (BPC)/United Kingdom

Description of the legal entity

Withdrawn from the consortium in M1.

Main tasks that will be performed

-

How your organisation is suited to these tasks

-

CV of the persons who will be primarily responsible for carrying out the work

-

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

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A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal

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Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

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Participant No.	15	**Name (short name)/ Country**	Eskisehir Chamber of Industry (ECI) / Turkey

Description of the legal entity

Eskişehir Chamber of Industry (ECI) was founded in 1968 by the law no: 5174 and is a professional public non-governmental organization. ECI has been established with the objectives of accelerating industrialization and dealing with the problems of the industry in the region. ECI has 700 members from various sectors including the aviation, rail systems, machinery production, metal working, white goods, food, textile industries etc. The total exports of the members summed up to 2.2 billion USD in 2013, their turnover is approximately 8 billion USD and total work force is around 45.000 people.

The services that ECI provides to its member companies can be summarized as below;

- To pioneer and work for development of the industry for local and public benefits,
- To ensure vocational information to its members,
- To make researches, surveys and statistical studies,
- To take compulsory decisions and give vocational opinions to related units,
- To execute courses, seminars and make informative activities through working groups,
- To ensure foundation of organized industrial sites and their administration,
- To prepare and implement projects in order to develop the regional industry and sectors.

Eskisehir Chamber of Industry is the operator of Eskisehir Organized Industrial which is the largest OIZ of Turkey with its 32 million m2 area. Today, 18 million m2 is allocated to the investors, 550 companies are registered in the Zone and 400 of them are actively working in different industrial fields.

Main tasks you would propose to perform in the proposal

Eskisehir Chamber of Industry will lead the field testing work and trials in Turkey and will actively contribute to WP5 and WP6. They will work with industry, deploying the platform, gathering feedback and identifying opportunities. Eskisehir Chamber of Industry will provide information at the phase of developing the new software within the project and when the new software is available, ECI will attend the tests and implement the software and populate with its companies' data.

How your organisation is suited to these tasks

Eskisehir Organized Industrial Zone (EOIZ) was founded in 1969 under the leadership of Eskişehir Chamber of Industry. The management of EOIZ has been carrying out its duties which are included in the coverage of the related law and its district regulations. The activities are as follows:

- To serve establishments that are supposed to make investments in the district, gathering the supplementary facilities of technical and business management,
- To present investment areas of which infrastructure is completed,
- To develop EOIZ in a modern and environmentally friendly manner,
- To support the development of modern industries and industrial approaches in terms of energy efficiency, environment and green production,
- To enable the main and the supplementary industries corporate,
- To help increase the economic growth and employment in the province,
- To complete the tasks given by certain institutions of the state and do studies on given subjects, for the country benefits,
- To carry out the tasks by the law and the Ministry of Commerce and Industry.

Eskisehir Organized Industrial Zone is one the best environment friendly zones of Turkey and was awarded by the Ministry of Environment in 2011. EOIZ is already carrying out activities about energy and waste management for the factories located in the industrial zone. Eskisehir Chamber of Industry has got also some other partnerships in organized industrial zone such as technology development area, natural gas power plant, logistic center, internet service provider, waste water treatment center, clusters and vocational training units. These facilities make ECI a suitable partner for the project especially in the data collection and test application of the new software during the implementation.

CV of the persons who will be primarily responsible for carrying out the work

ISMAIL OZTURK (Male) was born in 1981 and graduated from Uludag University, Econometrics Department in 2002. He used to work in a consulting company as a Business Plan Advisor in 2004. After completing his language education in England, he worked in an EU Regional Development Program as a procurement expert in 2006. He has been working in the Eskişehir Chamber of Industry as the Deputy Secretary General for 8 years. He is responsible for preparing and implementing national, international, EU projects, organizing fairs, trade missions to foreign countries and providing consultancy to members about foreign trade procedures.

SUZAN E. ÖNPEKER (Female) was born in 1978 and graduated from Trakya University, Environmental Engineering Department in 2000. She received her Master of Business Administration degree in 2004. She used to work in a holding company producing ceramic tiles as an environmental, quality, health & safety engineer until 2009. She has been working

in the Eskişehir Organized Industrial Zone as an Environmental Engineer for 5 years. She is responsible for environmental management unit, implementing waste management and legislation, controlling, prepared and implementing an environmental project with Development Agency (BEBKA), attended seminar and meetings about industrial symbiosis & energy efficiency. She is also making an academic research to prepare thesis at Anadolu University, Environmental Engineering MSc Programme.

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. ECI Magazine, Year: 2014, No: 8, Cover Subject: Industrial Symbiosis and Industry 4.0
2. Complete infrastructure services
3. Water, electricity, natural gas, steam and sewage lines, telephone and support services
4. Clustering activities
5. Cooperation with relevant organizations such as TUBITAK, BEBKA, TTGV etc.

A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. Clean Aerospace Regions (CARE) Project, www.care-aero.eu, FP7-REGIONS-2011-1, ID: 286560
2. Enterprise Europe Network (EEN) Project, COS-EEN-2014-2-01, ID: 649503
3. Internationalize SMEs Project, ETCF-II, ID: TR0604.03-02/002
4. Trainings and seminars provided for member companies

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Eskisehir Organized Industrial Zone is the subsidiary of Eskisehir Chamber of Industry. Eskisehir Organized Industrial Zone has significant infrastructure and facilities such as waste water treatment center, technology development area, natural gas power plant, logistic center, internet service provider, clusters and vocational training units which are relevant to proposed work. As an umbrella organization of manufacturers located in Eskisehir, Chamber of Industry has got a network and communication with companies which are useful for the implementation of the project.

Participant No.	16	**Name (short name)/ Country**	THE NORTH EAST OF ENGLAND PROCESS INDUSTRY CLUSTER (NEPIC)/ UNITED KINGDOM

Description of the legal entity

The North East of England Process Industry Cluster (NEPIC) is a stand-alone company, limited by guarantee, which was created and is owned by its member companies to represent the companies and supply chain of the Process Industry in the region. NEPIC's members are Chemical, Petrochemical, Polymer, Specialty Chemical, Fine Chemical, Pharmaceutical, Biotechnology, Composites, Bioresources, Biofuels and Renewable Materials and Energy Companies. As well as companies and organisations in their various supply chains all working in partnership through NEPIC to improve the economic environment in which they operate. Working through the cluster they are improving productivity & efficiency, continuously updating the infrastructure of this industrial locality, informing stakeholders and growing the skills base. They cooperate through the Cluster to promote the regions capabilities as well as the capability of individual member companies. All of this work is aimed at the rejuvenation and growth of these key hi-tech advanced manufacturing industries while leading them into a lower carbon sustainable future. Through a high level of engagement with its member companies and the credibility and support NEPIC receives from both public and private sector stakeholders, it has become a unified voice for the Process Industry in the North East England, where a substantial part of the UKs chemical, petrochemical, speciality chemical, pharmaceutical, polymer, biotechnology industries are based.

Main tasks that will be performed

Main role is to assist the consortium in defining the system specifications for the SHAREBOX platform in WP1 by contributing their knowledge of the requirements of their process industry members (which will be gathered through a dedicated workshop). They will also set up symbiosis using the SHAREBOX platform in WP5 and perform field trials in WP6 with the connected companies. During WP7, NEPIC will contribute to dissemination activities.

How your organisation is suited to these tasks

NEPIC is a process industry cluster with over 720 active participants. Sector engagement and connectivity across differing sectors and disciplines is a core activity of the cluster.

CV of the persons who will be primarily responsible for carrying out the work

JOHN BRADY, Male (NEPIC Project Director Industry) – created a new division for NEPIC in 2011 www.nebr.co.uk to exploit business growth opportunities in a developing low carbón economy. Responsible for international development focussing upon India, Brazil and China. Successful project experience at both ERDF and FP7 level. Active member of

ECRN, ECCP and SPIRE.
DR STAN HIGGINS, Male (NEPIC CEO) – established NEPIC as an internationally recognised private company, to develop the North East of England process industry. NEPIC is the only EU gold label accredited cluster in the UK. Since 2004, working in conjunction with industry, NEPIC has generated 2.5Bn GBP of added value (GVA) for the regional economy. He is the current European Cluster Manager of the Year.

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. Partnerships For Business – Dr Stan Higgins, Chemical News, November 2013.
2. Understanding and Supporting Supply Chains - Dr Stan Higgins, NEPIC Directory 2013 (40-48).
3. The Power of the Collaborative **–** Dr Stan Higgins - NEPIC Directory 2014 (Page 21-23).
4. Building a Region's Process Industry Innovation Strategy – Dr Stan Higgins/John Brady - Chemical News, May 2013 (pages 16-20)
5. A Sustainable Future (A Clustering Powerhouse) –John Brady - The NEPIC Directory (page 15) – June 2015.

A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal

1. DEELOCSI (Delivering Energy Efficient Low Carbon Sustainable Industry) - a part funded ERDF Project focussed upon the "greening up" of consumer products arising from manufacturing and its supply chain. SMEs are under pressure to assess and submit their resource efficiency and carbon footprint for products going into consumer markets. In order for companies to maintain and grow their businesses into the future, the manufacturing and its supply chain must be able to demonstrate and report a creditable assessment of their contribution to the green credentials. Companies who can clearly demonstrate an understanding and deliver this data will place themselves ahead of competitors that cannot provide such information and ultimately win new business.

2. LOCIMAP (Low Carbon Industrial Manufacturing Parks) – funded through FP7. With NEPIC as project coordinator, this resulted in a number of White Papers published. Objectives included developing a future model for Industrial Parks whilst understanding the opportunities European industrial parks offer to improve resource efficiency across industry sectors and along industry supply chains.
3. BASME – (Business Acceleration For SMEs) – a 3 year project funded by UK National Government under the Regional Growth Fund (RGF).

4. Go Global – a project on behalf of UK Trade and Investment (UKTI) to develop international trade through sector engagement.

5. www.nebr.co.uk NEPIC's dedicated bioresources division where industrial symbiosis is a major component in helping reduce carbon impacts.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

-

Participant No.	17	**Name (short name)/ Country**	University of Strathclyde (USTRATH) /UK

Description of the legal entity

University / Higher Education Institute. The University of Strathclyde has the largest engineering faculty in Scotland. It is a research-intensive university, ranked in the UKs top 10 for engineering and physical sciences. The University has a multi-disciplinary and internationally leading energy research community spanning technical, economic and business aspects of the energy space.

Main tasks you would propose to perform in the proposal

As part of WP2, working closely with partners at Leeds University we will assess, catalogue and derive schemes for optimisation of new energy transfer, storage and conversion infrastructure. These will be required to realise the IS opportunities identified by the SHAREBOX programme. We will also identify and develop business opportunities and plans for additional actors in the form energy service companies within the industrial ecosystem.

How your organisation is suited to these tasks

Energy is one of the University's four main research themes and as such activity cuts across all departments and schools in the organization. Strathclyde is a member of the **Energy Technology Partnership** (ETP), an alliance of Scottish universities chaired by Strathclydes Principal, Professor Sir Jim McDonald. With more than 250 academics and 700 researchers, the ETP is one of the largest power and energy research networks in Europe.
Strathclyde is home to two energy Centres for Doctoral Training in Wind and Marine Systems and Future Power Networks

and Smart Grids. The Department of Mechanical and Aerospace Engineering is home to the Energy Systems Research Group whose research is focused on energy efficiency and new and renewable energy system deployment. The group is a Building Research Establishment Centre of Excellence in Energy Utilisation. Strathclyde also has a Centre for Energy Policy Research which focusses on policies related to energy production, security and sustainability of supply and the economic and environmental impacts of energy policies.
CV of the persons who will be primarily responsible for carrying out the work
PROFESSOR ANDREW L HEYES (Male) PROFESSOR OF MECHANICAL ENGINEERING**.** Professor Andrew Heyes, is Professor of Mechanical Engineering and Head of Department in Mechanical and Aerospace Engineering. He has held the post since October 2015 having previously been Chair of Energy Technology and Environment at the University of Leeds from August 2013 and having spent eighteen years as an academic member of staff at Imperial College London where he was Deputy Director of the Energy Futures Lab. Professor Heyes is an expert in experimental thermofluids, heat transfer and thermodynamics. He has published over eighty papers in this field at international symposia and in archival journals, has five patents and is a founder and director of a spin-out company set-up to develop technology developed by his research group. In the field of energy technology, he has authored a number of papers on the novel thermodynamics of chemical looping combustion systems for efficient energy conversion with inherent carbon capture. In 2009 one such paper was awarded the International Gas Turbine Institutes, Cycle Innovations Committee best paper award.
List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content
1. McGlashan N.R, Childs P.R.N, Heyes A.L, "Chemical Looping Combustion Using The Direct Combustion Of Liquid Metals In A Gas Turbine Based Cycle" Trans. ASME: J. Eng. Gas Turb. And Pow, 2011, Vol. 133, Issue 3.2.
A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal
-
Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work
-

4.2. Third parties involved in the project (including use of third party resources)

Participant no. 11- KERAFRIT / Spain

Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted)	Y
If yes, please describe and justify the tasks to be subcontracted Environmental Certificate Organisation for measuring and evaluating flow and temperature that we cannot measure. Amount: € 4K.	
Does the participant envisage that part of its work is performed by linked third parties	N
If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party	
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement)	N
If yes, please describe the third party and their contributions	

5. Ethics and Security

5.1. Ethics

In keeping with the required Ethics Appraisal Procedure, an Ethics Self-assessment was performed with the completion of the Ethical Issues Table. It can be concluded that this project does not involve any ethics issues.

5.2 Security

Please indicate if your project will involve:

- activities or results raising security issues: (NO)
- 'EU-classified information' as background or results: (/NO)

ESTIMATED BUDGET FOR THE ACTION (page 1 of 2)

	Estimated eligible[1] costs (per budget category)									EU contribution			Additional information		
	A. Direct personnel costs A.1 Employees (or equivalent) A.2 Natural persons under direct contract A.3 Seconded persons [A.6 Personnel for providing access to research infrastructure]		A.4 SME owners without salary A.5 Beneficiaries that are natural persons without salary		B. Direct costs of subcontracting	[C. Direct costs of fin. support]	D. Other direct costs D.1 Travel D.2 Equipment D.3 Other goods and services D.4 Costs of large research infrastructure	E. Indirect costs[2]	Total costs	Reimbursement rate %	Maximum EU contribution[3]	Maximum grant amount[4]	Information for indirect costs Estimated costs of in-kind contributions not used on premises	Information for auditors Declaration of costs under Point D.4	Other information: Estimated costs of beneficiaries/ linked third parties not receiving EU funding
Form of costs[6]	Actual	Unit[7]	Unit[8]		Actual	Actual	Actual	Flat-rate[9] 25%							
	(a)	Total (b)	No hours	Total (c)	(d)	(e)	(f)	(g)=0,25x ((a)+(b)+ (c)+(f) +[(h1)+(h2)]- (m))	(i)= (a)+(b)+(c)+ (d)+(e)+(f)+ (g)+(h1)+(h2)+(h3)	(j)	(k)	(l)	(m)	Yes/No	
1. IRIS	777920.00	0.00	0	0.00	0.00	0.00	109600.00	221880.00	1109400.00	100.00	1109400.00	1109400.00	0.00	No	
2. ISL	941380.00	0.00	0	0.00	0.00	0.00	78180.00	254890.00	1274450.00	100.00	1274450.00	1274450.00	0.00	No	
3. UT	545536.00	0.00	0	0.00	0.00	0.00	34200.00	144934.00	724670.00	100.00	724670.00	724670.00	0.00	No	
4. UNIVLEEDS	49724.00	0.00	0	0.00	0.00	0.00	23547.00	18317.75	91588.75	100.00	91588.75	91588.75	0.00	No	
5. ZHAW[13]									493502.50						493502.50
6. AICE	262432.00	0.00	0	0.00	0.00	0.00	26200.00	72158.00	360790.00	100.00	360790.00	360790.00	0.00	No	
7. UPC	199111.00	0.00	0	0.00	0.00	0.00	28200.00	56827.75	284138.75	100.00	284138.75	284138.75	0.00	No	
8. DECHEMA	103772.00	0.00	0	0.00	0.00	0.00	35000.00	34693.00	173465.00	100.00	173465.00	173465.00	0.00	No	
9. CCB	165078.00	0.00	0	0.00	0.00	0.00	26200.00	47819.50	239097.50	100.00	239097.50	239097.50	0.00	No	
10. KEROS	134100.00	0.00	0	0.00	0.00	0.00	30200.00	41075.00	205375.00	100.00	205375.00	205375.00	0.00	No	
11. KERAFRIT,S.A.	139840.00	0.00	0	0.00	4000.00	0.00	36000.00	43960.00	223800.00	100.00	223800.00	223800.00	0.00	No	
12. ISSA	37317.00	0.00	0	0.00	0.00	0.00	31000.00	17079.25	85396.25	100.00	85396.25	85396.00	0.00	No	
13. GUZMAN GLOBAL	96370.00	0.00	0	0.00	0.00	0.00	14800.00	27792.50	138962.50	100.00	138962.50	138962.50	0.00	No	
14. Bris-Port	0.00	0.00	0	0.00	0.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00	0.00	No	
15. ESO (ECI)	67200.00	0.00	0	0.00	0.00	0.00	26200.00	23350.00	116750.00	100.00	116750.00	116750.00	0.00	No	
16. NEPIC	101200.00	0.00	0	0.00	0.00	0.00	6000.00	26800.00	134000.00	100.00	134000.00	134000.00	0.00	No	
17. USTRATH	159300.00	0.00	0	0.00	0.00	0.00	44429.00	50932.25	254661.25	100.00	254661.25	254661.25	0.00	No	
Total consortium	3780280.00	368602.00		0.00	4000.00	0.00	575956.00	1181209.50	5910047.50		5416545.00	5416544.75	0.00		493502.50

ESTIMATED BUDGET FOR THE ACTION (page 2 of 2)

(1) See Article 6 for the eligibility conditions
(2) The indirect costs covered by the operating grant (received under any EU or Euratom funding programme; see Article 6.5.(b)) are ineligible under the GA. Therefore, a beneficiary that receives an operating grant during the action's duration cannot declare indirect costs for the year(s)/reporting period(s) covered by the operating grant (see Article 6.2.E).
(3) This is the theoretical amount of EU contribution that the system calculates automatically (by multiplying all the budgeted costs by the reimbursement rate). This theoretical amount is capped by the 'maximum grant amount' (that the Commission/Agency decided to grant for the action) (see Article 5.1).
(4) The 'maximum grant amount' is the maximum grant amount decided by the Commission/Agency. It normally corresponds to the requested grant, but may be lower.
(5) Depending on its type, this specific cost category will or will not cover indirect costs. Specific unit costs that include indirect costs are: costs for energy efficiency measures in buildings, access costs for providing trans-national access to research infrastructure and costs for clinical studies.
(6) See Article 5 for the forms of costs
(7) Unit : hours worked on the action; costs per unit (hourly rate) : calculated according to beneficiary's usual accounting practice
(8) See Annex 2a 'Additional information on the estimated budget' for the details (costs per hour (hourly rate)).
(9) Flat rate : 25% of eligible direct costs, from which are excluded: direct costs of subcontracting, costs of in-kind contributions not used on premises, direct costs of financial support, and unit costs declared under budget category F if they include indirect costs
(10) See Annex 2a 'Additional information on the estimated budget' for the details (units, costs per unit).
(11) See Annex 2a 'Additional information on the estimated budget' for the details (units, costs per unit, estimated number of units, etc)
(12) Only specific unit costs that do not include indirect costs
(13) See Article 9 for beneficiaries not receiving EU funding
(14) Only for linked third parties that receive EU funding